

MAD LAB REPORT

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INTRODUCTION TO ANDROID

PROBLEM STATEMENT 1

Develop an Android application for the following program

1. Create an Android application to show the demo of displaying text with justifying elements, changing text colors ,fonts etc.
2. Find the “hello word” text in the XML document and modify the text.

SOLUTION

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">
    <TextView
        android:id="@+id/titleText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Text Formatting Demo"
        android:textSize="24sp"
        android:textColor="#FF5722"
        android:textStyle="bold"
        android:fontFamily="sans-serif-black"
        android:layout_marginBottom="20dp"/>
    <TextView
        android:id="@+id/helloText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```

        android:text="Hello Android World!"
        android:textSize="20sp"
        android:textColor="#3F51B5"
        android:fontFamily="cursive"
        android:layout_marginBottom="20dp"/>
<TextView
    android:id="@+id/justifiedText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="This is a long paragraph intended to
demonstrate how text justification works in Android. The text
should spread evenly across the width of the screen, creating
clean edges on both the left and right sides."
    android:textSize="16sp"
    android:textColor="#000000"
    android:justificationMode="inter_word"
    android:fontFamily="serif"/>
</LinearLayout>

```

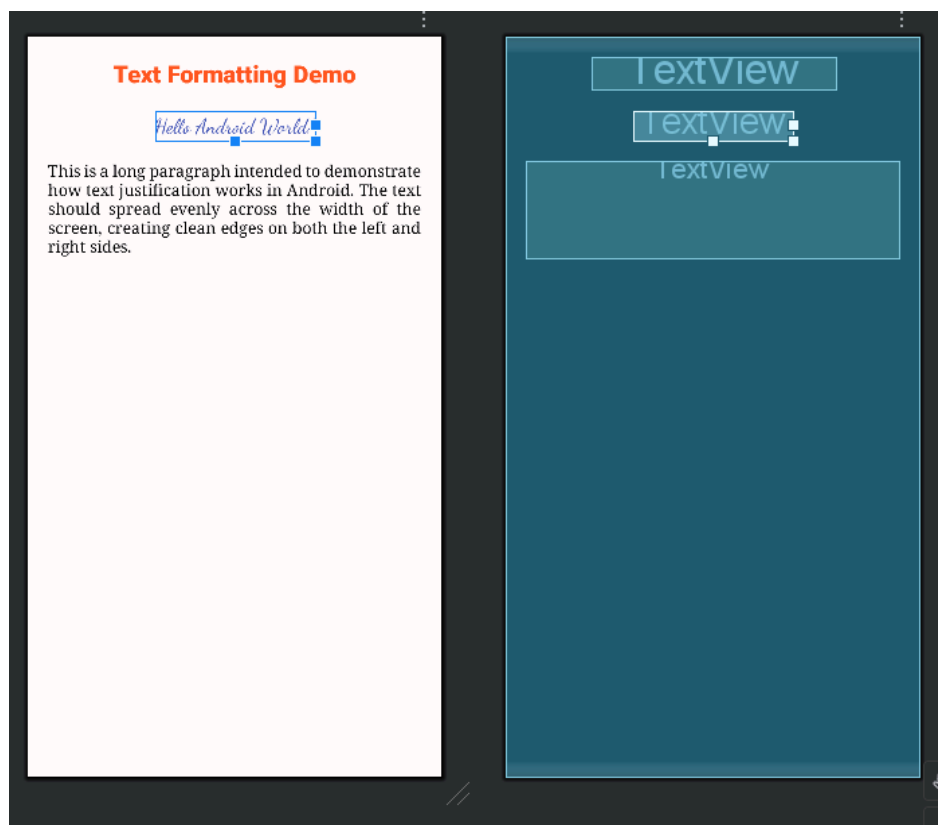


Fig 1.1: activity_main.xml

MainActivity.java

```
package com.example.textformattingdemo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

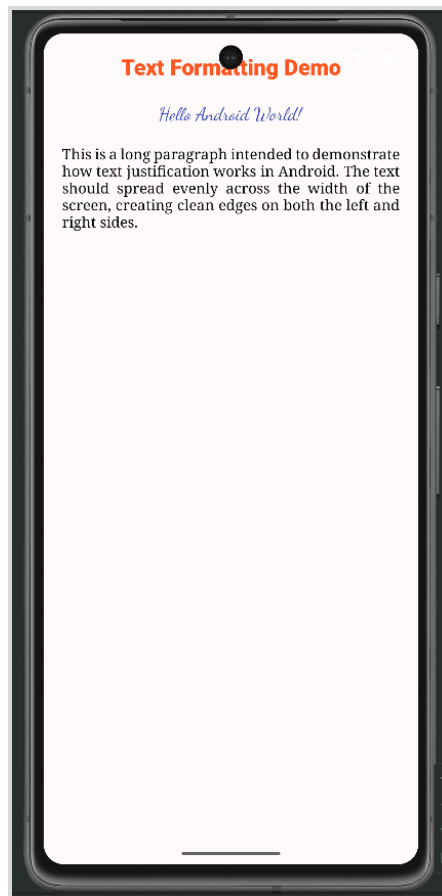


Fig 1.2: screenshot of the application

PRACTICE PROBLEM

Develop UI for Google sign in page.

SOLUTION

Colors.xml

```
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="google_blue">#1a73e8</color>
</resources>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    android:background="@android:color/white">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <ImageView
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:src="@drawable/google_logo"
            android:layout_gravity="center_horizontal"
            android:layout_marginBottom="24dp"/>
```

```

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
    android:textSize="24sp"
    android:textStyle="bold"
    android:textColor="@android:color/black"
    android:layout_gravity="center_horizontal"
    android:layout_marginBottom="6dp"/>

```

```

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Enter your email below to login"
    android:textSize="14sp"
    android:textColor="#71717A"
    android:layout_gravity="center_horizontal"
    android:layout_marginBottom="32dp"/>

```

```

<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Email"

```

```

style="@style/Widget.MaterialComponents.TextInputLayout.Outline
dBox"

```

```

    app:boxStrokeColor="@android:color/black"
    app:hintTextColor="@android:color/black"
    android:layout_marginBottom="16dp">

```

```

<com.google.android.material.textfield.TextInputEditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:inputType="textEmailAddress"/>

```

```

</com.google.android.material.textfield.TextInputLayout>

```

```

        <com.google.android.material.textfield.TextInputLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Password"

style="@style/Widget.MaterialComponents.TextInputLayout.Outline
dBox"

            app:boxStrokeColor="@android:color/black"
            app:hintTextColor="@android:color/black"
            app:endIconMode="password_toggle"
            android:layout_marginBottom="10dp">

        <com.google.android.material.textfield.TextInputEditText
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:inputType="textPassword"/>

    </com.google.android.material.textfield.TextInputLayout>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Forgot password?"
        android:textSize="14sp"
        android:textColor="@android:color/black"
        android:textStyle="bold"
        android:layout_gravity="end"
        android:layout_marginBottom="24dp"/>

    <Button
        android:layout_width="match_parent"
        android:layout_height="55dp"
        android:text="Sign In"
        android:textAllCaps="false"
        android:textSize="16sp"
        android:backgroundTint="@android:color/black"
        android:textColor="@android:color/white"
        app:cornerRadius="6dp"/>

```



```
</LinearLayout>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

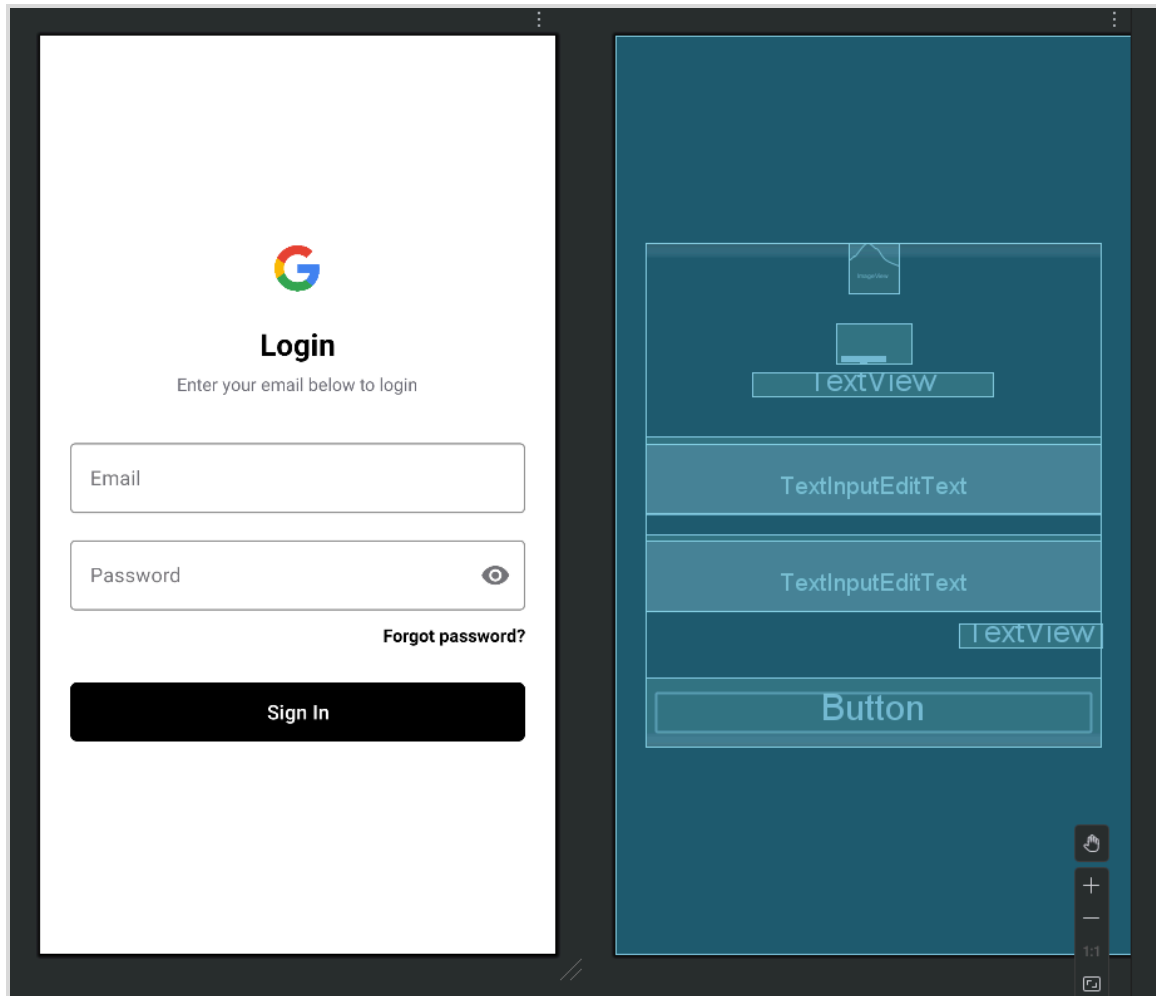


Fig 1.3: activity layout

MainActivity.java

```
package com.example.googleloginui;  
  
import android.os.Bundle;  
import androidx.appcompat.app.AppCompatActivity;
```

```

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

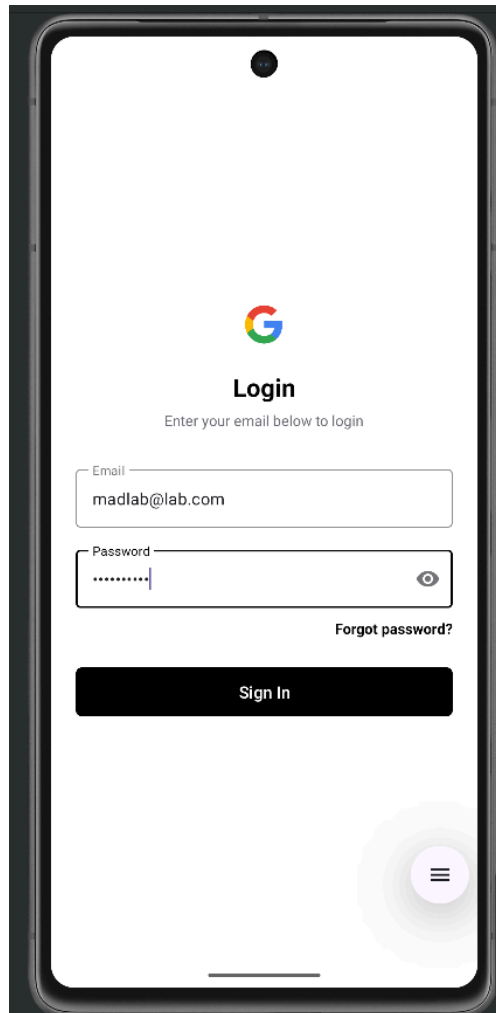


Fig 1.4: Application screenshot

INTRODUCTION TO ACTIVITY AND LAYOUTS IN ANDROID

PROBLEM STATEMENT 1

Create an app that illustrates that the activity lifecycle method being triggered by various action. Understand when onCreate(),onStart(),onResume event occur.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check Logcat for tag: LifecycleEvent"
        android:textSize="18sp" />

</LinearLayout>
```

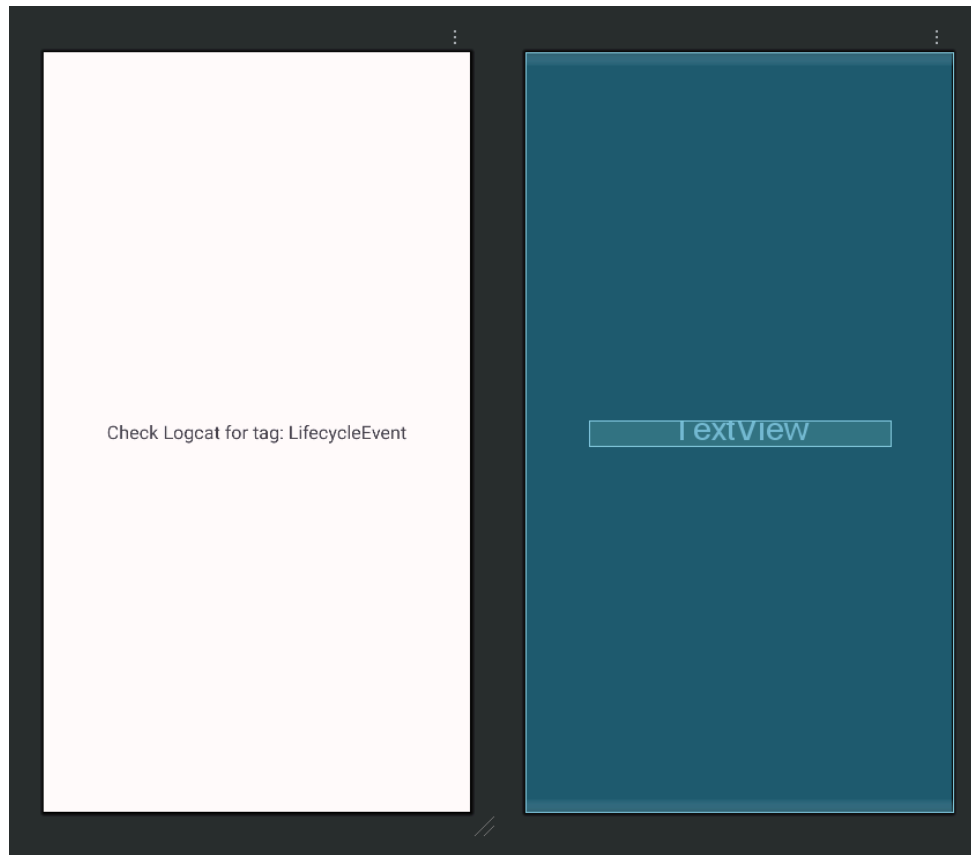


Fig 2.1: Application layout

MainActivity.java

```
package com.example.lifecycledemo;

import android.os.Bundle;
import android.util.Log;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private static final String TAG = "LifecycleEvent"; //
    Filter tag
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

        Log.d(TAG, "onCreate invoked"); // App created
    }
    @Override
    protected void onStart() {
        super.onStart();
        Log.d(TAG, "onStart invoked"); // App visible
    }
    @Override
    protected void onResume() {
        super.onResume();
        Log.d(TAG, "onResume invoked"); // App interactive
    }
    @Override
    protected void onPause() {
        super.onPause();
        Log.d(TAG, "onPause invoked"); // Partially obscured or
Leaving
    }
    @Override
    protected void onStop() {
        super.onStop();
        Log.d(TAG, "onStop invoked"); // Fully backgrounded
    }
    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d(TAG, "onRestart invoked"); // Returning from stop
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy invoked"); // App closed
    }
}

```

PROBLEM STATEMENT 2

Create a Calculator app that does the function of multiplication, addition, division, subtraction but displays the result in the format of:-Num1 operator num2 = result. Back button on the next activity should get back to the calculator activity again.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/etNum1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number 1"
        android:inputType="numberDecimal"/>

    <EditText
        android:id="@+id/etNum2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number 2"
        android:inputType="numberDecimal"/>

    <Button
        android:id="@+id/btnAdd"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Add (+)" />

    <Button
```

```
android:id="@+id/btnSub"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:text="Subtract (-)" />
```

```
<Button
```

```
    android:id="@+id/btnMul"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Multiply (*)" />
```

```
<Button
```

```
    android:id="@+id/btnDiv"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Divide (/)" />
```

```
</LinearLayout>
```

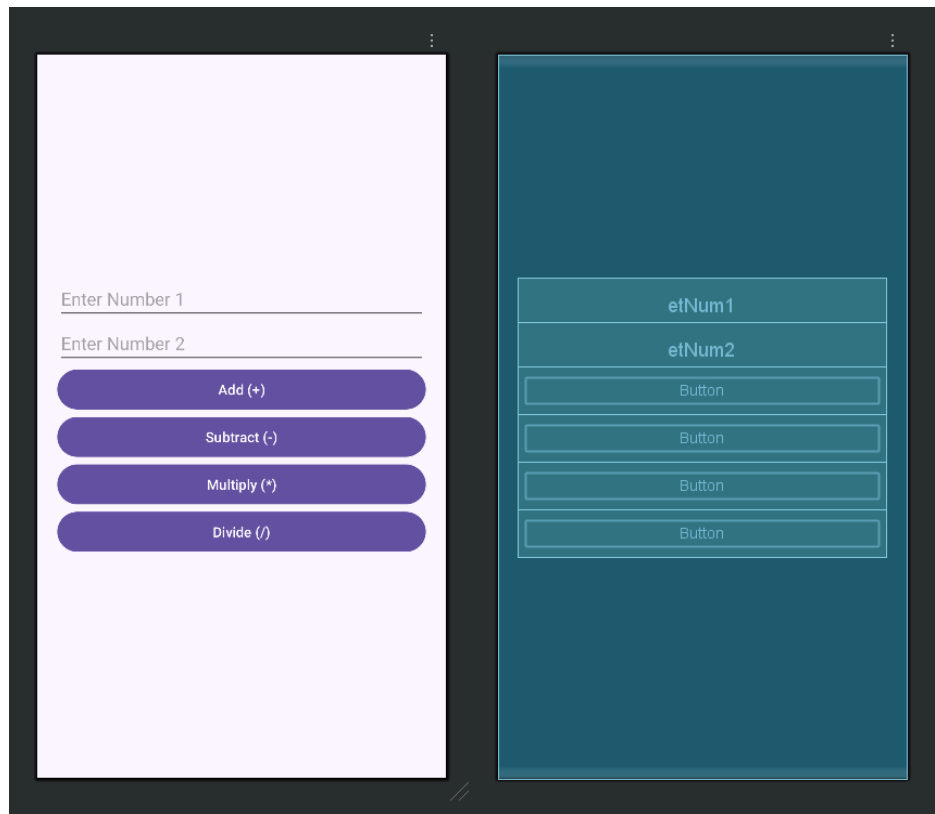


Fig 2.2: activity main layout

MainActivity.java

```
package com.example.simplecalculator;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etNum1, etNum2;
    Button btnAdd, btnSub, btnMul, btnDiv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Link Java variables to XML components
        etNum1 = findViewById(R.id.etNum1);
        etNum2 = findViewById(R.id.etNum2);
        btnAdd = findViewById(R.id.btnAdd);
        btnSub = findViewById(R.id.btnSub);
        btnMul = findViewById(R.id.btnMul);
        btnDiv = findViewById(R.id.btnDiv);

        // Set click listeners for each button
        btnAdd.setOnClickListener(v -> calculate("+"));
        btnSub.setOnClickListener(v -> calculate("-"));
        btnMul.setOnClickListener(v -> calculate("*"));
        btnDiv.setOnClickListener(v -> calculate("/"));
    }

    // Helper function to calculate and switch screens
    void calculate(String operator) {
```



```

String s1 = etNum1.getText().toString();
String s2 = etNum2.getText().toString();

// Check if inputs are empty
if (s1.isEmpty() || s2.isEmpty()) {
    Toast.makeText(this, "Please enter numbers",
Toast.LENGTH_SHORT).show();
    return;
}

double n1 = Double.parseDouble(s1);
double n2 = Double.parseDouble(s2);
double result = 0;

// Perform Math
if (operator.equals("+")) result = n1 + n2;
else if (operator.equals("-")) result = n1 - n2;
else if (operator.equals("*")) result = n1 * n2;
else if (operator.equals("/")) result = n1 / n2;

// Format: Num1 operator num2 = result
String finalOutput = n1 + " " + operator + " " + n2 + "
= " + result;

// Navigate to ResultActivity
Intent intent = new Intent(MainActivity.this,
ResultActivity.class);
intent.putExtra("RESULT_DATA", finalOutput); // Send
data
startActivity(intent);
}
}

```

Activity_result.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"

```

```
android:layout_width="match_parent"  
android:layout_height="match_parent"  
android:gravity="center"  
android:orientation="vertical">
```

```
<TextView  
    android:id="@+id/tvResult"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:textSize="24sp"  
    android:textStyle="bold"  
    android:text="Result will appear here"/>
```

```
<TextView  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_marginTop="20dp"  
    android:text="Press Back button to calculate again"/>
```

```
</LinearLayout>
```

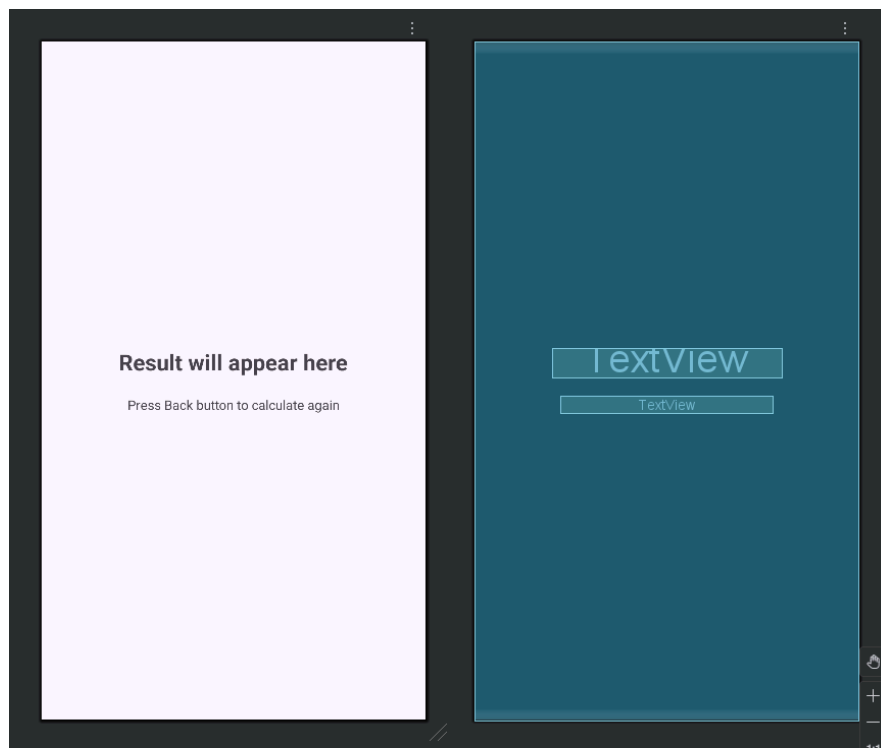


Fig 2.3: activity result layout

ResultActivity.java

```
package com.example.simplecalculator;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class ResultActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_result);

        TextView tvResult = findViewById(R.id.tvResult);

        // Get the data passed from the previous activity
        String data =
        getIntent().getStringExtra("RESULT_DATA");

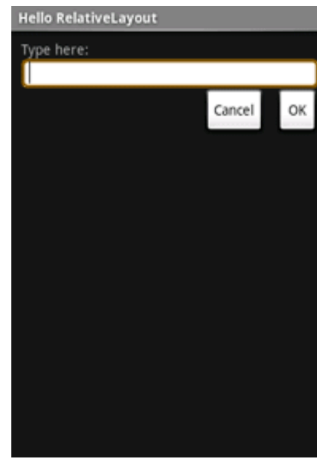
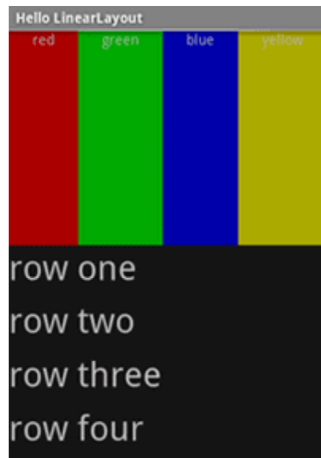
        // Display the data
        if(data != null) {
            tvResult.setText(data);
        }
    }
}
```



Fig 2.4: Application screenshots

PROBLEM STATEMENT 3

Create the following given scenario using linear and relative layout concept.



SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#333333">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="200dp"
        android:orientation="horizontal">

        <View
            android:layout_width="0dp"
            android:layout_height="match_parent"
            android:layout_weight="1"
            android:background="#AA0000" /> <View
```

```

        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:background="#00AA00" /> <View
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:background="#0000AA" /> <View
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:background="#AAAA00" /> </LinearLayout>

```

```

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="row one"
    android:textSize="24sp"
    android:textColor="#FFFFFF"
    android:padding="10dp" />

```

```

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="row two"
    android:textSize="24sp"
    android:textColor="#FFFFFF"
    android:padding="10dp" />

```

```

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="row three"
    android:textSize="24sp"
    android:textColor="#FFFFFF"
    android:padding="10dp" />

```

```

<TextView
    android:layout_width="match_parent"

```

```
android:layout_height="wrap_content"  
android:text="row four"  
android:textSize="24sp"  
android:textColor="#FFFFFF"  
android:padding="10dp" />
```

```
</LinearLayout>
```

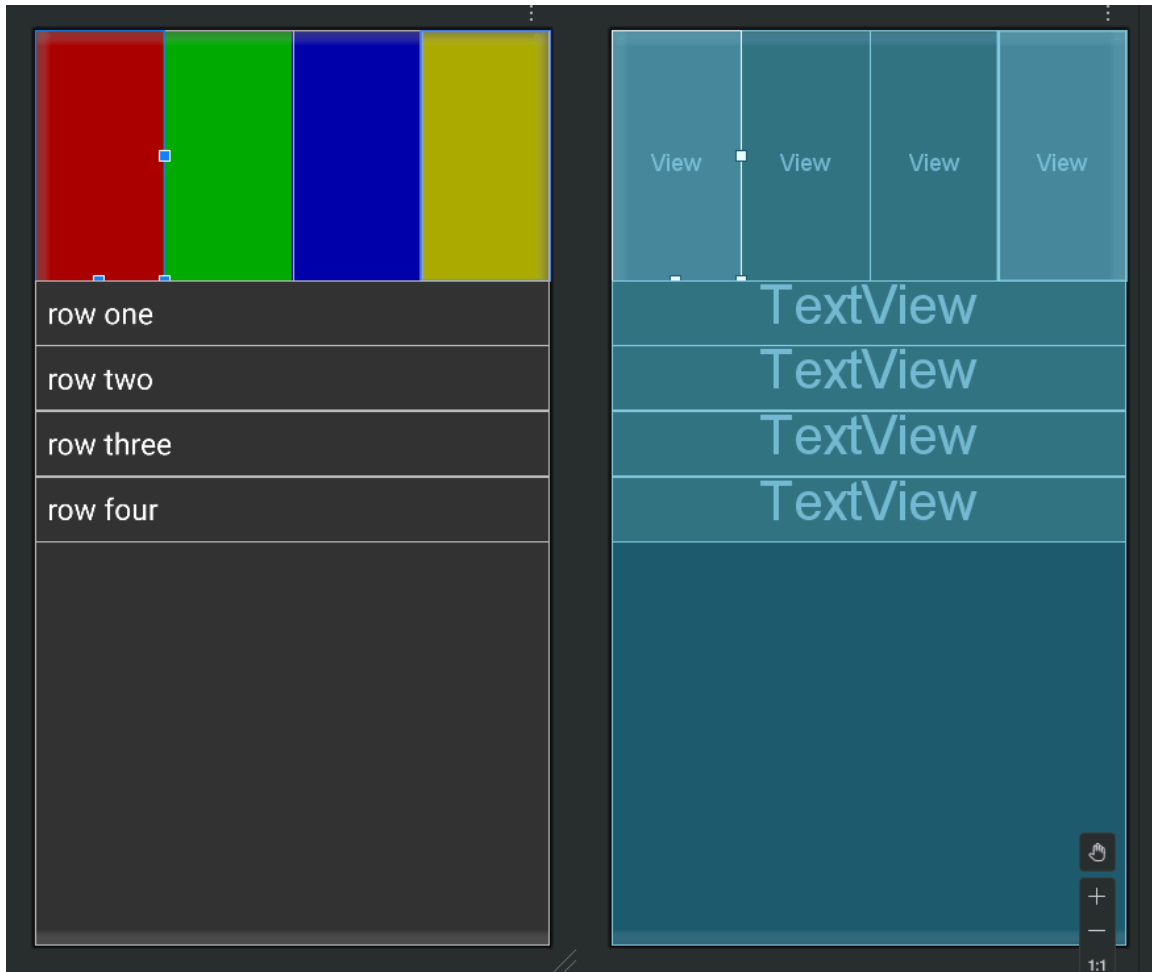


Fig 2.5: activity main layout

Activity_form.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:background="#222222">

    <TextView
        android:id="@+id/tv_label"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Type here:"
        android:textColor="#FFFFFF"
        android:textSize="18sp" />

    <EditText
        android:id="@+id/et_input"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/tv_label"
        android:layout_marginTop="5dp"
        android:background="#FFFFFF"
        android:padding="10dp"
        android:textColor="#000000" />

    <Button
        android:id="@+id/btn_ok"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="OK"
        android:layout_below="@id/et_input"
        android:layout_alignParentRight="true"
        android:layout_marginTop="10dp" />

    <Button
        android:id="@+id/btn_cancel"
```



```
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="Cancel"  
android:layout_toLeftOf="@id/btn_ok"  
android:layout_alignTop="@id/btn_ok"  
android:layout_marginRight="10dp" />
```

```
</RelativeLayout>
```

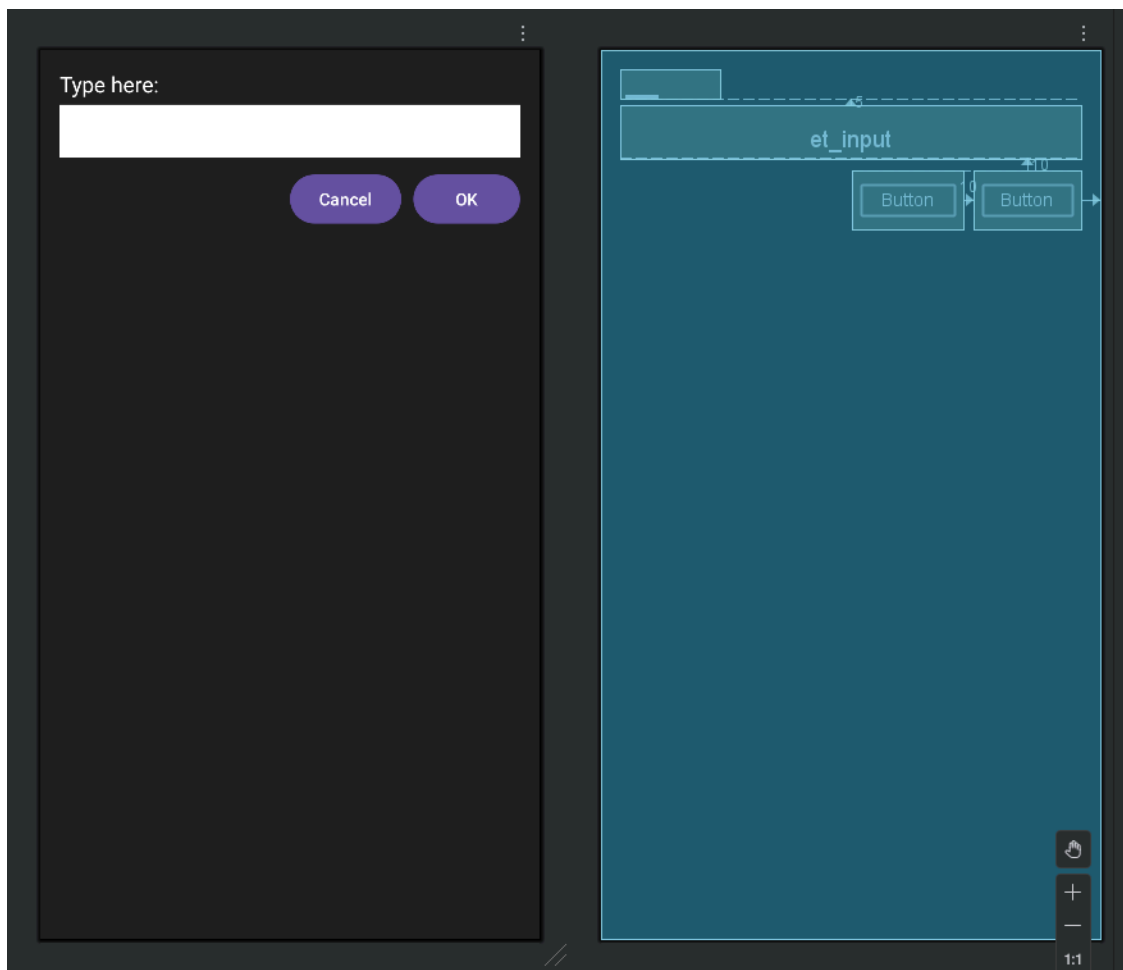


Fig 2.6: activity form layout

MainActivity.java

```
package com.example.layoutassignment;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Scenario 1: Load the Linear Layout (Left Image)
        setContentView(R.layout.activity_main);

        // Scenario 2: Load Relative Layout (Right Image),
        setContentView(R.layout.activity_form);
    }
}
```

PROBLEM STATEMENT 4

Create the following given scenario using linear and relative layout concept.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
```

```
android:id="@+id/etUrl"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:hint="Type URL (e.g., google.com)"  
android:inputType="textUri" />
```

```
<Button
```

```
    android:id="@+id/btnOpen"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_marginTop="20dp"  
    android:text="Open Website" />
```

```
</LinearLayout>
```

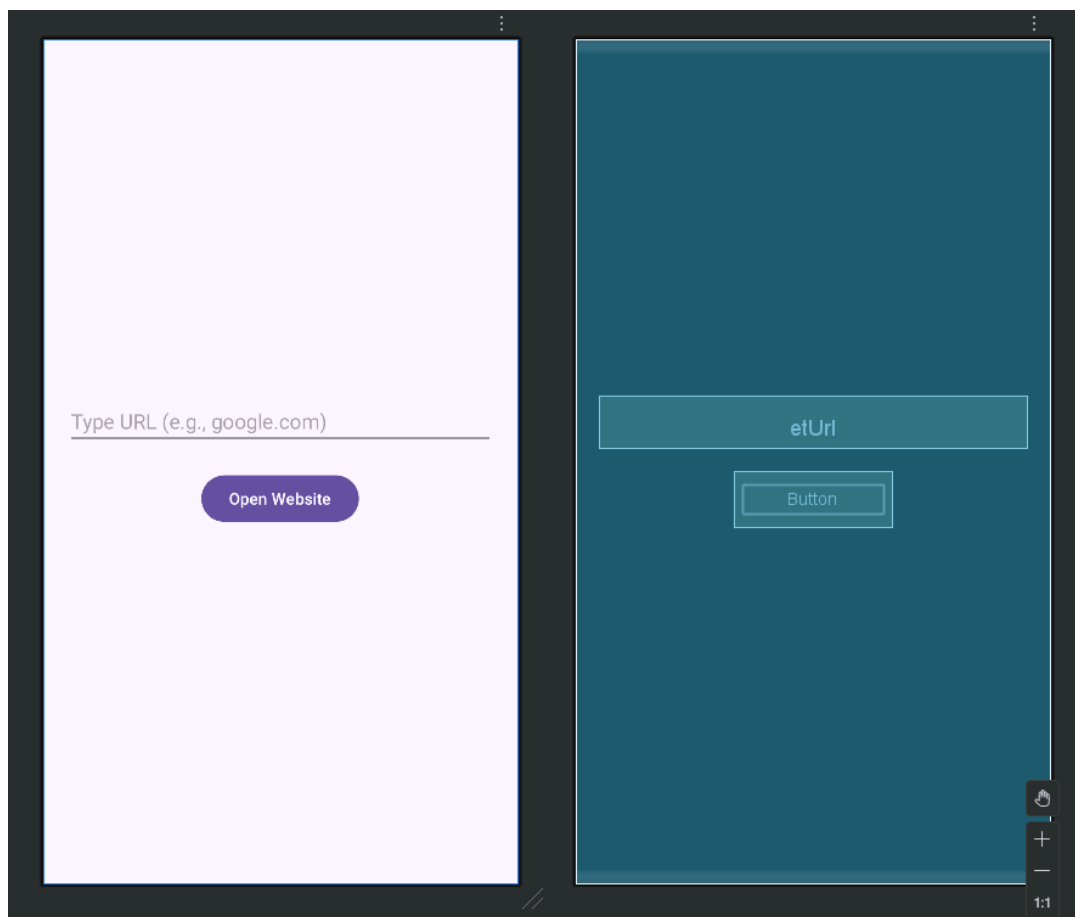


Fig 2.7: activity main layout

MainActivity.java

```
package com.example.urloper;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Link code to XML elements
        EditText urlInput = findViewById(R.id.etUrl);
        Button openBtn = findViewById(R.id.btnOpen);

        // Set button click action
        openBtn.setOnClickListener(v -> {
            String url = urlInput.getText().toString();

            // Simple check to ensure http protocol exists
            if (!url.startsWith("http://") &&
!url.startsWith("https://")) {
                url = "https://" + url;
            }

            // Create intent to open web browser
            Intent intent = new Intent(Intent.ACTION_VIEW,
Uri.parse(url));
            startActivity(intent);
        });
    }
}
```

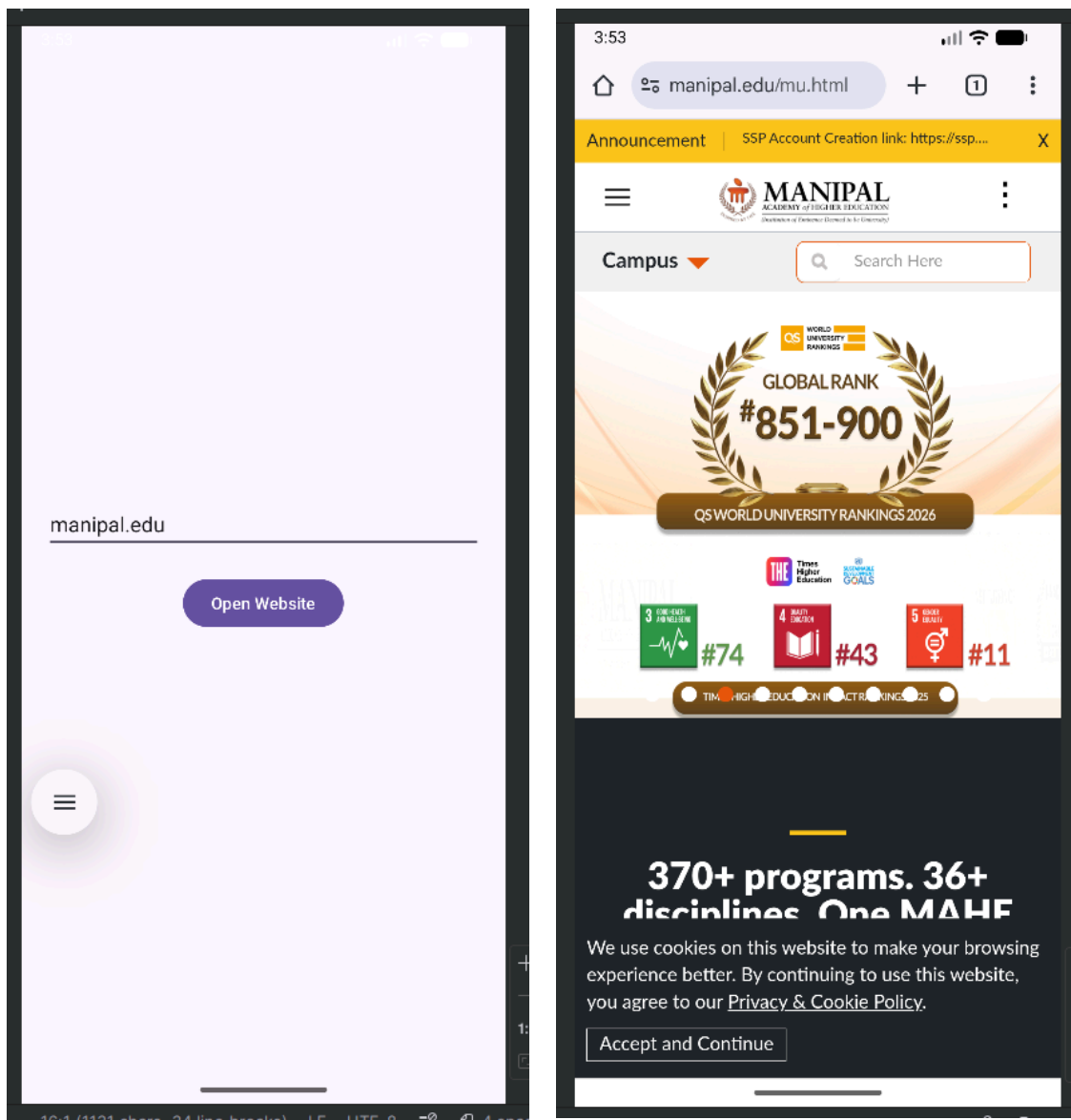


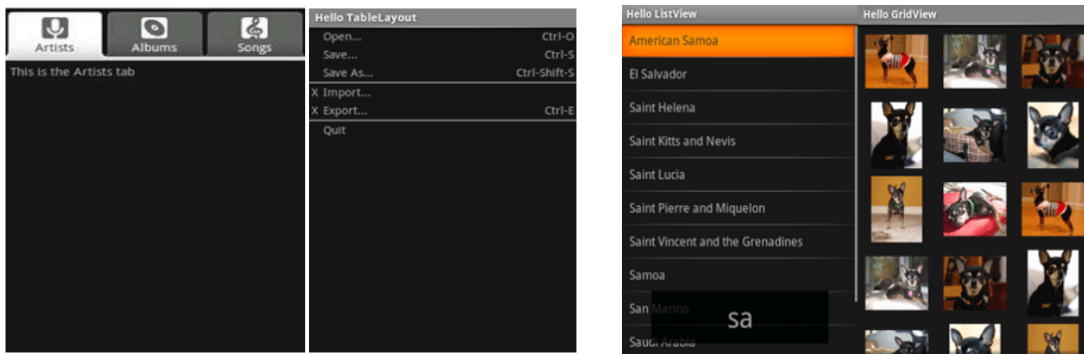
Fig 2.8: Application Screenshots

ACTIVITY AND LAYOUT CONTINUED

PROBLEM STATEMENT 1

Using given scenario develop an application to perform following layout operations -

- List view
- Grid view
- Tab layout
- Table layout



SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">
```

```

        android:padding="20dp">
        <Button
            android:id="@+id/btnList"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="1. List View" />
        <Button
            android:id="@+id/btnGrid"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="2. Grid View" />
        <Button
            android:id="@+id/btnTab"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="3. Tab Layout" />
        <Button
            android:id="@+id/btnTable"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="4. Table Layout" />
    </LinearLayout>

```

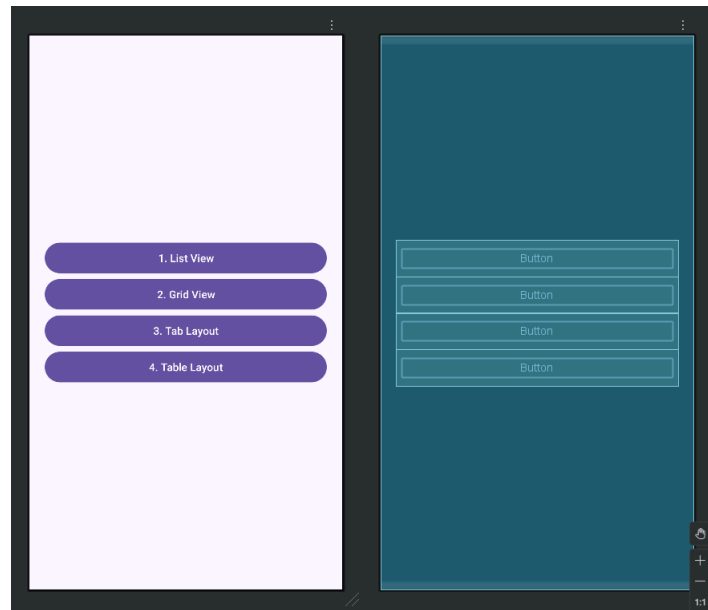


Fig 3.1: activity main layout

MainActivity.java

```
package com.example.layoutdemo;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Link buttons and set click listeners to open
        // specific activities
        findViewById(R.id.btnList).setOnClickListener(v ->
startActivity(new Intent(this, ListViewActivity.class)));
        findViewById(R.id.btnGrid).setOnClickListener(v ->
startActivity(new Intent(this, GridViewActivity.class)));
        findViewById(R.id.btnTab).setOnClickListener(v ->
startActivity(new Intent(this, TabLayoutActivity.class)));
        findViewById(R.id.btnTable).setOnClickListener(v ->
startActivity(new Intent(this, TableLayoutActivity.class)));
    }
}
```

Activity_list_view.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ListView
        android:id="@+id/myListView"
```



```

        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>

```

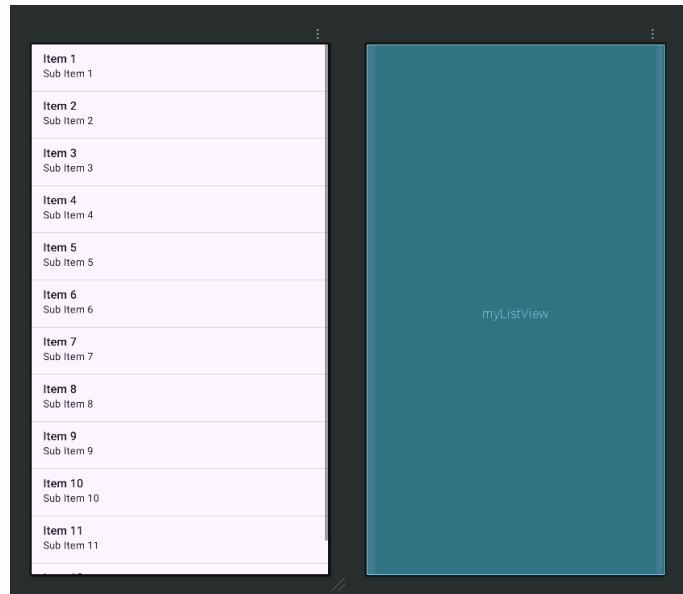


Fig 3.2: activity list layout

ListViewActivity.java

```

package com.example.layoutdemo;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;

public class ListViewActivity extends AppCompatActivity {

    // Data for the list
    String[] countries = {"American Samoa", "El Salvador",
        "Saint Helena", "Saint Kitts", "Saint Lucia", "Samoa", "San
        Marino", "Saudi Arabia"};

```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_list_view);

    ListView listView = findViewById(R.id.myListView);

    // Use default Android layout for list items
    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
    android.R.layout.simple_list_item_1, countries);

    listView.setAdapter(adapter);
}
}

```

Activity_grid_view.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <GridView
        android:id="@+id/myGridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="3"
        android:verticalSpacing="10dp"
        android:horizontalSpacing="10dp"
        android:padding="10dp"
        android:gravity="center" />
</LinearLayout>

```

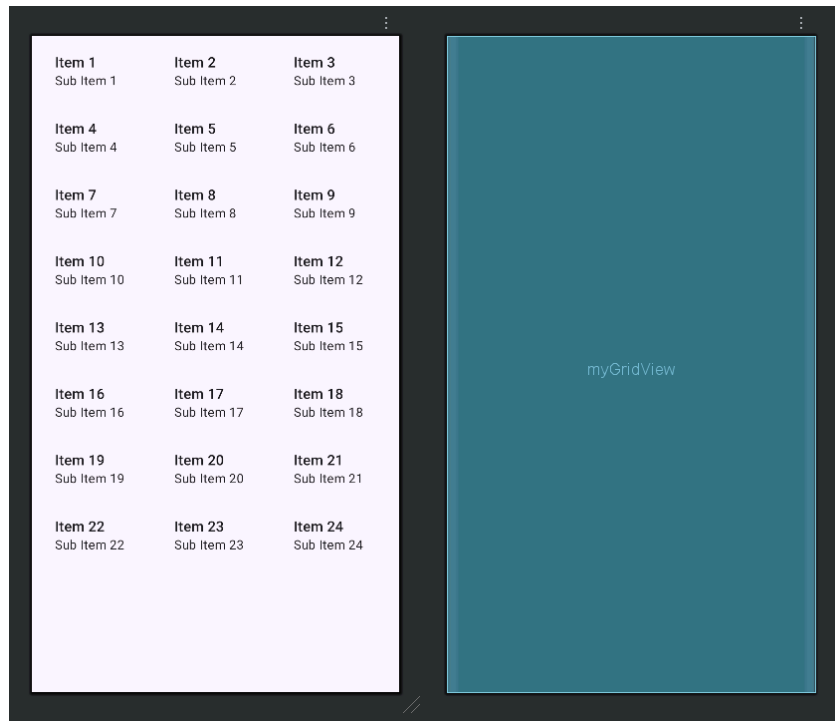


Fig 3.3: activity grid layout

GridViewActivity.java

```
package com.example.layoutdemo;

import android.os.Bundle;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.GridView;
import android.widget.ImageView;
import androidx.appcompat.app.AppCompatActivity;

public class GridViewActivity extends AppCompatActivity {

    // Using built-in system icons for simplicity
    int[] images = {
        android.R.drawable.ic_menu_camera,
        android.R.drawable.ic_menu_gallery,
        android.R.drawable.ic_menu_compass,
        android.R.drawable.ic_menu_call,
    }
}
```

```

    android.R.drawable.ic_menu_mapmode,
    android.R.drawable.ic_menu_my_calendar,
        android.R.drawable.ic_menu_camera,
    android.R.drawable.ic_menu_gallery,
    android.R.drawable.ic_menu_compass
    };
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_grid_view);

        GridView gridView = findViewById(R.id.myGridView);
        gridView.setAdapter(new ImageAdapter());
    }
    // Simple inner class adapter to display images
    private class ImageAdapter extends BaseAdapter {
        @Override
        public int getCount() { return images.length; }
        @Override
        public Object getItem(int position) { return
images[position]; }
        @Override
        public long getItemId(int position) { return 0; }
        @Override
        public View getView(int position, View convertView,
ViewGroup parent) {
            ImageView imageView = new
ImageView(getApplicationContext());
            imageView.setImageResource(images[position]);
            imageView.setLayoutParams(new
ViewGroup.LayoutParams(300, 300)); // Set image size

            imageView.setScaleType(ImageView.ScaleType.CENTER_CROP);
            return imageView;
        }
    }
}

```

Activity_tab_view.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <com.google.android.material.tabs.TabLayout
        android:id="@+id/tabLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="#333333"
        app:tabTextColor="#FFFFFF"
        app:tabSelectedTextColor="#FF9800">

        <com.google.android.material.tabs.TabItem
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Artists"
            android:icon="@android:drawable/ic_menu_edit"/>

        <com.google.android.material.tabs.TabItem
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Albums"
            android:icon="@android:drawable/ic_menu_gallery"/>

        <com.google.android.material.tabs.TabItem
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Songs"
            android:icon="@android:drawable/ic_menu_sort_by_size"/>
    </com.google.android.material.tabs.TabLayout>

    <TextView
```

```
android:id="@+id/tabContent"  
android:layout_width="match_parent"  
android:layout_height="match_parent"  
android:gravity="center"  
android:text="This is the Artists tab"  
android:textSize="24sp"  
android:background="#000000"  
android:textColor="#AAAAAA"/>
```

```
</LinearLayout>
```

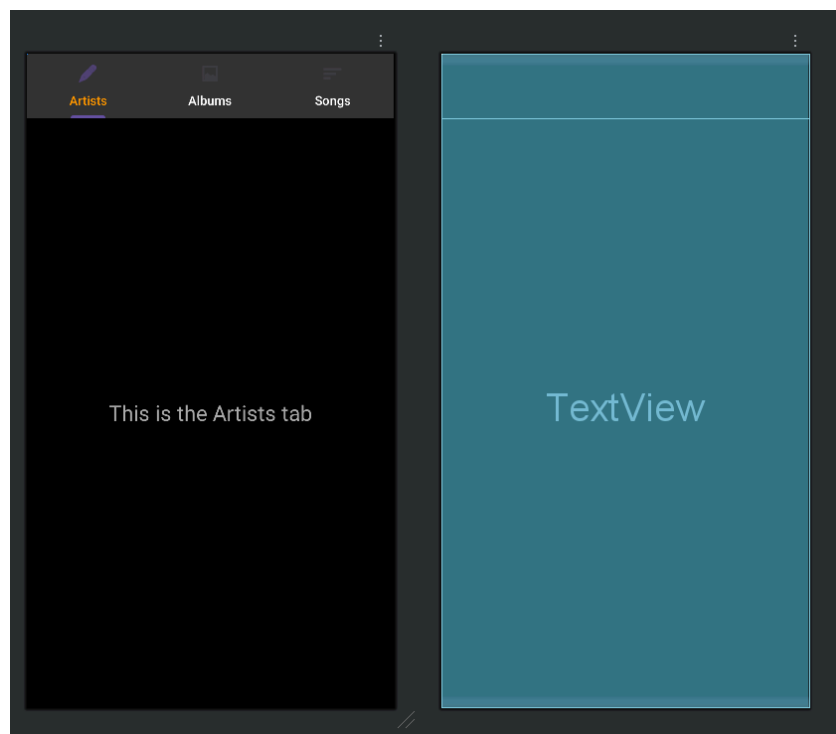


Fig 3.4: activity tab layout

TabLayoutActivity.java

```
package com.example.layoutdemo;  
  
import android.os.Bundle;  
import android.widget.TextView;
```

```

import androidx.appcompat.app.AppCompatActivity;
import com.google.android.material.tabs.TabLayout;

public class TabLayoutActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_tab_layout);

        TabLayout tabLayout = findViewById(R.id.tabLayout);
        TextView content = findViewById(R.id.tabContent);

        // Handle tab clicks
        tabLayout.addOnTabSelectedListener(new
        TabLayout.OnTabSelectedListener() {
            @Override
            public void onTabSelected(TabLayout.Tab tab) {
                content.setText("This is the " + tab.getText()
+ " tab");
            }
            @Override public void onTabUnselected(TabLayout.Tab
tab) {}
            @Override public void onTabReselected(TabLayout.Tab
tab) {}
        });
    }
}

```

Activity_table_layout.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#222222">

```

```

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Hello TableLayout"
    android:padding="10dp"
    android:background="#666666"
    android:textColor="#FFFFFF"
    android:textStyle="bold"/>
<TableLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:stretchColumns="1"> <TableRow
android:padding="10dp">
    <TextView android:text="Open..."
android:textColor="#CCCCCC"/>
    <TextView android:text="" /> <TextView
android:text="Ctrl-O" android:textColor="#CCCCCC"/>
</TableRow>
    <View android:layout_height="1dp"
android:background="#444444"/>
    <TableRow android:padding="10dp">
        <TextView android:text="Save..."
android:textColor="#CCCCCC"/>
        <TextView android:text="" />
        <TextView android:text="Ctrl-S"
android:textColor="#CCCCCC"/>
    </TableRow>
    <View android:layout_height="1dp"
android:background="#444444"/>
    <TableRow android:padding="10dp">
        <TextView android:text="Save As..."
android:textColor="#CCCCCC"/>
        <TextView android:text="" />
        <TextView android:text="Ctrl-Shift-S"
android:textColor="#CCCCCC"/>
    </TableRow>
    <View android:layout_height="1dp"
android:background="#FFFFFF"/>
    <TableRow android:padding="10dp">

```



```

        <TextView android:text="X Import..."
android:textColor="#CCCCCC"/>
    </TableRow>
    <TableRow android:padding="10dp">
        <TextView android:text="X Export..."
android:textColor="#CCCCCC"/>
        <TextView android:text="" />
        <TextView android:text="Ctrl-E"
android:textColor="#CCCCCC"/>
    </TableRow>
    <View android:layout_height="1dp"
android:background="#FFFFFF"/>
    <TableRow android:padding="10dp">
        <TextView android:text="Quit"
android:textColor="#CCCCCC"/>
    </TableRow>
</TableLayout>
</LinearLayout>

```

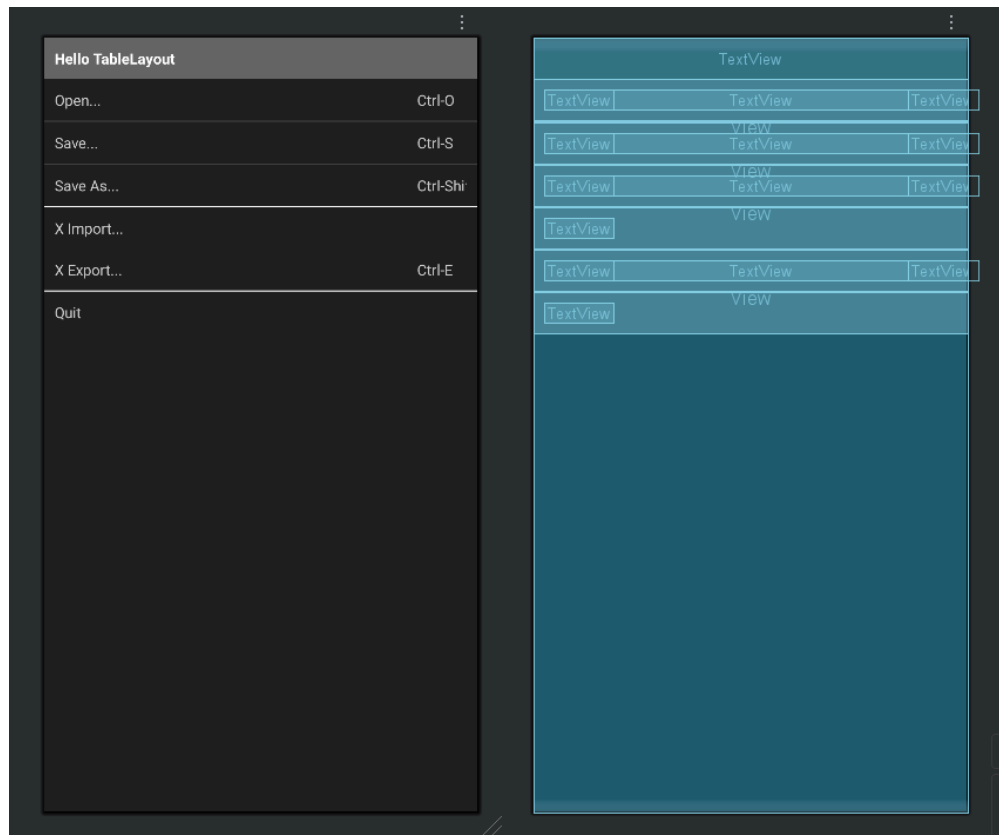


Fig 3.5: activity table layout

TableLayoutActivity.java

```
package com.example.layoutdemo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class TableLayoutActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_table_layout);
    }
}
```

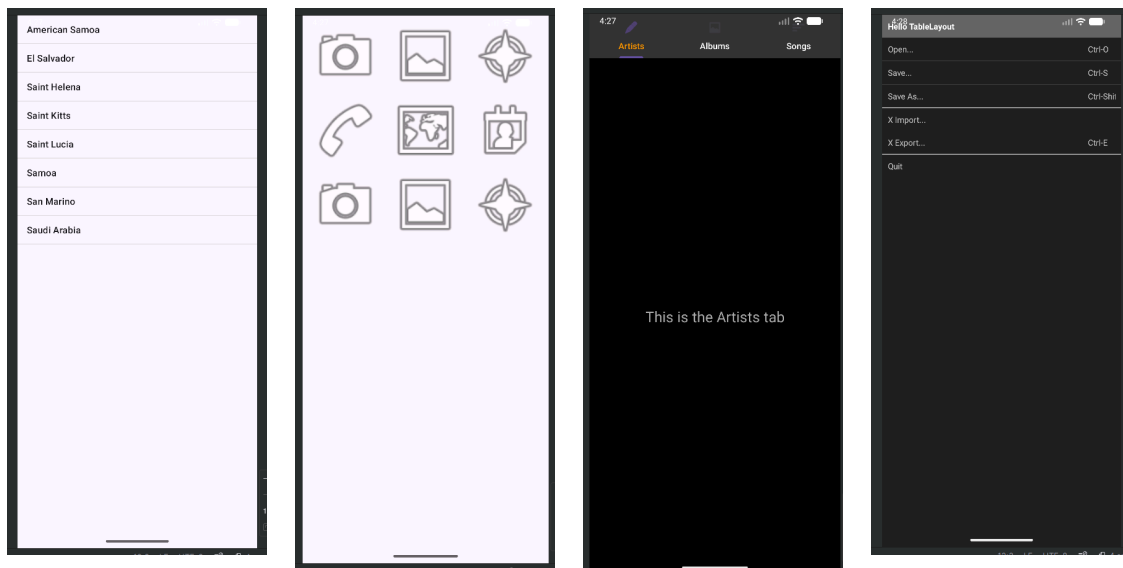


Fig 3.6: Application Screenshots

PROBLEM STATEMENT 2

Write a program to list different sports, and when a sport is selected, display a message showing the selected sport.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <ListView
        android:id="@+id/sportsListView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

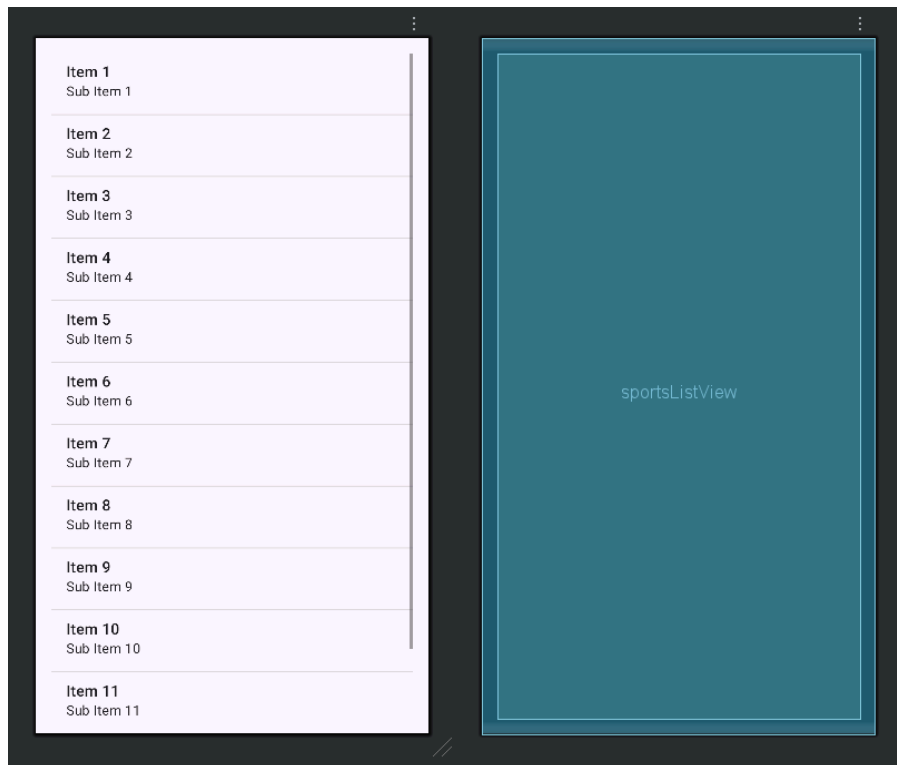


Fig 3.7: activity layout

MainActivity.java

```
package com.example.sportslistapp;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    // 1. Data source
    String[] sports = {"Football", "Basketball", "Tennis",
        "Cricket", "Badminton", "Rugby", "Golf"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // 2. Initialize ListView
        ListView listView = findViewById(R.id.sportsListView);

        // 3. Create adapter using default Android layout
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
            sports
        );

        // 4. Set adapter to list
        listView.setAdapter(adapter);

        // 5. Handle clicks
        listView.setOnItemClickListener((parent, view,
            position, id) -> {
            String selectedSport = sports[position];
            Toast.makeText(getApplicationContext(), "Selected:
```

```
" + selectedSport, Toast.LENGTH_SHORT).show();
    });
}
```

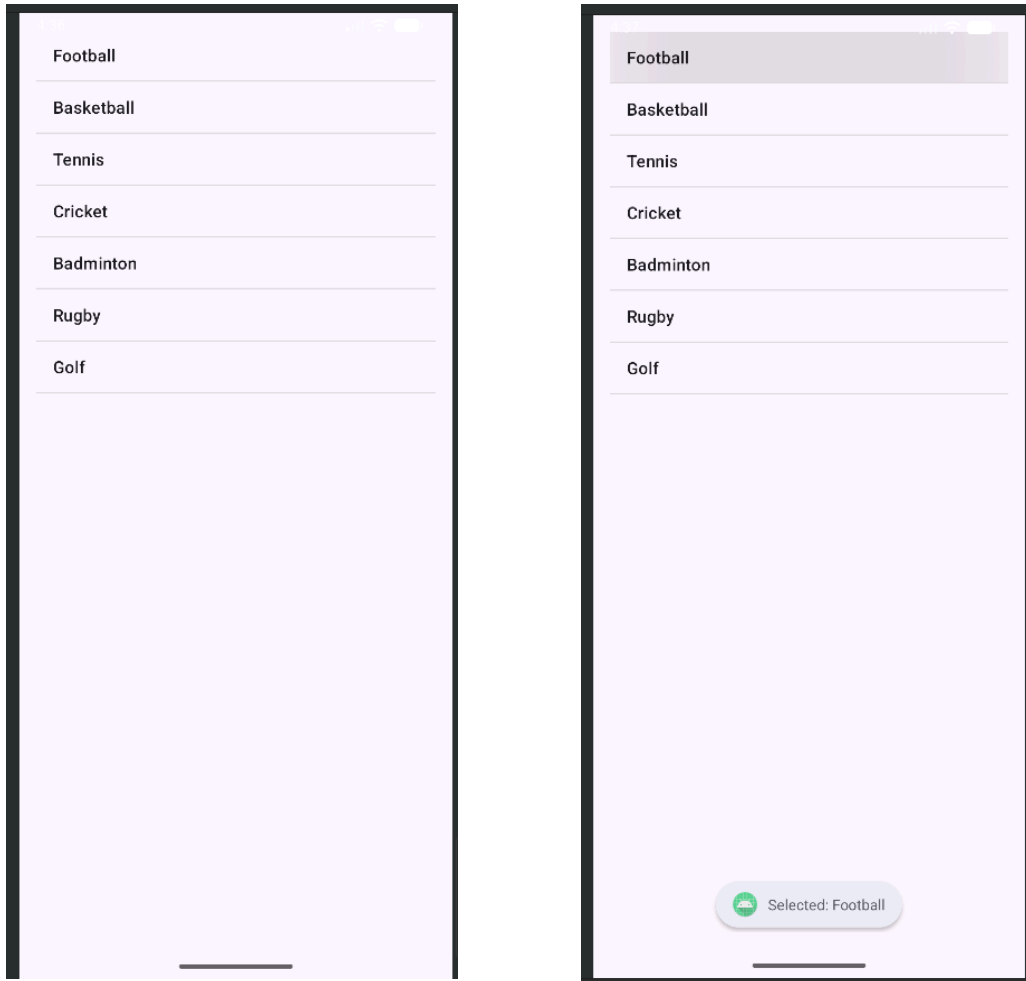


Fig 3.8: Application Screenshot

PROBLEM STATEMENT 3

Design an news application and Implement the navigation between sections like\ Top Stories, Sports, and Entertainment using tab layout.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
    <com.google.android.material.tabs.TabLayout
        android:id="@+id/tabLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:tabMode="fixed" />
    <androidx.viewpager2.widget.ViewPager2
        android:id="@+id/viewPager"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

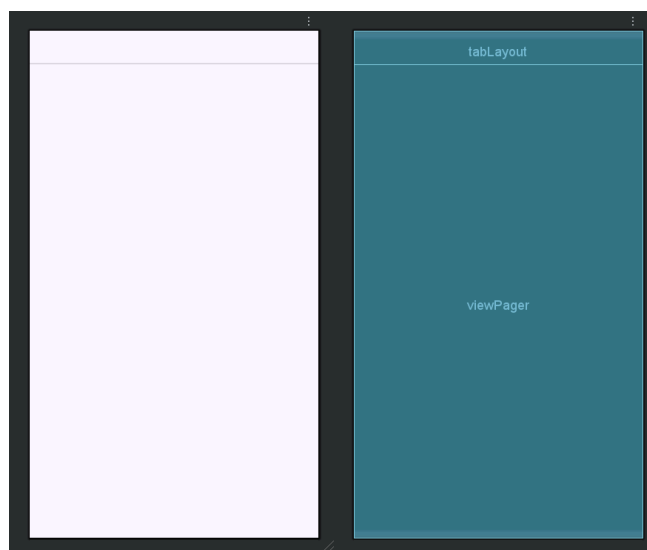


Fig 3.9: Activity Layout

Fragment_news.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/sectionLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="News Section"
        android:textSize="24sp"
        android:textStyle="bold" />

</LinearLayout>
```

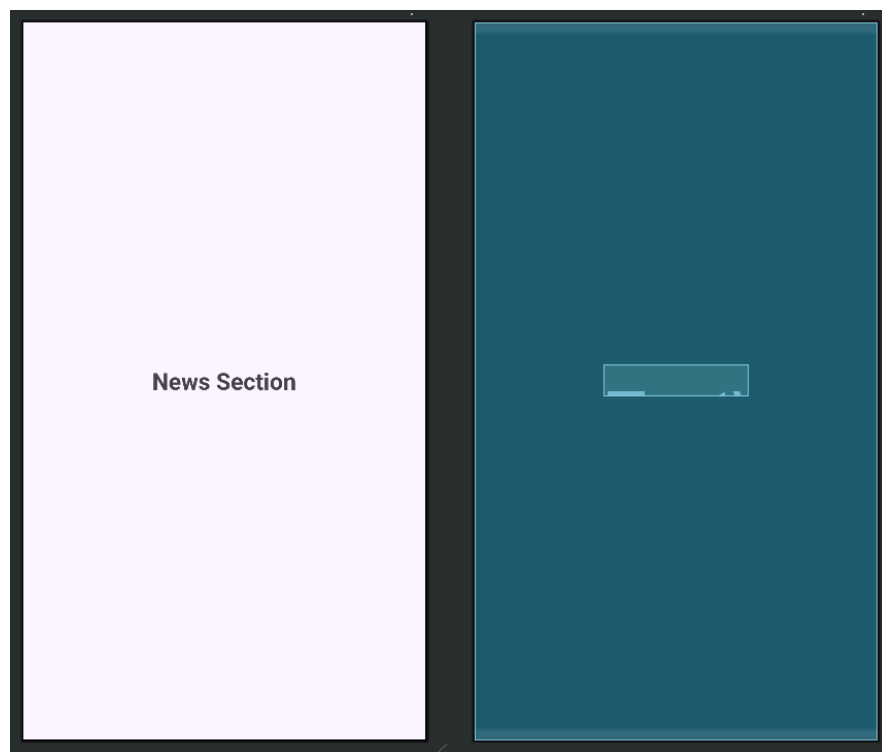


Fig 3.10: Activity Layout

NewsFragment.java

```
package com.example.newsapp;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class NewsFragment extends Fragment {

    private String sectionName;

    // Constructor to pass the category name (e.g., "Sports")
    public NewsFragment(String sectionName) {
        this.sectionName = sectionName;
    }

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
        @Nullable ViewGroup container, @Nullable Bundle
        savedInstanceState) {
        // Inflate the XML layout
        View view = inflater.inflate(R.layout.fragment_news,
            container, false);

        // Set the text based on the section
        TextView textView =
            view.findViewById(R.id.sectionLabel);
        textView.setText(sectionName);

        return view;
    }
}
```


ViewPagerAdapter.java

```
package com.example.newsapp;

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentActivity;
import androidx.viewpager2.adapter.FragmentStateAdapter;

public class ViewPagerAdapter extends FragmentStateAdapter {

    // Define your tab titles here
    private final String[] titles = new String[]{"Top Stories",
"Sports", "Entertainment"};

    public ViewPagerAdapter(@NonNull FragmentActivity
fragmentActivity) {
        super(fragmentActivity);
    }

    @NonNull
    @Override
    public Fragment createFragment(int position) {
        // Create a new fragment for each tab, passing the
title
        return new NewsFragment(titles[position]);
    }

    @Override
    public int getItemCount() {
        return titles.length; // Returns 3
    }
}
```

MainActivity.java

```
package com.example.newsapp;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.viewpager2.widget.ViewPager2;
import com.google.android.material.tabs.TabLayout;
import com.google.android.material.tabs.TabLayoutMediator;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // 1. Find views
        TabLayout tabLayout = findViewById(R.id.tabLayout);
        ViewPager2 viewPager = findViewById(R.id.viewPager);

        // 2. Set up the adapter
        ViewPagerAdapter adapter = new ViewPagerAdapter(this);
        viewPager.setAdapter(adapter);

        // 3. Connect TabLayout and ViewPager2
        String[] titles = new String[]{"Top Stories", "Sports",
"Entertainment"};

        new TabLayoutMediator(tabLayout, viewPager, (tab,
position) -> {
            tab.setText(titles[position]);
        }).attach();
    }
}
```

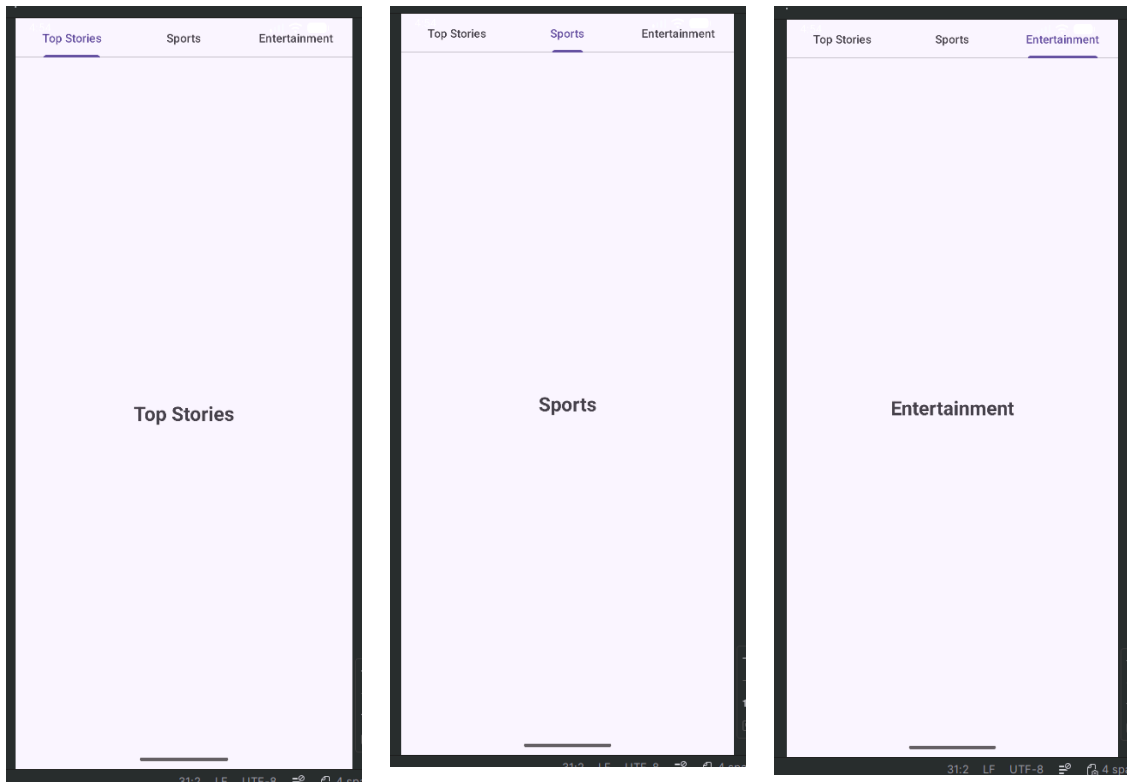


Fig 3.11: Application Screenshot

INPUT CONTROLS IN ANDROID

PROBLEM STATEMENT 1

Develop a "Test App" that includes a layout with a **Button** and a **ToggleButton**. When each button is clicked, a custom **Toast message** should be displayed with different images as their content.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/myButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me" />

    <ToggleButton
        android:id="@+id/myToggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Toggle Me" />

</LinearLayout>
```

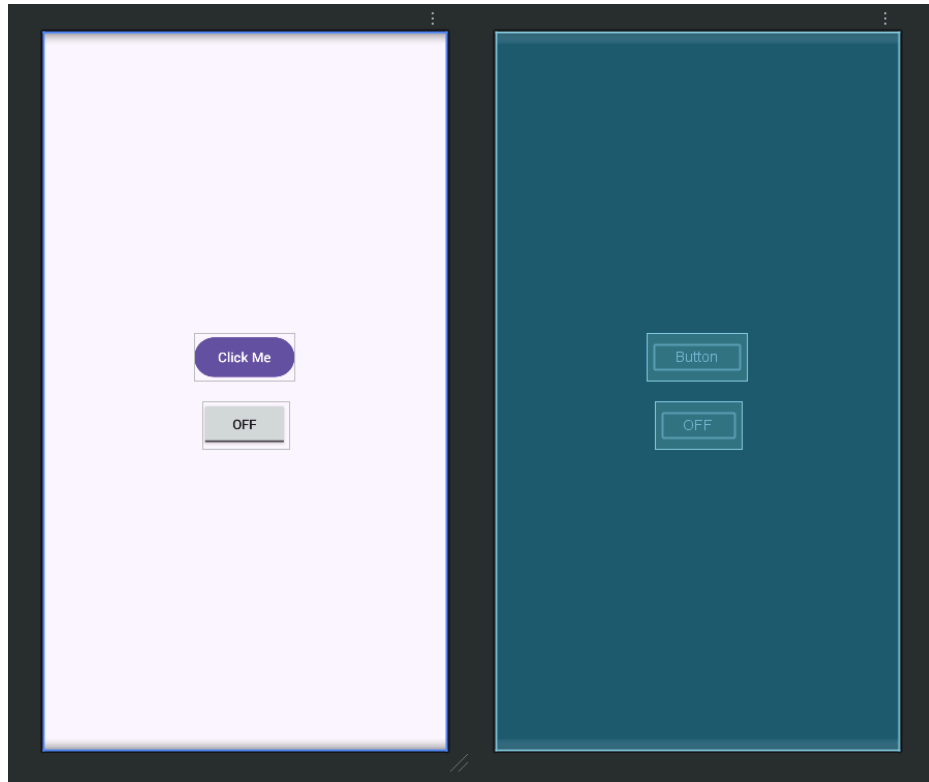


Fig 4.1: Activity layout

Custom_toast_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/toast_root"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:background="#DDDDDD"
    android:padding="20dp"
    android:orientation="vertical"
    android:gravity="center">

    <ImageView
        android:id="@+id/toastImage"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:src="@drawable/ic_launcher_foreground" />
```

```

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Custom Toast!"
    android:textSize="16sp"
    android:textStyle="bold"/>

```

```

</LinearLayout>

```

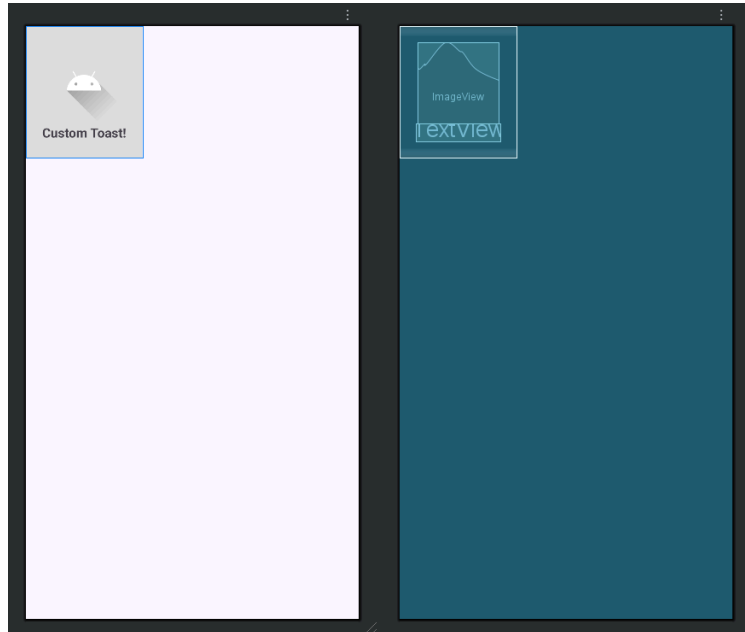


Fig 4.2: Custom toast layout

MainActivity.java

```

package com.example.testapp;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;

```

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Find buttons by ID
        Button simpleBtn = findViewById(R.id.myButton);
        ToggleButton toggleBtn =
        findViewById(R.id.myToggleButton);

        // Set click listener for standard button
        simpleBtn.setOnClickListener(v ->
        showCustomToast(R.drawable.star));

        // Set click listener for toggle button
        toggleBtn.setOnClickListener(v ->
        showCustomToast(R.drawable.heart));
    }

    // Helper method to show custom toast
    private void showCustomToast(int imageResId) {
        // 1. Inflate the custom layout
        LayoutInflater inflater = getLayoutInflater();
        View layout =
        inflater.inflate(R.layout.custom_toast_layout,
        findViewById(R.id.toast_root));

        // 2. Set the specific image
        ImageView image = layout.findViewById(R.id.toastImage);
        image.setImageResource(imageResId);

        // 3. Create and show the Toast
        Toast toast = new Toast(getApplicationContext());
        toast.setDuration(Toast.LENGTH_SHORT);
        toast.setView(layout);
        toast.show();
    }
}

```

}

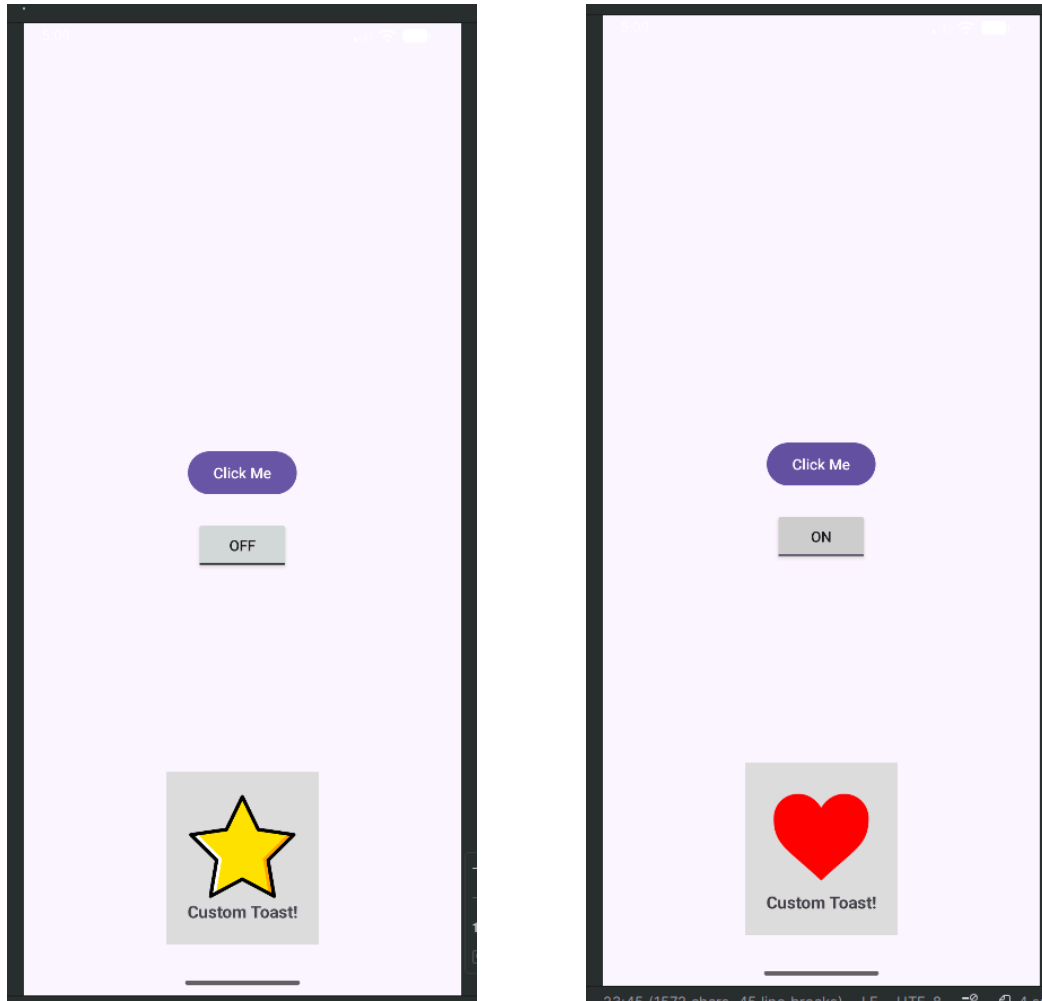


Fig 4.3: Application Screenshot

PROBLEM STATEMENT 2

Create an app that contains a view with **multiple buttons**, each labeled with different Android versions. When a button is clicked, a **Toast message** should appear, displaying the corresponding Android version's name along with its associated icon

SOLUTION

Activity_main.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btn_pie"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Android 9.0 (Pie)" />

    <Button
        android:id="@+id/btn_oreo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Android 8.0 (Oreo)" />

    <Button
        android:id="@+id/btn_nougat"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Android 7.0 (Nougat)" />

</LinearLayout>
```

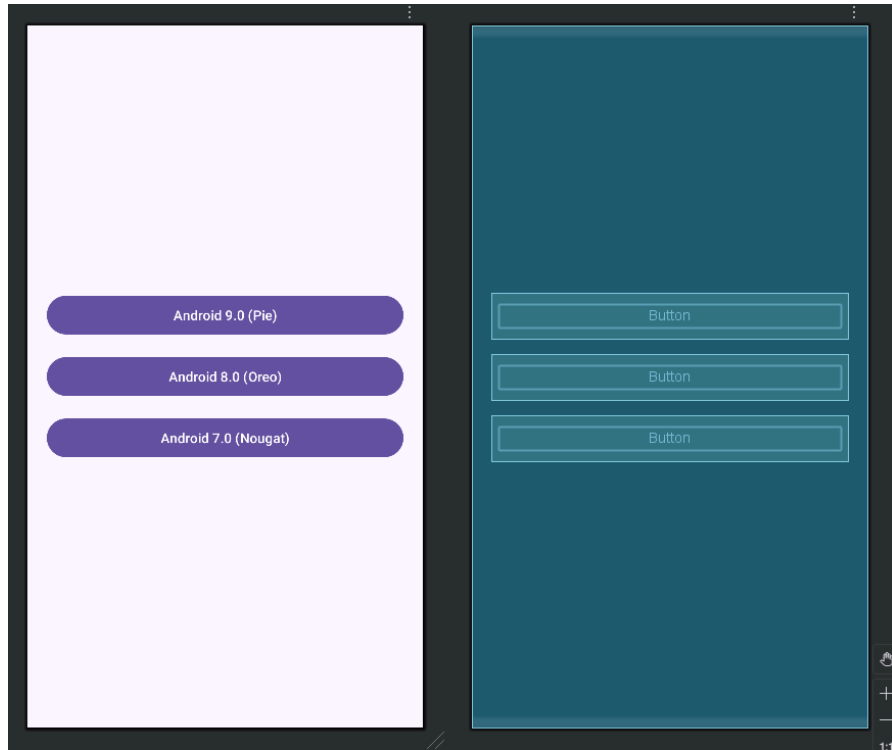


Fig 4.4: Activity layout

Custom_toast.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/toast_root"
    android:orientation="horizontal"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:padding="16dp"
    android:background="#444444"
    android:gravity="center_vertical">

    <ImageView
        android:id="@+id/toast_image"
        android:layout_width="40dp"
        android:layout_height="40dp"
        android:layout_marginEnd="10dp"
        android:contentDescription="icon" />
```

```

<TextView
    android:id="@+id/toast_text"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textColor="#FFFFFF"
    android:textStyle="bold" />
</LinearLayout>

```



Fig 4.5: Activity layout

MainActivity.java

```

package com.example.versiontoastapp;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

```

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Find buttons
        Button btnPie = findViewById(R.id.btn_pie);
        Button btnOreo = findViewById(R.id.btn_oreo);
        Button btnNougat = findViewById(R.id.btn_nougat);

        // Set listeners
        btnPie.setOnClickListener(v -> showCustomToast("Android
Pie", R.drawable.img_pie));
        btnOreo.setOnClickListener(v ->
showCustomToast("Android Oreo", R.drawable.img_oreo));
        btnNougat.setOnClickListener(v ->
showCustomToast("Android Nougat", R.drawable.img_nougat));
    }

    private void showCustomToast(String message, int
imageResId) {
        // Inflate custom layout
        LayoutInflater inflater = getLayoutInflater();
        View layout = inflater.inflate(R.layout.custom_toast,
findViewById(R.id.toast_root));

        // Set image and text
        ImageView image =
layout.findViewById(R.id.toast_image);
        image.setImageResource(imageResId);

        TextView text = layout.findViewById(R.id.toast_text);
        text.setText(message);

        // Create and show toast
        Toast toast = new Toast(getApplicationContext());
        toast.setDuration(Toast.LENGTH_SHORT);
    }
}

```

```
        toast.setView(layout);  
        toast.show();  
    }  
}
```



Fig 4.6: Application Screenshot

PROBLEM STATEMENT 3

Develop a view with a ToggleButton labeled "Current Mode" that has two states: "Wi-Fi" and "Mobile Data." Based on the state of the toggle button, an image corresponding to the selected mode should appear, and a Toast message should display the current mode. Additionally, when the user clicks the "Change Mode" button, the app should switch to the corresponding mode and update the image accordingly.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Wi-Fi"
        android:textOff="Mobile Data"
        android:checked="false" />

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_marginTop="20dp"
        android:layout_marginBottom="20dp"
        android:src="@drawable/ic_data" />

    <Button
        android:id="@+id/btnChangeMode"
        android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"  
android:text="Change Mode" />
```

```
</LinearLayout>
```

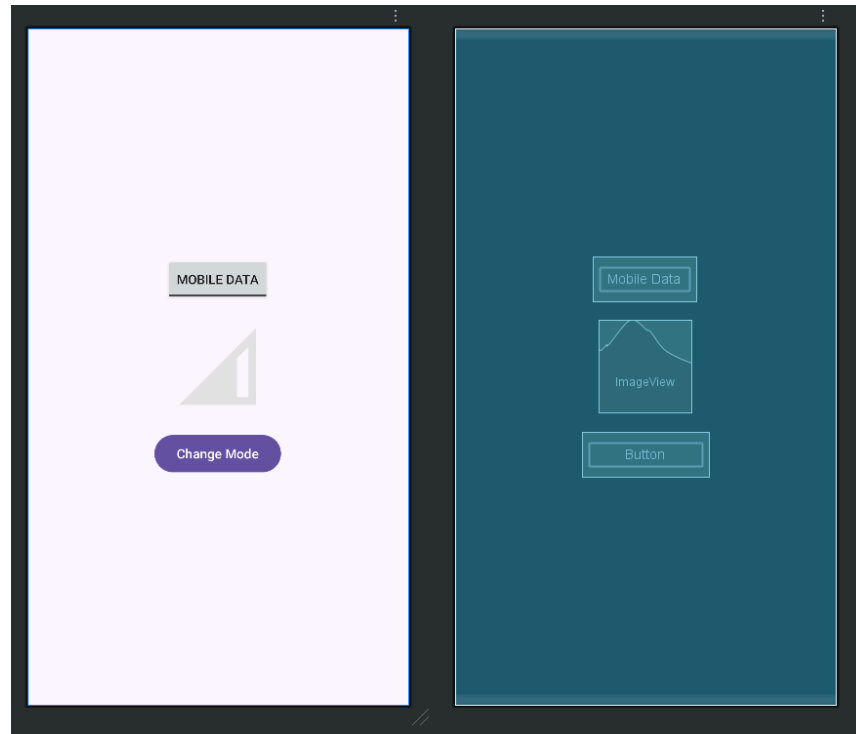


Fig 4.7: Activity layout

MainActivity.java

```
package com.example.modeswitcher;  
  
import android.os.Bundle;  
import android.widget.Button;  
import android.widget.ImageView;  
import android.widget.Toast;  
import android.widget.ToggleButton;  
import androidx.appcompat.app.AppCompatActivity;  
  
public class MainActivity extends AppCompatActivity {
```

```

ToggleButton toggleButton;
ImageView imageView;
Button btnChangeMode;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // 1. Initialize Views
    toggleButton = findViewById(R.id.toggleButton);
    imageView = findViewById(R.id.imageView);
    btnChangeMode = findViewById(R.id.btnChangeMode);

    // 2. Set listener for ToggleButton state changes
    toggleButton.setOnCheckedChangeListener((buttonView,
isChecked) -> {
        if (isChecked) {
            // Wi-Fi State
            imageView.setImageResource(R.drawable.ic_wifi);
            Toast.makeText(MainActivity.this, "Current
Mode: Wi-Fi", Toast.LENGTH_SHORT).show();
        } else {
            // Mobile Data State
            imageView.setImageResource(R.drawable.ic_data);
            Toast.makeText(MainActivity.this, "Current
Mode: Mobile Data", Toast.LENGTH_SHORT).show();
        }
    });

    // 3. Set listener for "Change Mode" button
    btnChangeMode.setOnClickListener(v -> {
        // Simply flip the current state; the listener
above will handle the rest
        toggleButton.toggle();
    });
}
}

```

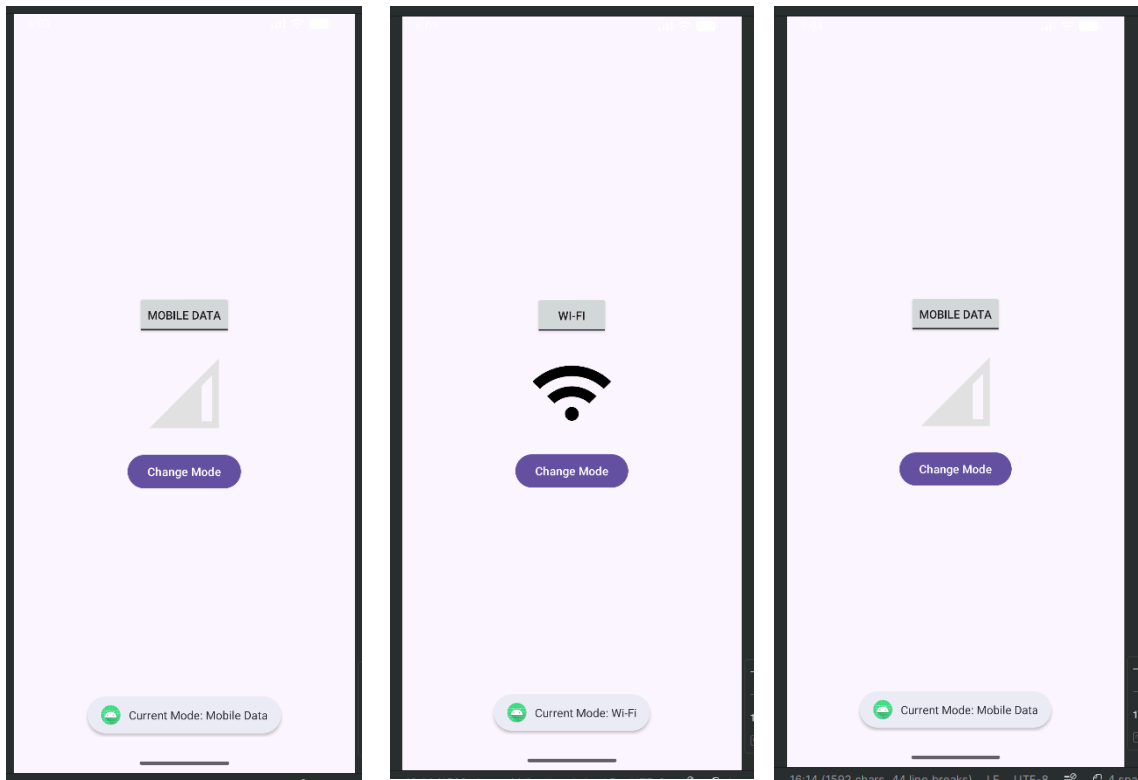



Fig 4.8: Application Screenshot

PROBLEM STATEMENT 4

Create a “Food Ordering App” which **lists** food items with **check boxes**. Once the user checks /unchecks the item and click on the submit button display the items ordered along with cost of each item and total cost in a new activity. Once the user clicks on the submit button he/she should not be allowed to change the order i.e. he should not be allowed to change the state of the items checked.

SOLUTION

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <CheckBox
        android:id="@+id/cbPizza"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Pizza ($10)"/>

    <CheckBox
        android:id="@+id/cbBurger"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Burger ($5)"/>
```

```

<CheckBox
    android:id="@+id/cbCoffee"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Coffee ($3)"/>

<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Place Order"
    android:layout_marginTop="30dp"/>

</LinearLayout>

```

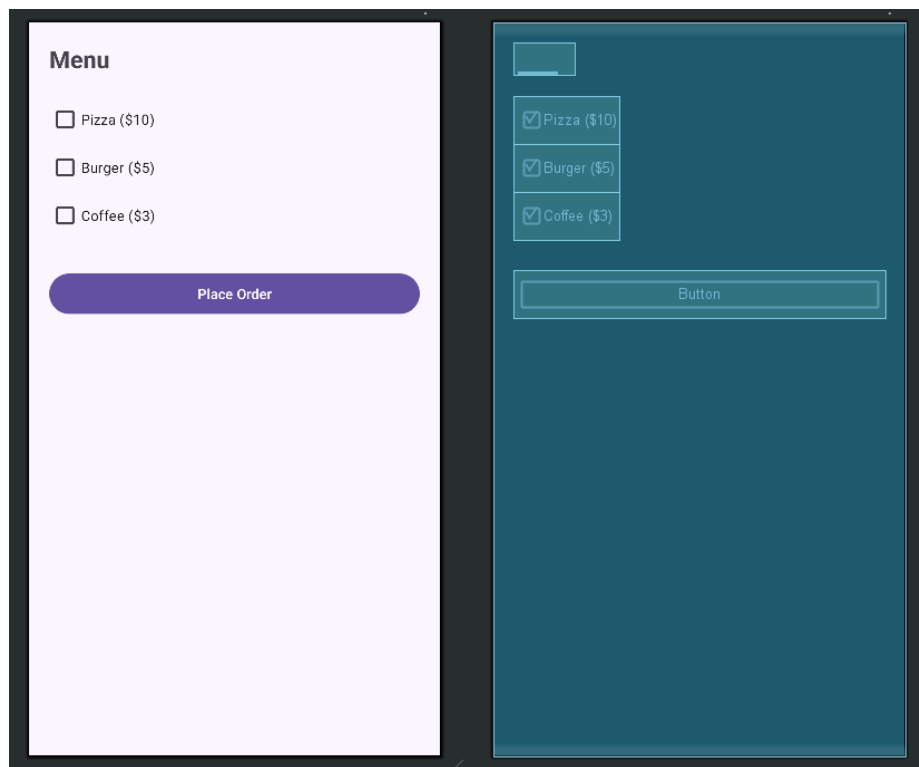


Fig 4.9: Activity Layout

Activity_order_summary.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

```

```

xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Order Summary"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <TextView
        android:id="@+id/tvOrderDetails"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"/>

</LinearLayout>

```

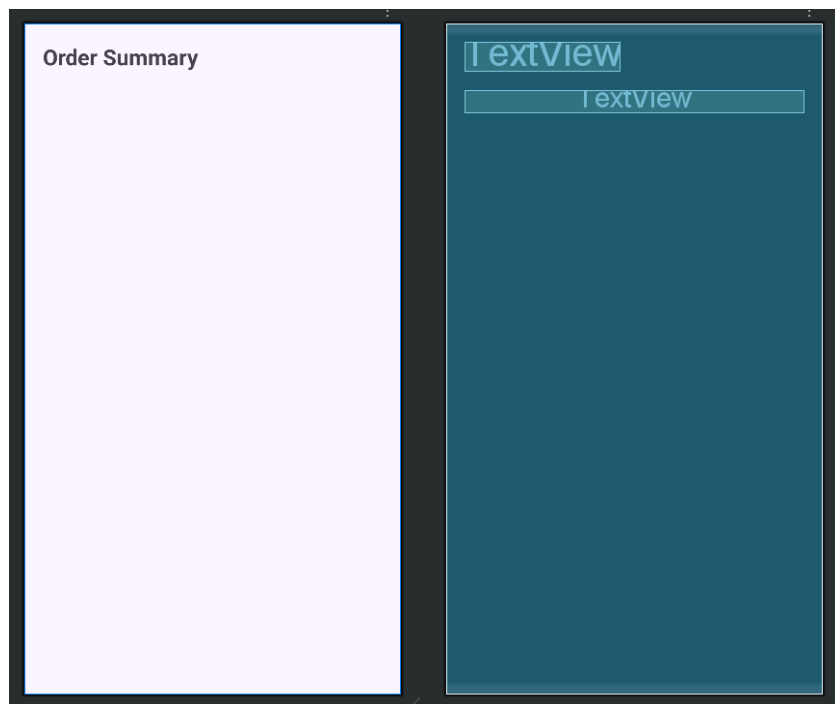


Fig 4.10: Activity Layout

OrderSummaryActivity.java

```
package com.example.foodorderingapp;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class OrderSummaryActivity extends AppCompatActivity {

    TextView tvOrderDetails;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_order_summary);

        tvOrderDetails = findViewById(R.id.tvOrderDetails);

        // Get data passed from MainActivity
        String orderData =
        getIntent().getStringExtra("ORDER_DATA");

        // Display data
        if (orderData != null) {
            tvOrderDetails.setText(orderData);
        }
    }
}
```

MainActivity.java

```
package com.example.foodorderingapp;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
```

```

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    CheckBox cbPizza, cbBurger, cbCoffee;
    Button btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Initialize Views
        cbPizza = findViewById(R.id.cbPizza);
        cbBurger = findViewById(R.id.cbBurger);
        cbCoffee = findViewById(R.id.cbCoffee);
        btnSubmit = findViewById(R.id.btnSubmit);
        btnSubmit.setOnClickListener(new View.OnClickListener()
{
    @Override
    public void onClick(View v) {
        StringBuilder order = new StringBuilder();
        int total = 0;
        // Check items and calculate cost
        if (cbPizza.isChecked()) {
            order.append("Pizza: $10\n");
            total += 10;
        }
        if (cbBurger.isChecked()) {
            order.append("Burger: $5\n");
            total += 5;
        }
        if (cbCoffee.isChecked()) {
            order.append("Coffee: $3\n");
            total += 3;
        }
        order.append("\nTotal Cost: $").append(total);
        // Disable checkboxes so order cannot be
changed
        cbPizza.setEnabled(false);
    }
});
}

```

```

        cbBurger.setEnabled(false);
        cbCoffee.setEnabled(false);
        btnSubmit.setEnabled(false);
        // Navigate to next activity
        Intent intent = new Intent(MainActivity.this,
OrderSummaryActivity.class);
        intent.putExtra("ORDER_DATA",
order.toString());
        startActivity(intent);
    }
});
}
}

```

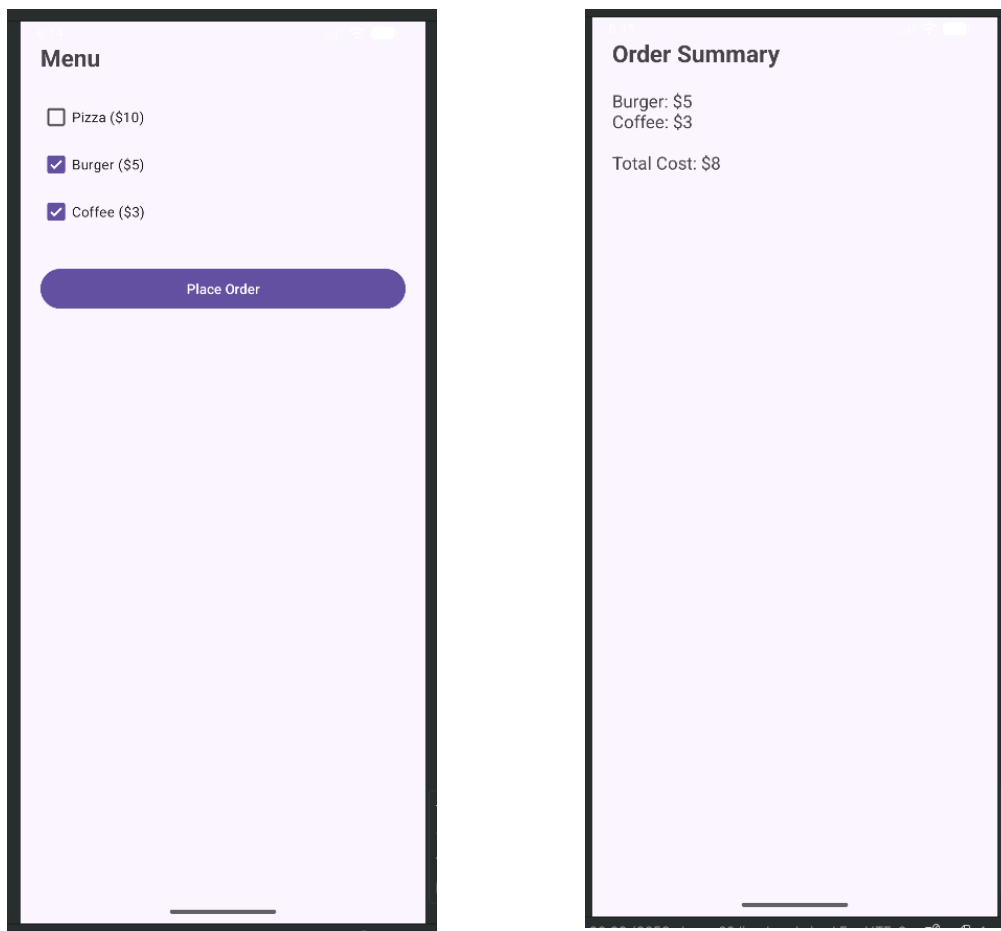


Fig 4.11: Application Screenshot

INPUT CONTROLS-SPINNERS, PICKERS

PROBLEM STATEMENT 1

Develop an application named "Vehicle Parking Registration" where the user can register their vehicle for parking. The app should include a spinner that allows users to select the type of vehicle (e.g., car, bike, etc.) and text fields for entering the vehicle number and RC number. Upon clicking the submit button, the entered details should be displayed in a separate view, providing the user with options to either confirm the details or edit them. Once the user confirms the information, a toast message should appear showing a unique serial number to confirm the parking allotment.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Parking Registration"
        android:textSize="24sp"
```



```
        android:textStyle="bold"  
        android:layout_marginBottom="20dp"/>
```

```
<Spinner  
    android:id="@+id/spinnerVehicleType"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:minHeight="48dp"  
    android:layout_marginBottom="10dp"/>
```

```
<EditText  
    android:id="@+id/etVehicleNumber"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="Vehicle Number"  
    android:layout_marginBottom="10dp"/>
```

```
<EditText  
    android:id="@+id/etRcNumber"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="RC Number"  
    android:layout_marginBottom="20dp"/>
```

```
<Button  
    android:id="@+id/btnSubmit"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Submit"/>
```

```
</LinearLayout>
```

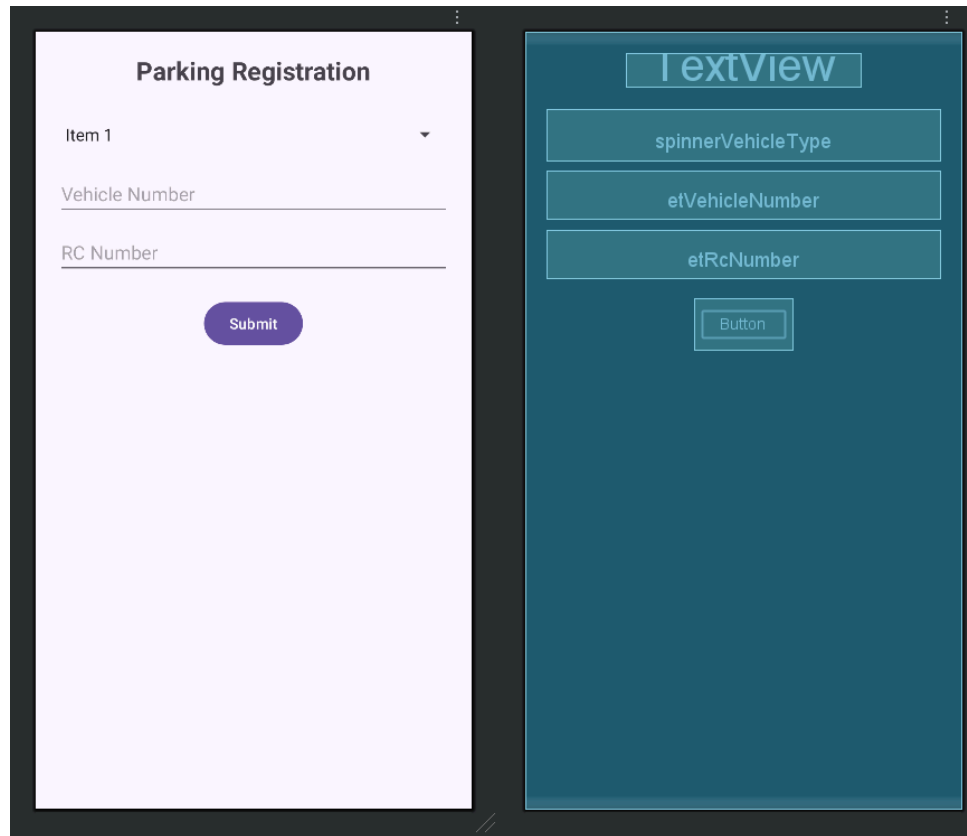


Fig 5.1: activity_main.xml

MainActivity.java

```
package com.example.vehicleparkingregistration;

import android.content.Intent;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Spinner spinnerType;
    EditText etVehicleNo, etRcNo;
    Button btnSubmit;
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // Link views
    spinnerType = findViewById(R.id.spinnerVehicleType);
    etVehicleNo = findViewById(R.id.etVehicleNumber);
    etRcNo = findViewById(R.id.etRcNumber);
    btnSubmit = findViewById(R.id.btnSubmit);

    // Setup Spinner
    String[] types = {"Car", "Bike", "Truck", "Other"};
    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
    android.R.layout.simple_spinner_dropdown_item, types);
    spinnerType.setAdapter(adapter);

    // Submit Logic
    btnSubmit.setOnClickListener(v -> {
        String type =
    spinnerType.getSelectedItem().toString();
        String vehicleNo =
    etVehicleNo.getText().toString();
        String rcNo = etRcNo.getText().toString();

        // Pass data to DetailsActivity
        Intent intent = new Intent(MainActivity.this,
    DetailsActivity.class);
        intent.putExtra("TYPE", type);
        intent.putExtra("V_NO", vehicleNo);
        intent.putExtra("RC_NO", rcNo);
        startActivity(intent);
    });
}
}

```

Activity_details.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Confirm Details"
        android:textSize="22sp"
        android:layout_marginBottom="20dp"/>

    <TextView
        android:id="@+id/tvDisplayDetails"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:layout_marginBottom="30dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center">

        <Button
            android:id="@+id/btnEdit"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Edit"
            android:layout_marginEnd="20dp"/>

        <Button
            android:id="@+id/btnConfirm"
```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Confirm"/>
    </LinearLayout>

</LinearLayout>

```

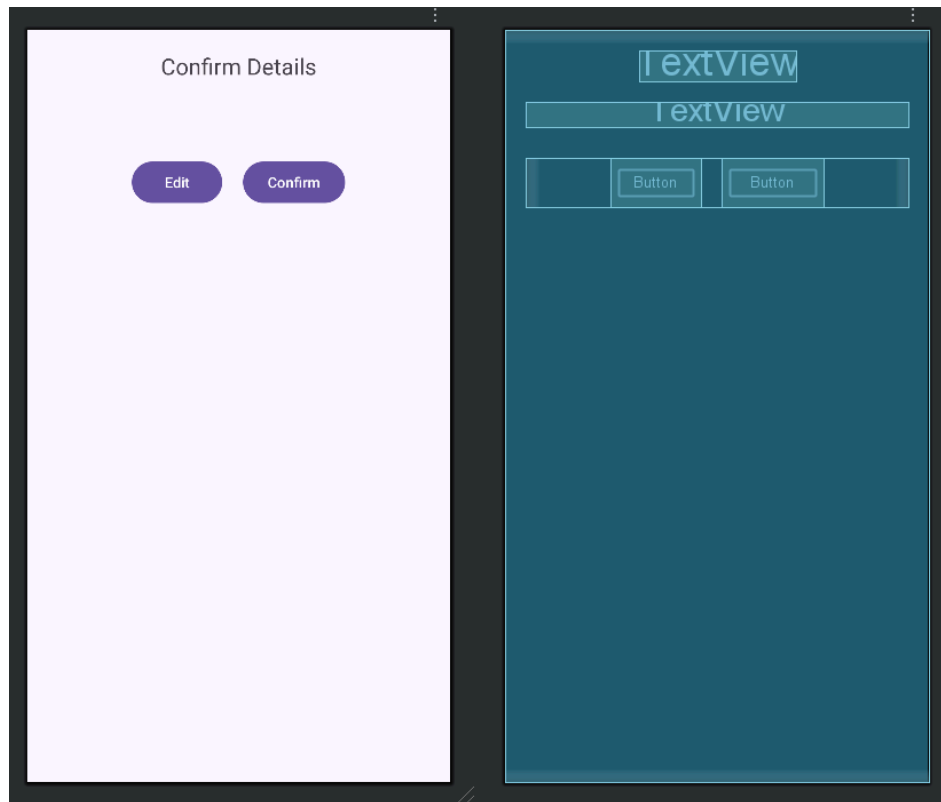


Fig 5.2: activity_details.xml

DetailsActivity.java

```

package com.example.vehicleparkingregistration;

import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

```

```

import java.util.UUID;

public class DetailsActivity extends AppCompatActivity {

    TextView tvDetails;
    Button btnConfirm, btnEdit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_details);

        // Link views
        tvDetails = findViewById(R.id.tvDisplayDetails);
        btnConfirm = findViewById(R.id.btnConfirm);
        btnEdit = findViewById(R.id.btnEdit);

        // Get data from intent
        String type = getIntent().getStringExtra("TYPE");
        String vNo = getIntent().getStringExtra("V_NO");
        String rcNo = getIntent().getStringExtra("RC_NO");

        // Show data
        tvDetails.setText("Type: " + type + "\nVehicle No: " +
vNo + "\nRC No: " + rcNo);

        // Edit Action: Close this activity to go back
        btnEdit.setOnClickListener(v -> finish());

        // Confirm Action: Generate ID and show Toast
        btnConfirm.setOnClickListener(v -> {
            String serial = "SN-" +
UUID.randomUUID().toString().substring(0, 5).toUpperCase();
            Toast.makeText(this, "Allocated! Serial: " +
serial, Toast.LENGTH_LONG).show();
        });
    }
}

```

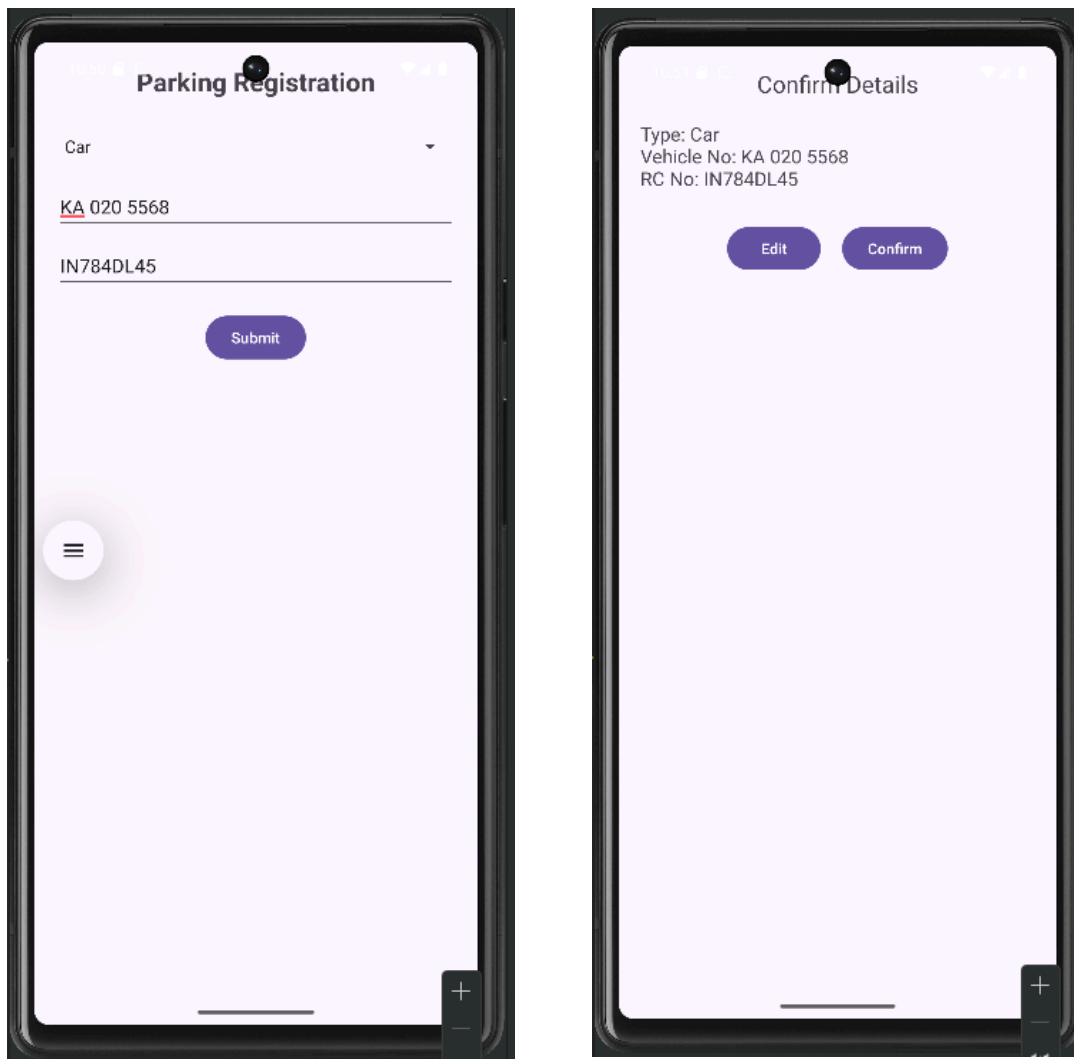


Fig 5.3: screenshot of the application

PROBLEM STATEMENT 2

Design and develop a "Travel Ticket Booking" app where users can book tickets by selecting the source and destination from dropdown lists (spinners) and choosing the date of travel using a date picker. Include a toggle button to let users specify whether they want a one-way ticket or a round-trip ticket. The app should have "Submit" and "Reset" buttons. When the "Submit" button is clicked, display the entered details on a new screen in a structured format. The "Reset" button should clear all input fields and set the date picker to the current system date and time.

SOLUTION

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/select_source"
        android:textSize="16sp"
        android:textStyle="bold"/>

    <Spinner
        android:id="@+id/spinnerSource"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/select_dest"
```



```
android:textSize="16sp"  
android:textStyle="bold"/>
```

<Spinner

```
    android:id="@+id/spinnerDest"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:layout_marginBottom="16dp"/>
```

<Button

```
    android:id="@+id/btnDate"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="@string/select_date"  
    android:layout_marginBottom="16dp"/>
```

<ToggleButton

```
    android:id="@+id/toggleTripType"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:textOff="@string/trip_type_off"  
    android:textOn="@string/trip_type_on"  
    android:checked="false"  
    android:layout_marginBottom="24dp"/>
```

<LinearLayout

```
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:orientation="horizontal"  
    android:gravity="center">
```

<Button

```
    android:id="@+id/btnSubmit"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="@string/submit"  
    android:layout_marginEnd="20dp"/>
```

<Button

```

        android:id="@+id/btnReset"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/reset"
        android:backgroundTint="#FF5722"/>
    </LinearLayout>

```

```

</LinearLayout>

```

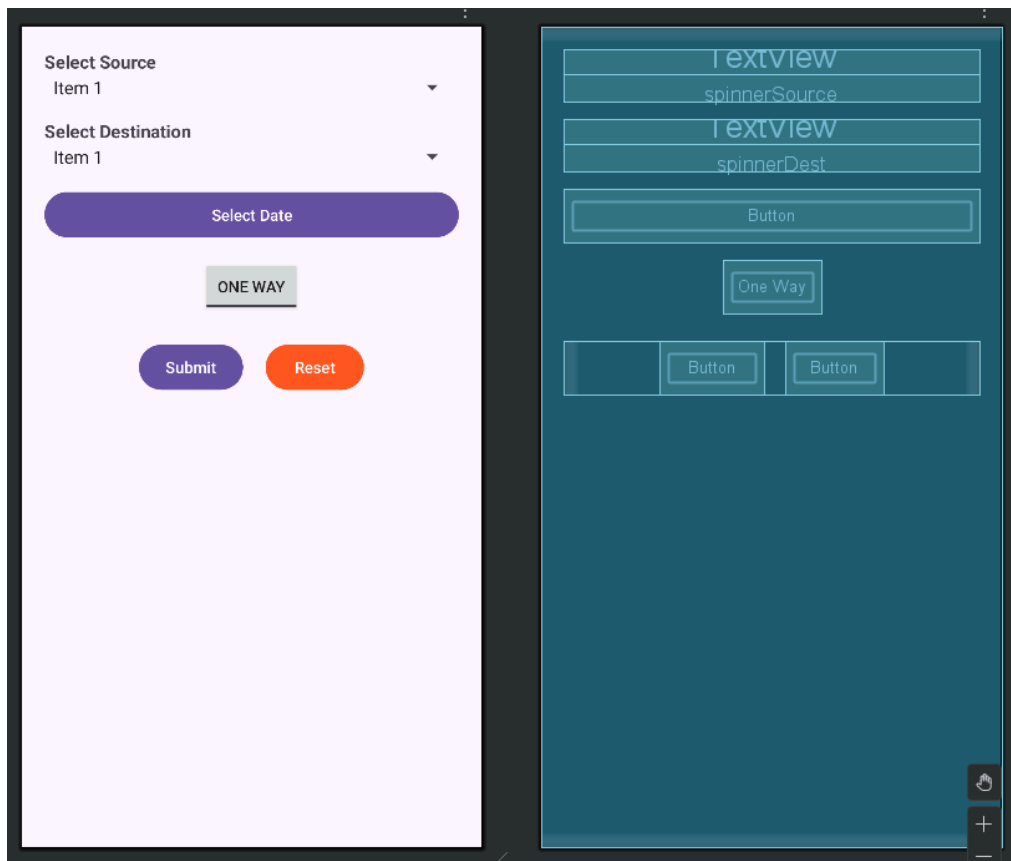


Fig 5.4: activity_main.xml

MainActivity.java

```

package com.example.travelbookingapp; // Check your package
name!

import android.app.DatePickerDialog;
import android.content.Intent;

```

```

import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.Button;
import android.widget.Spinner;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    Spinner spinSource, spinDest;
    Button btnDate, btnSubmit, btnReset;
    ToggleButton toggleTrip;
    String selectedDate = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Init views
        spinSource = findViewById(R.id.spinnerSource);
        spinDest = findViewById(R.id.spinnerDest);
        btnDate = findViewById(R.id.btnDate);
        btnSubmit = findViewById(R.id.btnSubmit);
        btnReset = findViewById(R.id.btnReset);
        toggleTrip = findViewById(R.id.toggleTripType);

        // Setup Spinners with data from strings.xml
        ArrayAdapter<CharSequence> adapter =
        ArrayAdapter.createFromResource(this,
            R.array.cities,
            android.R.layout.simple_spinner_item);

        adapter.setDropDownViewResource(android.R.layout.simple_spinner
        _dropdown_item);
        spinSource.setAdapter(adapter);
        spinDest.setAdapter(adapter);
    }
}

```

```

// DatePicker Logic
btnDate.setOnClickListener(v -> {
    final Calendar c = Calendar.getInstance();
    int year = c.get(Calendar.YEAR);
    int month = c.get(Calendar.MONTH);
    int day = c.get(Calendar.DAY_OF_MONTH);

    DatePickerDialog datePickerDialog = new
DatePickerDialog(this,
                (view, year1, month1, dayOfMonth) -> {
                    selectedDate = dayOfMonth + "/" +
(month1 + 1) + "/" + year1;
                    btnDate.setText(selectedDate);
                }, year, month, day);
    datePickerDialog.show();
});

// Submit Button Logic
btnSubmit.setOnClickListener(v -> {
    if (selectedDate.isEmpty()) {
        Toast.makeText(this, "Please select a date",
Toast.LENGTH_SHORT).show();
        return;
    }

    String source =
spinSource.getSelectedItem().toString();
    String dest =
spinDest.getSelectedItem().toString();
    String type = toggleTrip.isChecked() ? "Round Trip"
: "One Way";

    // Pass data to next activity
    Intent intent = new Intent(MainActivity.this,
TicketDetailsActivity.class);
    intent.putExtra("SOURCE", source);
    intent.putExtra("DEST", dest);
    intent.putExtra("DATE", selectedDate);

```

```

        intent.putExtra("TYPE", type);
        startActivity(intent);
    });

    // Reset Button Logic
    btnReset.setOnClickListener(v -> {
        spinSource.setSelection(0);
        spinDest.setSelection(0);
        toggleTrip.setChecked(false);

        // Reset date to current system date
        final Calendar c = Calendar.getInstance();
        selectedDate = c.get(Calendar.DAY_OF_MONTH) + "/" +
            (c.get(Calendar.MONTH) + 1) + "/" +
            c.get(Calendar.YEAR);
        btnDate.setText("Select Date"); // Or set to
current date string
        selectedDate = ""; // Clear selection tracking
    });
}
}

```

Activity_ticket_details.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Ticket Summary"
        android:textSize="24sp"
        android:textStyle="bold"

```

```
        android:layout_marginBottom="20dp"/>

<TextView
    android:id="@+id/txtDetails"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    android:lineSpacingExtra="8dp"/>

</LinearLayout>
```

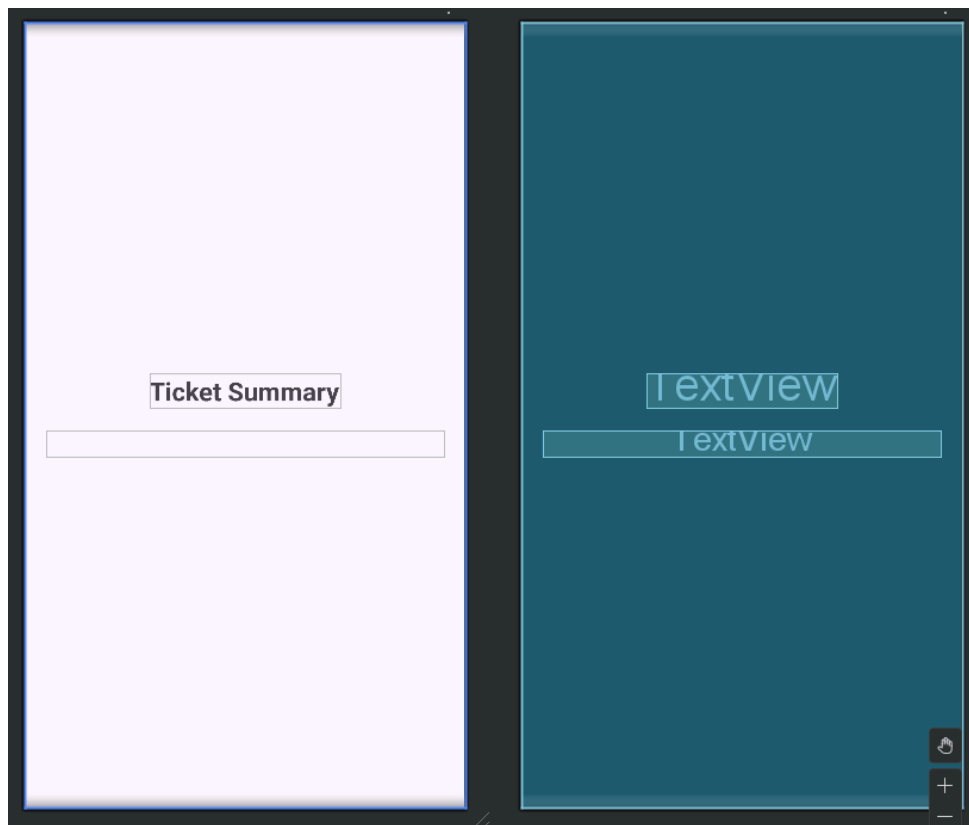


Fig 5.4: activity_ticket_details.xml

TicketDetailsActivity.java

```
package com.example.travelbookingapp; // Check your package name!

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class TicketDetailsActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_ticket_details);

        TextView txtDetails = findViewById(R.id.txtDetails);

        // Get data passed from MainActivity
        String source = getIntent().getStringExtra("SOURCE");
        String dest = getIntent().getStringExtra("DEST");
        String date = getIntent().getStringExtra("DATE");
        String type = getIntent().getStringExtra("TYPE");

        // Format and display
        String summary = "Source: " + source + "\n\n" +
            "Destination: " + dest + "\n\n" +
            "Date of Travel: " + date + "\n\n" +
            "Trip Type: " + type;

        txtDetails.setText(summary);
    }
}
```

Strings.xml

```
<resources>
    <string name="app_name">Travel Booking</string>
    <string name="select_source">Select Source</string>
```

```

<string name="select_dest">Select Destination</string>
<string name="select_date">Select Date</string>
<string name="trip_type_off">One Way</string>
<string name="trip_type_on">Round Trip</string>
<string name="submit">Submit</string>
<string name="reset">Reset</string>

<string-array name="cities">
    <item>New York</item>
    <item>London</item>
    <item>Paris</item>
    <item>Tokyo</item>
    <item>Mumbai</item>
</string-array>
</resources>

```

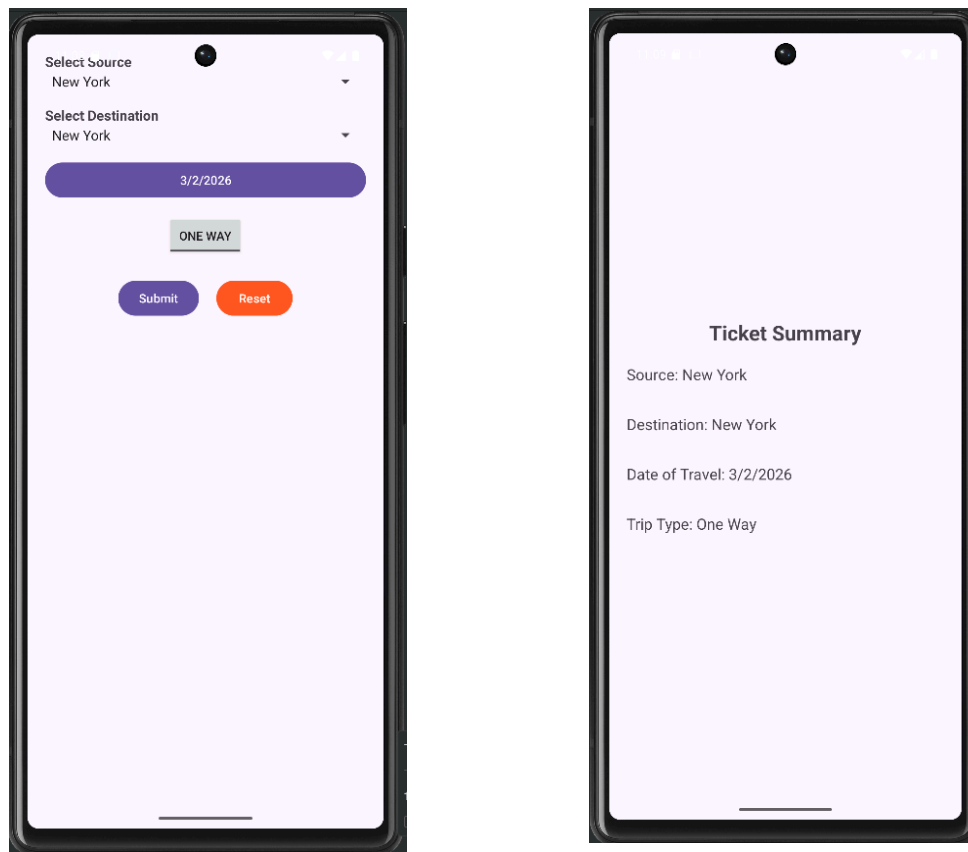


Fig 5.5: Application Screenshot

PROBLEM STATEMENT 3

Design and develop a "Movie Ticket Booking" app where users can book tickets by selecting the movie and theatre from dropdown menus (spinners), the date of the show using a date picker, and the preferred showtime using a time picker. Include a toggle button to let users choose between a standard ticket or a premium ticket. If the "premium" option is selected, ensure the "Submit" button becomes clickable only after 12:00 PM. The app should have "Book Now" and "Reset" buttons. When the user clicks "Book Now," display the entered details along with available seats for the selected show in a new screen in a well-organized format. The "Reset" button should clear all fields and reset the date picker to the current system date. Validate all inputs to ensure correct data entry.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center_horizontal">

        <TextView
            android:text="Movie Booking"
            android:textSize="24sp"
            android:textStyle="bold"
            android:layout_marginBottom="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <Spinner
            android:id="@+id/spinnerMovie"
```

```
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:minHeight="48dp"  
    android:layout_marginBottom="10dp"/>
```

<Spinner

```
    android:id="@+id/spinnerTheatre"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:minHeight="48dp"  
    android:layout_marginBottom="10dp"/>
```

<ToggleButton

```
    android:id="@+id/toggleTicketType"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:textOn="Premium"  
    android:textOff="Standard"  
    android:layout_marginBottom="10dp"/>
```

<TextView

```
    android:text="Select Date:"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

<DatePicker

```
    android:id="@+id/datePicker"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:datePickerMode="spinner"  
    android:calendarViewShown="false"/>
```

<TextView

```
    android:text="Select Time:"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

<TimePicker

```
    android:id="@+id/timePicker"
```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:timePickerMode="spinner"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center"
    android:layout_marginTop="20dp">

    <Button
        android:id="@+id/btnReset"
        android:text="Reset"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginEnd="20dp"/>

    <Button
        android:id="@+id/btnBook"
        android:text="Book Now"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>

</LinearLayout>
</ScrollView>

```

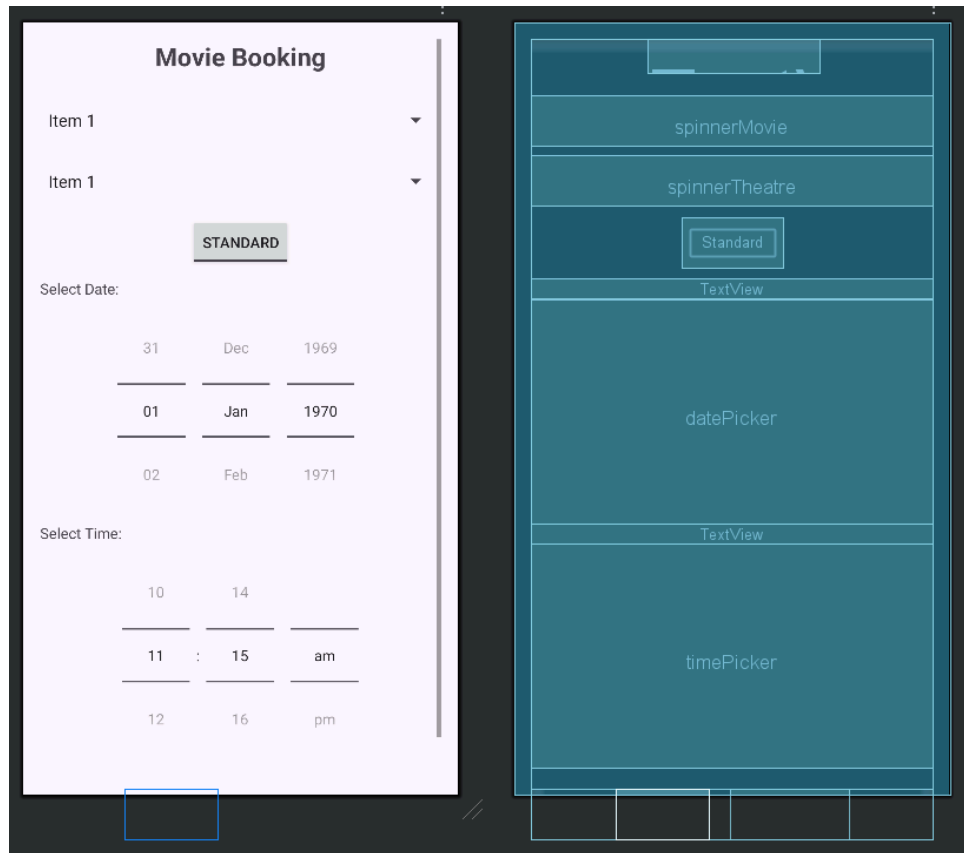


Fig 5.6: activity_main.xml

MainActivity.java

```
package com.example.moviebookingapp;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.Button;
import android.widget.DatePicker;
import android.widget.Spinner;
import android.widget.TimePicker;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;
```

```

public class MainActivity extends AppCompatActivity {

    Spinner spMovie, spTheatre;
    DatePicker datePicker;
    TimePicker timePicker;
    ToggleButton toggleTicket;
    Button btnBook, btnReset;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Init Views
        spMovie = findViewById(R.id.spinnerMovie);
        spTheatre = findViewById(R.id.spinnerTheatre);
        datePicker = findViewById(R.id.datePicker);
        timePicker = findViewById(R.id.timePicker);
        toggleTicket = findViewById(R.id.toggleTicketType);
        btnBook = findViewById(R.id.btnBook);
        btnReset = findViewById(R.id.btnReset);

        // Setup Spinners
        ArrayAdapter<CharSequence> movieAdapter =
        ArrayAdapter.createFromResource(this, R.array.movies_array,
        android.R.layout.simple_spinner_item);

        movieAdapter.setDropDownViewResource(android.R.layout.simple_sp
        inner_dropdown_item);
        spMovie.setAdapter(movieAdapter);

        ArrayAdapter<CharSequence> theatreAdapter =
        ArrayAdapter.createFromResource(this, R.array.theatres_array,
        android.R.layout.simple_spinner_item);

        theatreAdapter.setDropDownViewResource(android.R.layout.simple_
        spinner_dropdown_item);
        spTheatre.setAdapter(theatreAdapter);
    }
}

```

```

        // Listeners for Premium Logic
        timePicker.setOnTimeChangedListener((view, hour,
minute) -> validatePremiumTime());
        toggleTicket.setOnCheckedChangeListener((buttonView,
isChecked) -> validatePremiumTime());

        btnBook.setOnClickListener(v -> bookTicket());
        btnReset.setOnClickListener(v -> resetFields());
    }

    // Check if Premium is selected and Time is before 12 PM
    (12:00)
    private void validatePremiumTime() {
        boolean isPremium = toggleTicket.isChecked();
        int hour = timePicker.getHour(); // 24-hour format

        if (isPremium && hour < 12) {
            btnBook.setEnabled(false);
            Toast.makeText(this, "Premium tickets only
available after 12:00 PM", Toast.LENGTH_SHORT).show();
        } else {
            btnBook.setEnabled(true);
        }
    }

    private void bookTicket() {
        if (spMovie.getSelectedItemPosition() == 0 ||
spTheatre.getSelectedItemPosition() == 0) {
            Toast.makeText(this, "Please select Movie and
Theatre", Toast.LENGTH_SHORT).show();
            return;
        }

        String date = datePicker.getDayOfMonth() + "/" +
(datePicker.getMonth() + 1) + "/" + datePicker.getYear();
        String time = String.format("%02d:%02d",
timePicker.getHour(), timePicker.getMinute());
        String ticketType = toggleTicket.isChecked() ?
"Premium" : "Standard";

```

```

        Intent intent = new Intent(MainActivity.this,
DisplayActivity.class);
        intent.putExtra("MOVIE",
spMovie.getSelectedItem().toString());
        intent.putExtra("THEATRE",
spTheatre.getSelectedItem().toString());
        intent.putExtra("DATE", date);
        intent.putExtra("TIME", time);
        intent.putExtra("TYPE", ticketType);
        startActivity(intent);
    }

    private void resetFields() {
        spMovie.setSelection(0);
        spTheatre.setSelection(0);
        toggleTicket.setChecked(false);

        // Reset Date to current
        Calendar c = Calendar.getInstance();
        datePicker.updateDate(c.get(Calendar.YEAR),
c.get(Calendar.MONTH), c.get(Calendar.DAY_OF_MONTH));

        // Reset Time to current
        timePicker.setHour(c.get(Calendar.HOUR_OF_DAY));
        timePicker.setMinute(c.get(Calendar.MINUTE));

        btnBook.setEnabled(true);
    }
}

```

Activity_display.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"

```

```

    android:padding="20dp"
    android:gravity="center">
    <TextView
        android:text="Booking Confirmed!"
        android:textSize="26sp"
        android:textStyle="bold"
        android:textColor="#4CAF50"
        android:layout_marginBottom="30dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/tvDetails"
        android:textSize="18sp"
        android:lineSpacingExtra="8dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>

```

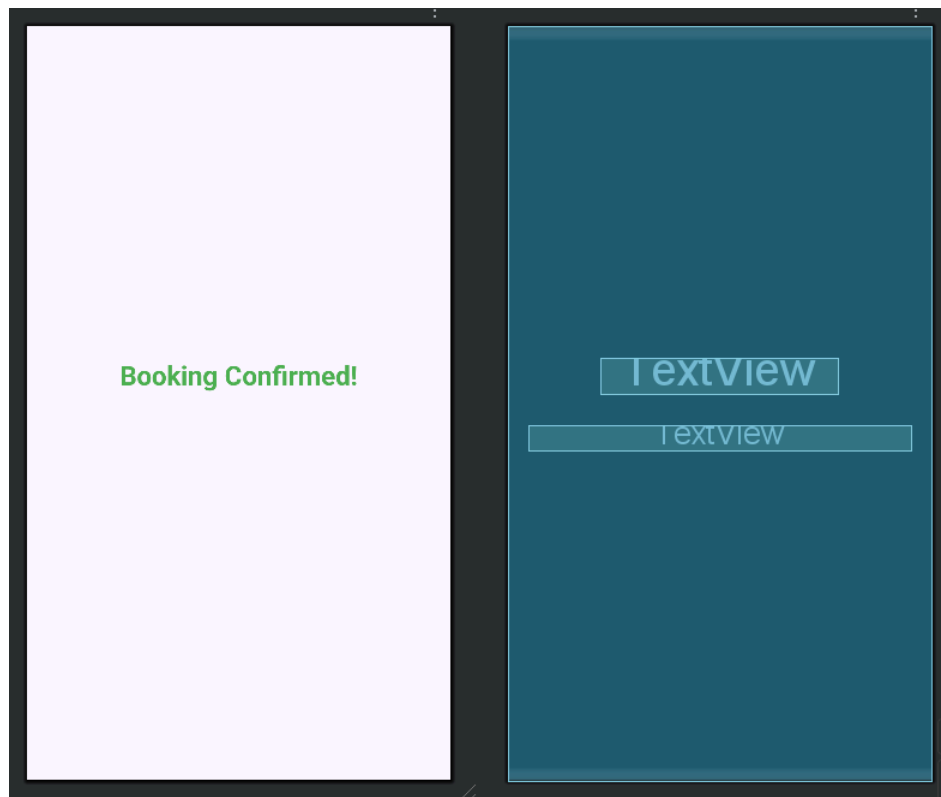


Fig 5.6: activity_display.xml

DisplayActivity.java

```
package com.example.moviebookingapp;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Random;

public class DisplayActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_display);
        TextView tvDetails = findViewById(R.id.tvDetails);
        Bundle extras = getIntent().getExtras();

        if (extras != null) {
            String movie = extras.getString("MOVIE");
            String theatre = extras.getString("THEATRE");
            String date = extras.getString("DATE");
            String time = extras.getString("TIME");
            String type = extras.getString("TYPE");

            // Simulate available seats
            int seats = new Random().nextInt(50) + 1;

            String result = "Movie: " + movie + "\n" +
                "Theatre: " + theatre + "\n" +
                "Date: " + date + "\n" +
                "Time: " + time + "\n" +
                "Type: " + type + "\n\n" +
                "Available Seats: " + seats;

            tvDetails.setText(result);
        }
    }
}
```

Strings.xml

```
<resources>
  <string name="app_name">MovieBookingApp</string>
  <string-array name="movies_array">
    <item>Select Movie</item>
    <item>Inception</item>
    <item>The Dark Knight</item>
    <item>Interstellar</item>
  </string-array>
  <string-array name="theatres_array">
    <item>Select Theatre</item>
    <item>IMAX Central</item>
    <item>PVR Cinemas</item>
    <item>Regal Movies</item>
  </string-array>
</resources>
```

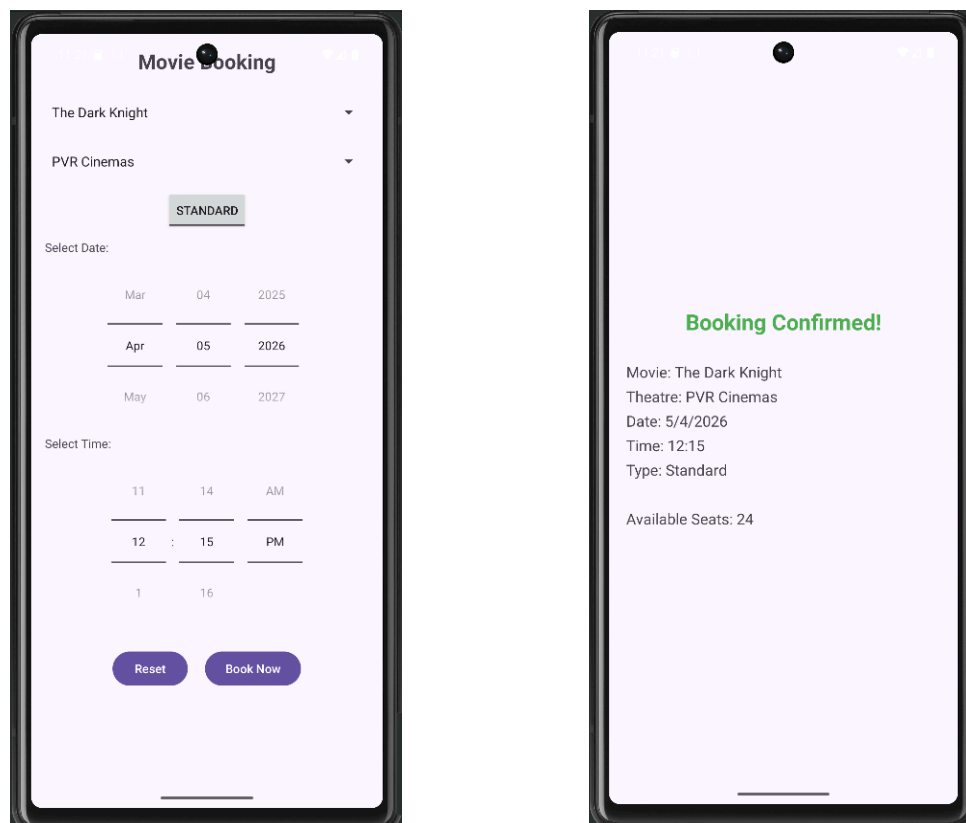


Fig 5.7: Application screenshot

INTRODUCTION TO MENU

PROBLEM STATEMENT 1

Design a home page for the "XYZ Fitness Center" using an Options Menu with the following Requirements.

- a. Create a simple option menus where in once you click the “menu options”, the option items should get displayed which are "Workout Plans," "Trainers," "Membership". "Workout Plans" should display the list of workout programs (e.g., Weight Loss, Cardio). "Trainers" should display the names and specializations of trainers with their photos. "Membership" should show the membership packages with pricing details.
- b. This option menu uses images/icons instead of textual content. This textual content should represent “Contact US”, “About Us”,” Homepage”. Once the user clicks on the corresponding icons it should display corresponding content.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal"
    android:fitsSystemWindows="true"> <TextView
    android:id="@+id/tvTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="XYZ Fitness Center"
    android:textSize="28sp"
```

```

        android:textStyle="bold"
        android:textColor="@android:color/holo_blue_dark"
        android:layout_marginBottom="20dp"/>

        <ImageView
            android:id="@+id/ivContent"
            android:layout_width="200dp"
            android:layout_height="200dp"
            android:src="@drawable/ic_launcher_background"
            android:visibility="gone"
            android:layout_marginBottom="20dp"/>

        <ScrollView
            android:layout_width="match_parent"
            android:layout_height="match_parent">
            <TextView
                android:id="@+id/tvContent"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Welcome! Select an option from the
menu."
                android:textSize="18sp"/>
            </ScrollView>

    </LinearLayout>

```

MainActivity.java

```

package com.example.xyzfitness;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;

```

```

import android.widget.ImageView;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private TextView tvTitle, tvContent;
    private ImageView ivContent;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Init views
        tvTitle = findViewById(R.id.tvTitle);
        tvContent = findViewById(R.id.tvContent);
        ivContent = findViewById(R.id.ivContent);
    }

    // Inflate the menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }

    // Handle clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();
        ivContent.setVisibility(View.GONE); // Hide image by
default

        if (id == R.id.action_home) {
            tvTitle.setText("XYZ Fitness Home");
            tvContent.setText("Welcome to XYZ Fitness
Center.\nYour journey starts here.");
            return true;
        }
    }
}

```

```

    }
    else if (id == R.id.action_about) {
        tvTitle.setText("About Us");
        tvContent.setText("Established in 2024.\nWe are
dedicated to your health.");
        return true;
    }
    else if (id == R.id.action_contact) {
        tvTitle.setText("Contact Us");
        tvContent.setText("Email:
contact@xyzfitness.com\nPhone: +1 555-0199");
        return true;
    }
    else if (id == R.id.opt_workout) {
        tvTitle.setText("Workout Plans");
        tvContent.setText("1. Weight Loss Program\n    -
High Intensity Interval Training\n    - Calorie Deficit
Plan\n\n2. Cardio Blast\n    - Treadmill & Cycling\n    -
Aerobics");
        return true;
    }
    else if (id == R.id.opt_trainers) {
        tvTitle.setText("Our Trainers");
        // Show image for trainers requirement
        ivContent.setVisibility(View.VISIBLE);

        ivContent.setImageResource(android.R.drawable.ic_menu_my_calend
ar); // Using system icon as placeholder photo
        tvContent.setText("Name: John Doe\nSpecialization:
Strength Training\n\nName: Sarah Smith\nSpecialization: Yoga &
Pilates");
        return true;
    }
    else if (id == R.id.opt_membership) {
        tvTitle.setText("Membership Packages");
        tvContent.setText("1. Gold Package - $50/mo\n    -
ALL access\n    - Personal Trainer\n\n2. Silver Package -
$30/mo\n    - Gym equipment only");
        return true;
    }

```

```

    }

    return super.onOptionsItemSelected(item);
}
}

```

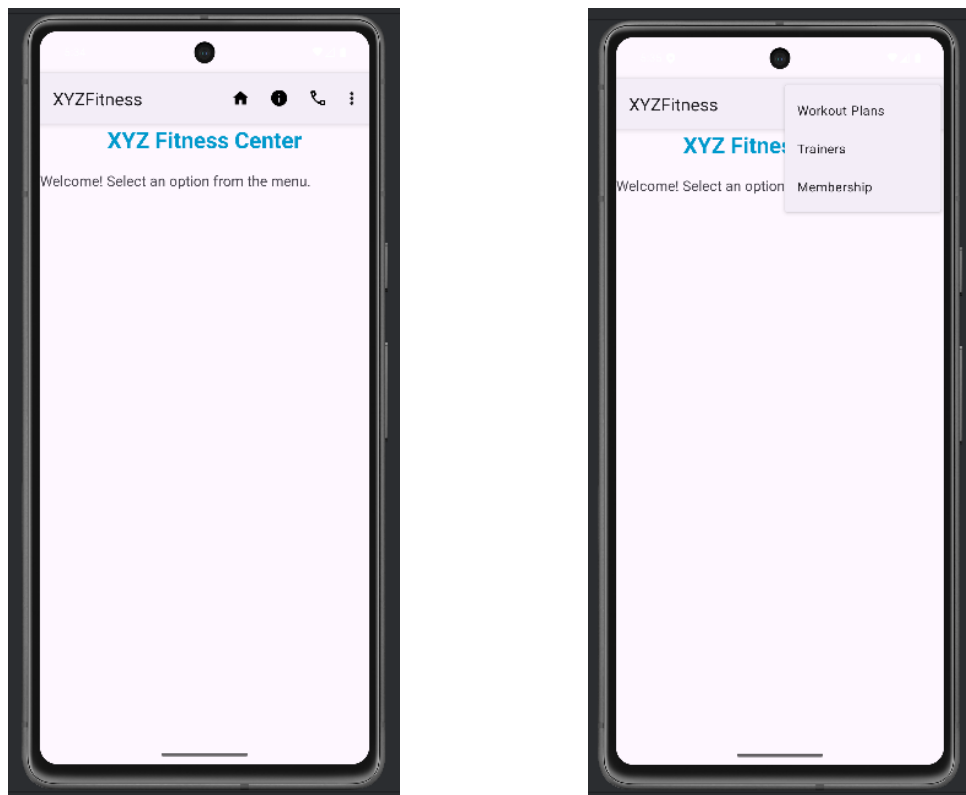


Fig 6.1: Main Page

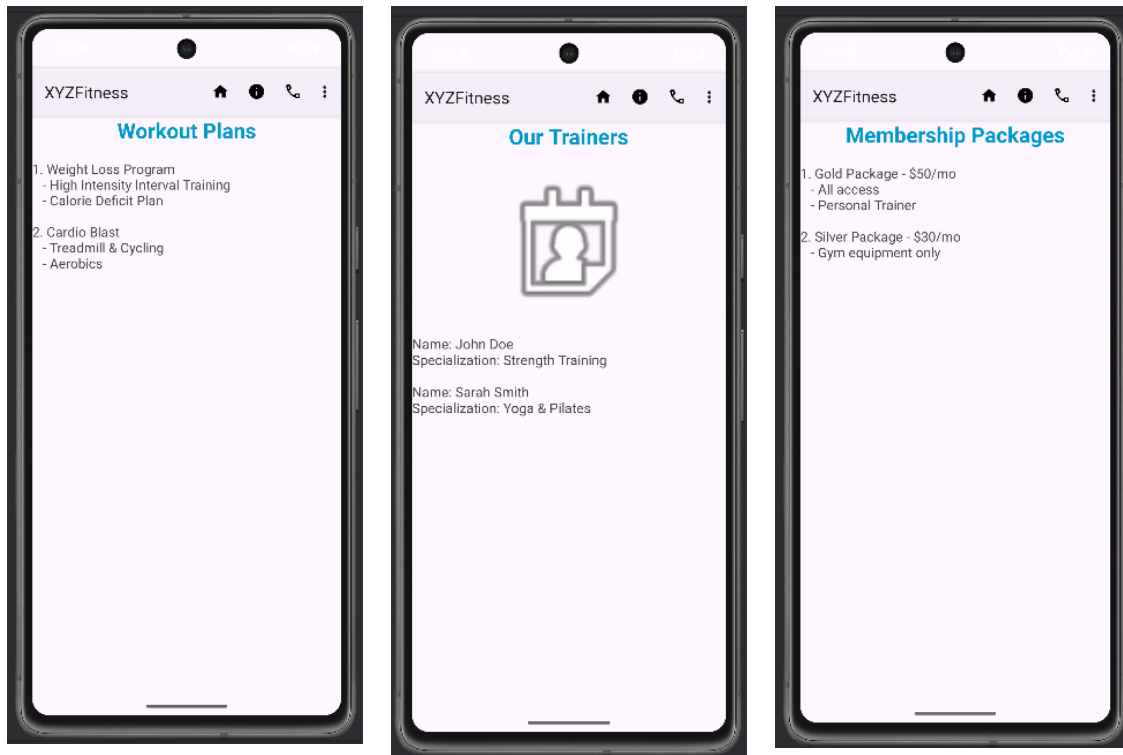


Fig 6.2: Menu options

CREATING CONTEXTUAL AND POP-UP MENUS

PROBLEM STATEMENT 1

Design and develop an application that displays a list of installed applications on your device. On long press of any application, display the following options:

- Show whether the application is a system app or user-installed.
- Provide options to open the app, uninstall it, or view its details (e.g., version, storage usage).
- Indicate whether the app has special permissions enabled, like location or camera access.

When an option like **"View Details"** is selected, navigate to a detailed view showing the app's permissions, size, and version. For the **"Uninstall"** option, prompt the user for confirmation before proceeding.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="8dp">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</RelativeLayout>
```

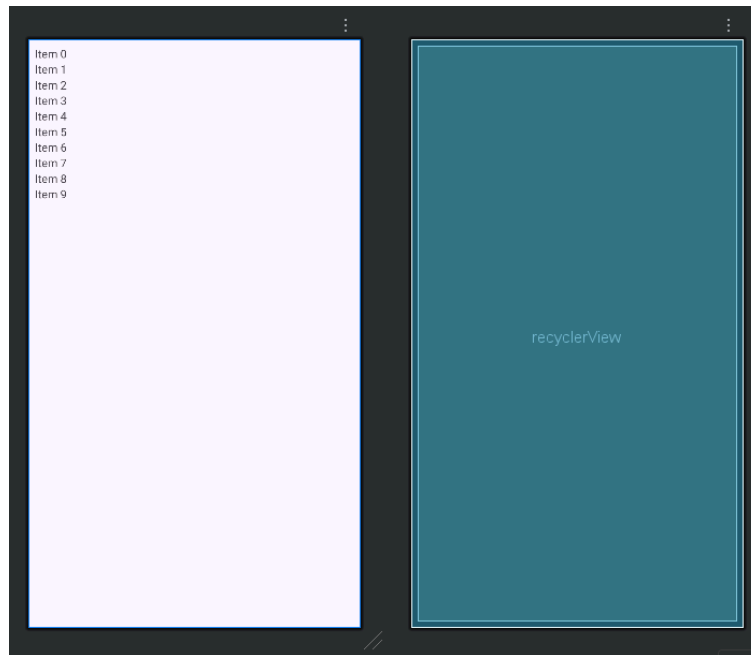


Fig 7.1: Main Activity

MainActivity.java

```
package com.example.appmanager;

import android.content.pm.ApplicationInfo;
import android.content.pm.PackageInfo;
import android.content.pm.PackageManager;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    RecyclerView recyclerView;
    AppAdapter adapter;
    List<AppModel> appList = new ArrayList<>();

    @Override
```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    recyclerView = findViewById(R.id.recyclerView);
    recyclerView.setLayoutManager(new
LinearLayoutManager(this));

    loadInstalledApps();
}

private void loadInstalledApps() {
    PackageManager pm = getPackageManager();
    // Get all packages
    List<PackageInfo> packages =
pm.getInstalledPackages(PackageManager.GET_META_DATA);

    for (PackageInfo p : packages) {
        ApplicationInfo a = p.applicationInfo;

        // Check if System App
        boolean isSystem = (a.flags &
ApplicationInfo.FLAG_SYSTEM) != 0;

        String name = a.loadLabel(pm).toString();
        AppModel app = new AppModel(name, p.packageName,
a.loadIcon(pm), isSystem);

        appList.add(app);
    }

    adapter = new AppAdapter(this, appList);
    recyclerView.setAdapter(adapter);
}

// Refresh list if an app was uninstalled
@Override
protected void onResume() {
    super.onResume();
}

```

```

        appList.clear();
        loadInstalledApps();
    }
}

```

activity_detail.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <ImageView android:id="@+id/detailIcon"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_gravity="center"
        android:layout_marginBottom="16dp"/>

    <TextView android:id="@+id/detailName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="24sp" android:textStyle="bold"
        android:gravity="center"/>

    <TextView android:id="@+id/detailVersion"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:textSize="16sp"/>

    <TextView android:id="@+id/detailSize"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:textSize="16sp"/>

```

```

<TextView android:layout_width="match_parent"
android:layout_height="wrap_content"
    android:text="Permissions:" android:textStyle="bold"
    android:layout_marginTop="24dp"
android:textSize="18sp"/>

<ScrollView android:layout_width="match_parent"
android:layout_height="match_parent">
    <TextView android:id="@+id/detailPermissions"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="8dp"/>
</ScrollView>
</LinearLayout>

```

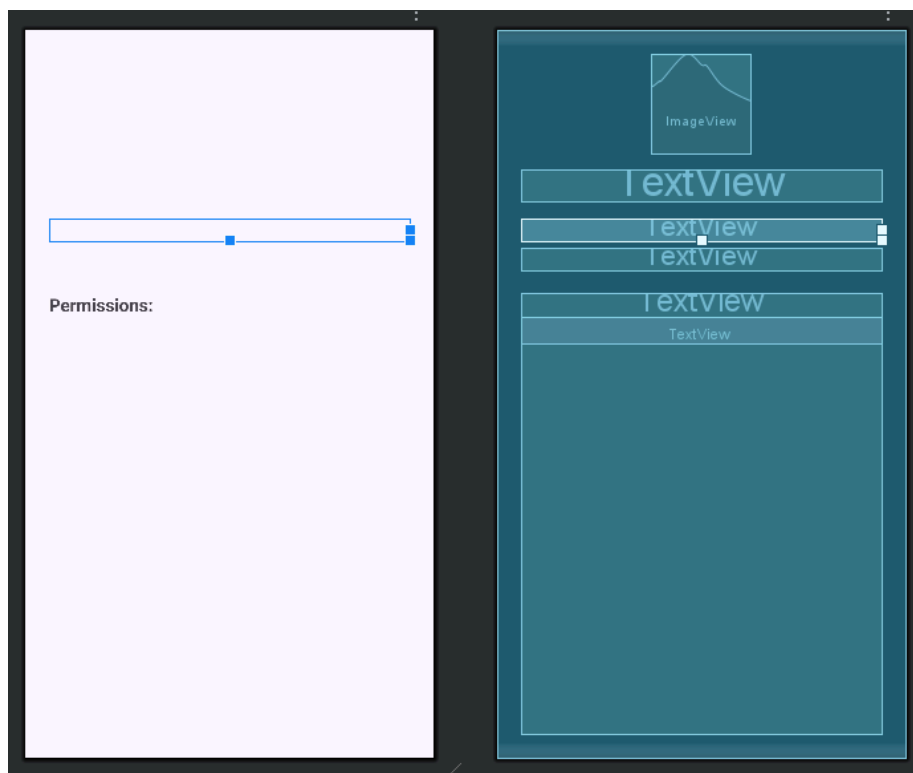


Fig 7.2: Details Activity

Detailactivity.java

```
package com.example.appmanager;

import android.content.pm.ApplicationInfo;
import android.content.pm.PackageInfo;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.widget.ImageView;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.io.File;
import java.util.Arrays;

public class DetailActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_detail);

        String pkg = getIntent().getStringExtra("pkg");
        PackageManager pm = getPackageManager();

        try {
            // Get full package info including permissions
            PackageInfo pInfo = pm.getPackageInfo(pkg,
PackageManager.GET_PERMISSIONS);
            ApplicationInfo appInfo = pInfo.applicationInfo;

            ImageView icon = findViewById(R.id.detailIcon);
            TextView name = findViewById(R.id.detailName);
            TextView version =
findViewById(R.id.detailVersion);
            TextView size = findViewById(R.id.detailSize);
            TextView perms =
findViewById(R.id.detailPermissions);

            icon.setImageDrawable(appInfo.loadIcon(pm));
```

```

name.setText(appInfo.loadLabel(pm));
version.setText("Version: " + pInfo.versionName);

// Calculate APK size
File file = new File(appInfo.publicSourceDir);
Long sizeInMb = file.length() / (1024 * 1024);
size.setText("App Size: ~" + sizeInMb + " MB");

// List Permissions & Check for Special ones
StringBuilder sb = new StringBuilder();
boolean hasCamera = false, hasLoc = false;

if (pInfo.requestedPermissions != null) {
    for (String p : pInfo.requestedPermissions) {
        sb.append(p).append("\n");
        if (p.contains("CAMERA")) hasCamera = true;
        if (p.contains("LOCATION")) hasLoc = true;
    }
} else {
    sb.append("No permissions requested.");
}

perms.setText(sb.toString());

// Optional: Show Toast for special permissions
detected
if (hasCamera || hasLoc) {
    String msg = "Special Access: " + (hasCamera ?
"Camera " : "") + (hasLoc ? "Location" : "");
    android.widget.Toast.makeText(this, msg,
android.widget.Toast.LENGTH_LONG).show();
}

} catch (PackageManager.NameNotFoundException e) {
    e.printStackTrace();
}
}
}

```

Item_app.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="16dp"
    android:gravity="center_vertical">

    <ImageView
        android:id="@+id/appIcon"
        android:layout_width="48dp"
        android:layout_height="48dp" />

    <TextView
        android:id="@+id/appName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:textSize="18sp"
        android:textStyle="bold" />
</LinearLayout>
```

AppModel.java

```
package com.example.appmanager;

import android.graphics.drawable.Drawable;

public class AppModel {
    String name;
    String packageName;
    Drawable icon;
    boolean isSystem;

    public AppModel(String name, String packageName, Drawable
    icon, boolean isSystem) {
```



```

        this.name = name;
        this.packageName = packageName;
        this.icon = icon;
        this.isSystem = isSystem;
    }
}

```

AppAdapter.java

```

package com.example.appmanager;

import android.app.AlertDialog;
import android.content.Context;
import android.content.Intent;
import android.net.Uri;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;
import java.util.List;

public class AppAdapter extends
RecyclerView.Adapter<AppAdapter.ViewHolder> {

    Context context;
    List<AppModel> appList;

    public AppAdapter(Context context, List<AppModel> appList)
    {
        this.context = context;
        this.appList = appList;
    }

    @NonNull
    @Override

```

```

    public ViewHolder onCreateViewHolder(@NonNull ViewGroup
parent, int viewType) {
        View view =
LayoutInflater.from(context).inflate(R.layout.item_app, parent,
false);
        return new ViewHolder(view);
    }

    @Override
    public void onBindViewHolder(@NonNull ViewHolder holder,
int position) {
        AppModel app = appList.get(position);
        holder.name.setText(app.name);
        holder.icon.setImageDrawable(app.icon);

        // Long Press Logic
        holder.itemView.setOnLongClickListener(v -> {
            showOptionsDialog(app);
            return true;
        });
    }

    @Override
    public int getItemCount() { return appList.size(); }

    private void showOptionsDialog(AppModel app) {
        String type = app.isSystem ? "System App" : "User
Installed";
        // Check for special perms flag in MainActivity logic,
or just navigate to details to see them.
        // For menu simplicity, we list actions.

        String[] options = {"Open App", "Uninstall", "View
Details"};

        new AlertDialog.Builder(context)
            .setTitle(app.name + " (" + type + ")")
            .setItems(options, (dialog, which) -> {
                if (which == 0) openApp(app.packageName);
            });
    }

```

```

        if (which == 1)
confirmUninstall(app.packageName);
        if (which == 2)
openDetails(app.packageName);
    })
    .show();
}

private void openApp(String packageName) {
    Intent launchIntent =
context.getPackageManager().getLaunchIntentForPackage(packageNa
me);
    if (launchIntent != null)
context.startActivity(launchIntent);
    else Toast.makeText(context, "Cannot open this app",
Toast.LENGTH_SHORT).show();
}

private void confirmUninstall(String packageName) {
    new AlertDialog.Builder(context)
        .setTitle("Confirm Uninstall")
        .setMessage("Are you sure you want to uninstall
this app?")
        .setPositiveButton("Yes", (d, w) -> {
            Intent intent = new
Intent(Intent.ACTION_DELETE);
            intent.setData(Uri.parse("package:" +
packageName));
            context.startActivity(intent);
        })
        .setNegativeButton("No", null)
        .show();
}

private void openDetails(String packageName) {
    Intent intent = new Intent(context,
DetailActivity.class);
    intent.putExtra("pkg", packageName);
    context.startActivity(intent);
}

```

```

    }

    public static class ViewHolder extends
RecyclerView.ViewHolder {
        ImageView icon;
        TextView name;
        public ViewHolder(@NonNull View itemView) {
            super(itemView);
            icon = itemView.findViewById(R.id.appIcon);
            name = itemView.findViewById(R.id.appName);
        }
    }
}

```

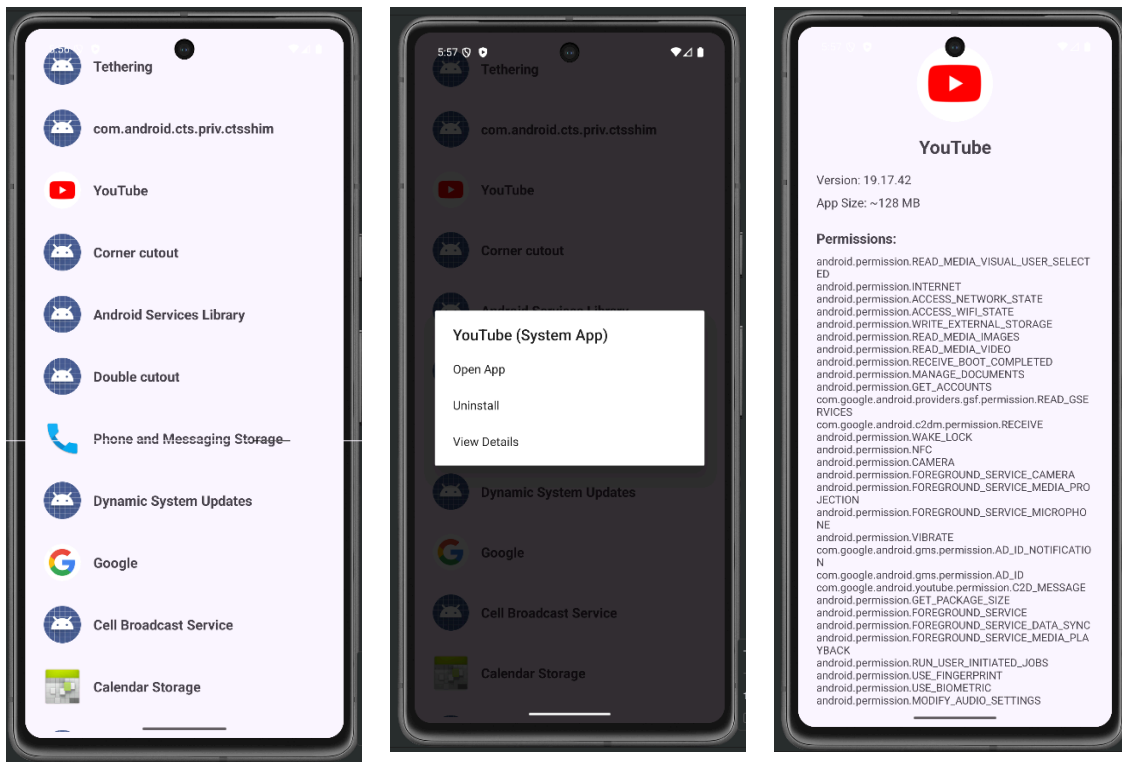


Fig 7.3: Application screenshots

PROBLEM STATEMENT 2

Create a View with the name “My Menu ” which contains an image as an icon. On click of that icon it shows a submenu as ”Image -1”,”Image -2” .On click of each item that particular image should be displayed with the corresponding content in the Toast.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <ImageView
        android:id="@+id/displayImageView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:src="@drawable/image1"
        android:contentDescription="Main Image" />

</RelativeLayout>
```

my_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">

    <item
        android:id="@+id/menu_master"
        android:title="My Menu"
        android:icon="@drawable/my_icon"
        app:showAsAction="always">

        <menu>
```

```

        <item
            android:id="@+id/action_image1"
            android:title="Image -1" />
        <item
            android:id="@+id/action_image2"
            android:title="Image -2" />
    </menu>
</item>
</menu>

```

MainActivity.java

```

package com.example.menuassignment; // KEEP YOUR PACKAGE NAME
HERE

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.ImageView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    // Declare the variable
    ImageView imageView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // 1. Load the layout FIRST. If this is missing, app
        crashes.
        setContentView(R.layout.activity_main);

        // 2. Find the View by ID.
        imageView = findViewById(R.id.displayImageView);
    }
}

```

```

// Load the menu
@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.my_menu, menu);
    return true;
}

// Handle menu clicks
@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.action_image1) {
        imageView.setImageResource(R.drawable.image1);
        Toast.makeText(this, "Showing Image 1",
Toast.LENGTH_SHORT).show();
        return true;
    }
    else if (id == R.id.action_image2) {
        imageView.setImageResource(R.drawable.image2);
        Toast.makeText(this, "Showing Image 2",
Toast.LENGTH_SHORT).show();
        return true;
    }

    return super.onOptionsItemSelected(item);
}
}

```

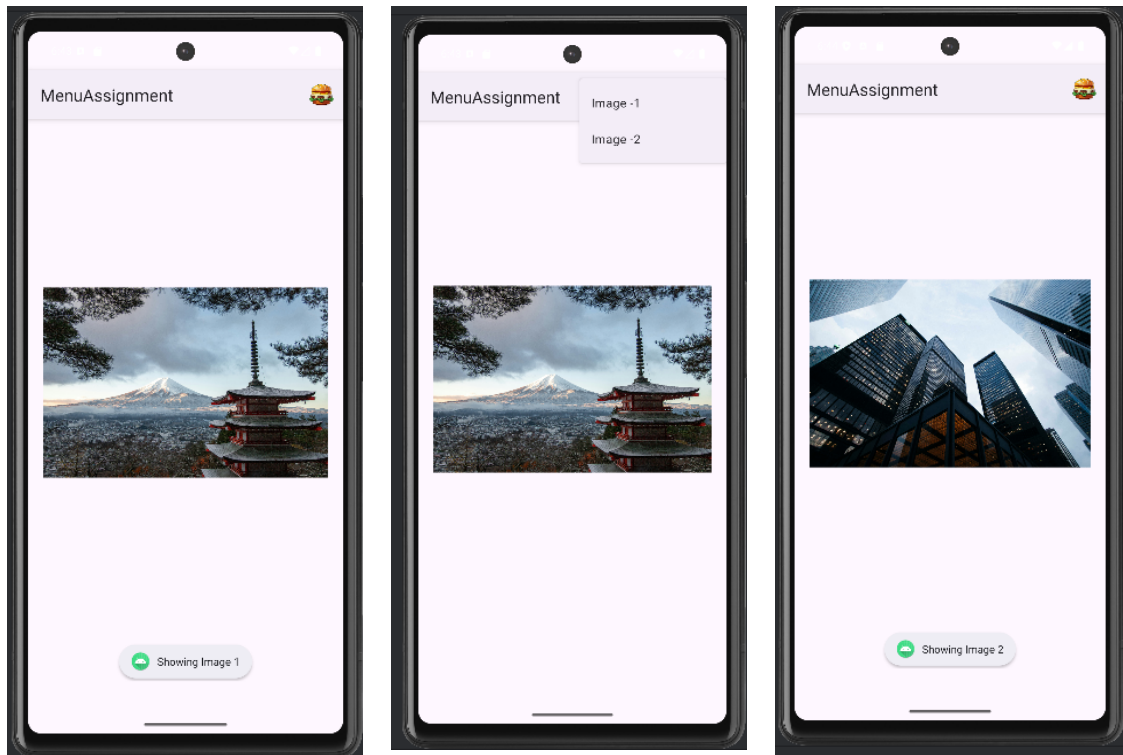


Fig 7.4: Application Screenshots

PROBLEM STATEMENT 3

Create a view that displays textual content providing a description of "Digital Transformation". Add a filter option with submenus:

- a. "Search Keywords" - Allow the user to input keywords that can be searched within the content.
- b. "Highlight" - Highlight specific words or phrases provided by the user in the document.
- c. "Sort" - Provide an option to sort the entire content either alphabetically or by relevance to the search keywords.

SOLUTION

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:fitsSystemWindows="true"> <TextView
    android:id="@+id/tvContent"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/article_content"
    android:textSize="18sp"
    android:lineSpacingExtra="4dp" />

</ScrollView>
```

main_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">

    <item android:title="Filter" app:showAsAction="ifRoom">
```

```

        <menu>
            <item android:id="@+id/action_search"
android:title="Search Keywords" />
            <item android:id="@+id/action_highlight"
android:title="Highlight" />
            <item android:title="Sort">
                <menu>
                    <item android:id="@+id/sort_alpha"
android:title="Alphabetically" />
                    <item android:id="@+id/sort_relevance"
android:title="By Relevance (to last search)" />
                </menu>
            </item>
        </menu>
    </item>
</menu>

```

ActivityMain.xml

```

package com.example.digitaltransformapp;

import android.graphics.Color;
import android.os.Bundle;
import android.text.Spannable;
import android.text.SpannableString;
import android.text.style.BackgroundColorSpan;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.Collections;

```

```

import java.util.Comparator;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    private TextView tvContent;
    private String fullText;
    private String currentKeyword = ""; // Stores last search
for relevance sorting

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvContent = findViewById(R.id.tvContent);
        fullText = tvContent.getText().toString(); // Get
initial text from XML
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.action_search) {
            showInputDialog("Search Keyword", input -> {
                currentKeyword = input;
                int count = countOccurrences(fullText, input);
                Toast.makeText(this, "Found " + count + "
matches.", Toast.LENGTH_SHORT).show();
            });
            return true;
        } else if (id == R.id.action_highlight) {

```

```

        showInputDialog("Highlight Word",
this::highlightText);
        return true;
    } else if (id == R.id.sort_alpha) {
        sortContent(true);
        return true;
    } else if (id == R.id.sort_relevance) {
        sortContent(false);
        return true;
    }

    return super.onOptionsItemSelected(item);
}

// Generic dialog to get text input
private void showInputDialog(String title, InputCallback
callback) {
    EditText input = new EditText(this);
    new AlertDialog.Builder(this)
        .setTitle(title)
        .setView(input)
        .setPositiveButton("OK", (dialog, which) ->
callback.onInput(input.getText().toString()))
        .setNegativeButton("Cancel", null)
        .show();
}

private void highlightText(String word) {
    if (word.isEmpty()) return;
    SpannableString spannable = new
SpannableString(tvContent.getText());
    String text = spannable.toString().toLowerCase();
    String target = word.toLowerCase();
    int start = 0;

    // Clear previous highlights by resetting text
    tvContent.setText(fullText);
    spannable = new SpannableString(fullText);

```

```

        while ((start = text.indexOf(target, start)) >= 0) {
            spannable.setSpan(new
BackgroundColorSpan(Color.YELLOW), start, start +
target.length(), Spannable.SPAN_EXCLUSIVE_EXCLUSIVE);
            start += target.length();
        }
        tvContent.setText(spannable);
    }

    private void sortContent(boolean alphabetical) {
        // Split by double newline to preserve paragraphs
        List<String> paragraphs = new
ArrayList<>(Arrays.asList(fullText.split("\n\n")));

        if (alphabetical) {
            Collections.sort(paragraphs);
        } else {
            // Sort by occurrence count of the last searched
keyword
            if (currentKeyword.isEmpty()) {
                Toast.makeText(this, "Search for a keyword
first!", Toast.LENGTH_SHORT).show();
                return;
            }
            Collections.sort(paragraphs, (p1, p2) -> {
                int count1 = countOccurrences(p1,
currentKeyword);
                int count2 = countOccurrences(p2,
currentKeyword);
                return Integer.compare(count2, count1); //
Descending order
            });
        }

        // Rejoin and display
        StringBuilder builder = new StringBuilder();
        for (String p : paragraphs)
builder.append(p).append("\n\n");
        tvContent.setText(builder.toString().trim());
    }

```

```

    }

    private int countOccurrences(String text, String keyword) {
        if (keyword.isEmpty()) return 0;
        return (text.toLowerCase().length() -
text.toLowerCase().replace(keyword.toLowerCase(), "").length())
/ keyword.length();
    }

    interface InputCallback {
        void onInput(String text);
    }
}

```

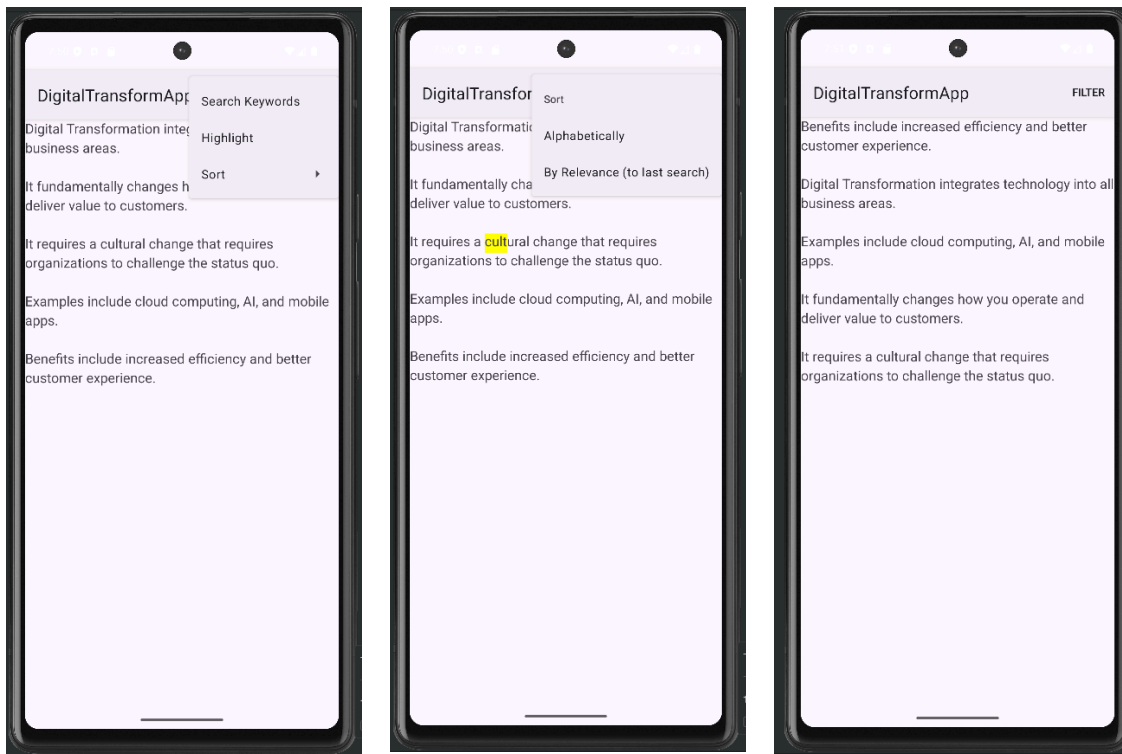


Fig 7.5: Application Screenshots

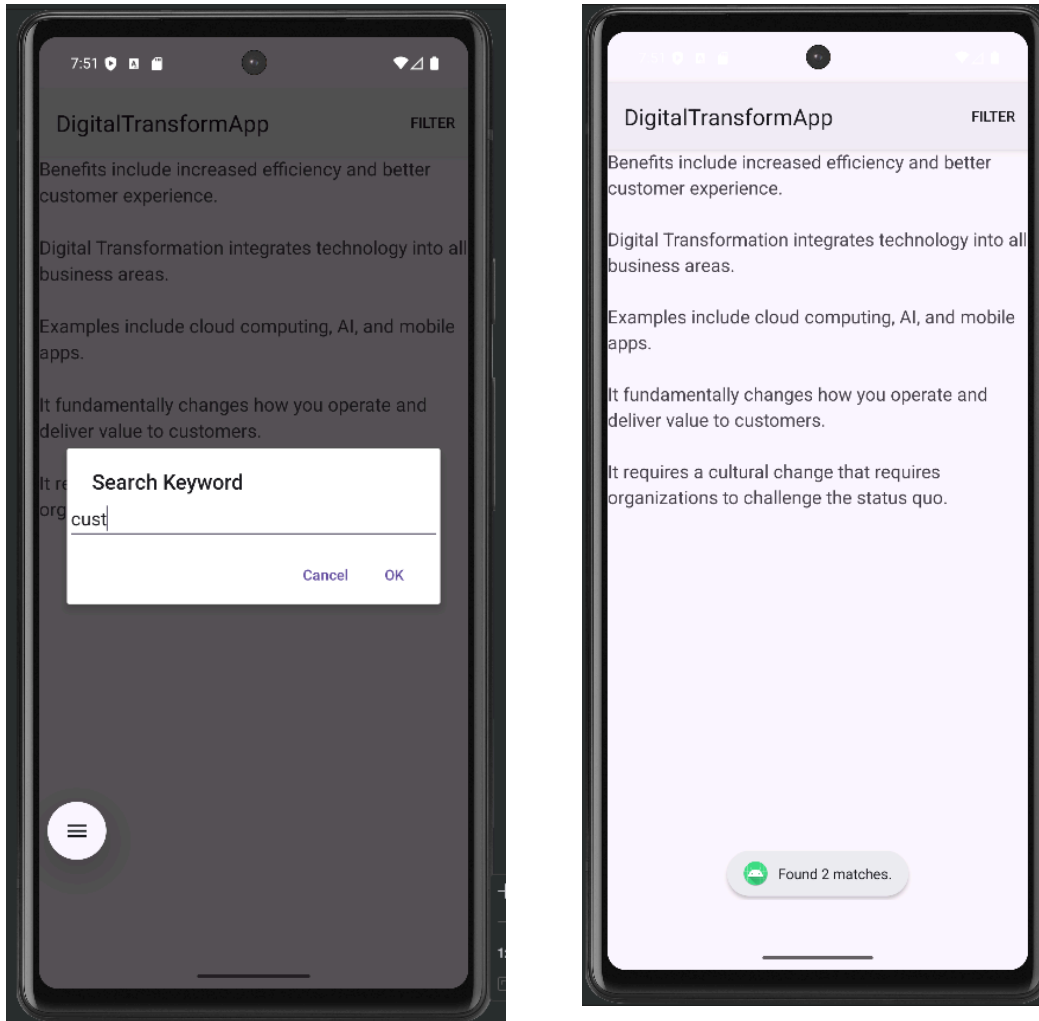


Fig 7.6: Application Screenshots

LAB 08

ANDROID SQLITE AND SHARED PREFERENCES

PROBLEM STATEMENT 1

Write a program to create a "Task Manager" application with the following features:

- a) A form to create and save tasks with fields like task name, due date, and priority level (High, Medium, Low).
- b) A list view to display all the saved tasks with their details.
- c) An option to edit or delete any task from the list.

SOLUTION

Activity_main.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/taskName"
        android:hint="Task Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/dueDate"
        android:hint="Due Date"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Spinner
        android:id="@+id/priority"
        android:layout_width="match_parent"
```



```

        android:layout_height="wrap_content"/>

<Button
    android:id="@+id/saveTask"
    android:text="Save Task"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<ListView
    android:id="@+id/taskList"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

</LinearLayout>

```

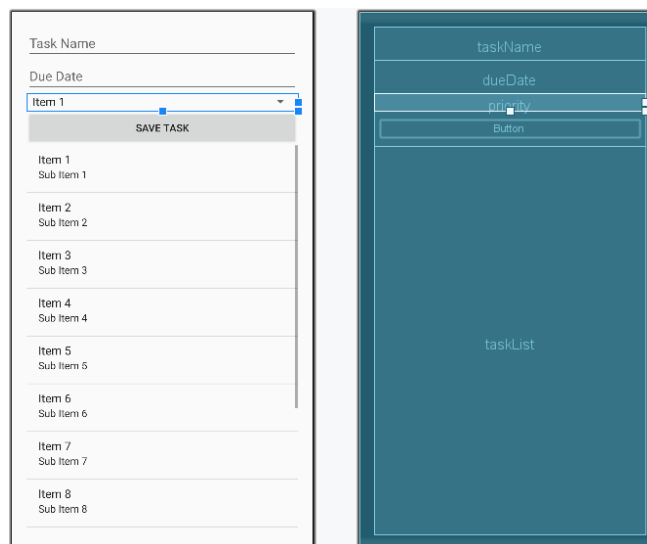


Fig 8.1.1: Activity LAYOUT

TaskDatabaseHelper.java

```

package com.example.lab8_q1;

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class TaskDatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME =
"TaskManager.db";
    private static final int DATABASE_VERSION = 1;

```

```

public static final String TABLE_NAME = "Tasks";

public static final String COL_ID = "id";
public static final String COL_NAME = "task_name";
public static final String COL_DATE = "due_date";
public static final String COL_PRIORITY = "priority";

public TaskDatabaseHelper(Context context) {
    super(context, DATABASE_NAME, null, DATABASE_VERSION);
}

@Override
public void onCreate(SQLiteDatabase db) {

    String CREATE_TABLE = "CREATE TABLE " + TABLE_NAME + "("
        + COL_ID + " INTEGER PRIMARY KEY AUTOINCREMENT,"
        + COL_NAME + " TEXT,"
        + COL_DATE + " TEXT,"
        + COL_PRIORITY + " TEXT)";

    db.execSQL(CREATE_TABLE);
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {

    db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
    onCreate(db);
}
}

```

MainActivity.java

```

package com.example.lab8_q1;

import android.content.ContentValues;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;

```

```

public class MainActivity extends AppCompatActivity {

    EditText taskName, dueDate;
    Spinner priority;
    Button saveTask;
    ListView taskList;

    TaskDatabaseHelper dbHelper;
    SQLiteDatabase db;

    ArrayList<String> tasks;
    ArrayAdapter<String> adapter;

    int selectedId = -1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        taskName = findViewById(R.id.taskName);
        dueDate = findViewById(R.id.dueDate);
        priority = findViewById(R.id.priority);
        saveTask = findViewById(R.id.saveTask);
        taskList = findViewById(R.id.taskList);

        dbHelper = new TaskDatabaseHelper(this);
        db = dbHelper.getWritableDatabase();

        String[] priorities = {"High", "Medium", "Low"};

        ArrayAdapter<String> spinnerAdapter =
            new ArrayAdapter<>(this,
                android.R.layout.simple_spinner_item, priorities);

        priority.setAdapter(spinnerAdapter);

        loadTasks();

        saveTask.setOnClickListener(v -> {

            String name = taskName.getText().toString();
            String date = dueDate.getText().toString();
            String pr = priority.getSelectedItem().toString();

            ContentValues values = new ContentValues();
            values.put("task_name", name);
            values.put("due_date", date);

```

```

        values.put("priority", pr);

        if(selectedId == -1)
        {
            db.insert("Tasks", null, values);
        }
        else
        {
            db.update("Tasks", values, "id=?",
                new
String[]{String.valueOf(selectedId)});
            selectedId = -1;
        }

        taskName.setText("");
        dueDate.setText("");

        loadTasks();
    });

    taskList.setOnItemClickListener((parent, view, position,
id) -> {

        Cursor cursor = db.rawQuery("SELECT * FROM Tasks",
null);

        cursor.moveToPosition(position);

        selectedId = cursor.getInt(0);
        taskName.setText(cursor.getString(1));
        dueDate.setText(cursor.getString(2));

        cursor.close();
    });

    taskList.setOnItemLongClickListener((parent, view,
position, id) -> {

        Cursor cursor = db.rawQuery("SELECT * FROM Tasks",
null);

        cursor.moveToPosition(position);

        int deleteId = cursor.getInt(0);

        db.delete("Tasks", "id=?",
            new String[]{String.valueOf(deleteId)});

        cursor.close();
    });

```

```

        loadTasks();

        return true;
    });
}

private void loadTasks()
{
    tasks = new ArrayList<>();

    Cursor cursor = db.rawQuery("SELECT * FROM Tasks", null);

    while(cursor.moveToNext())
    {
        String task = "Task: " + cursor.getString(1)
            + "\nDue: " + cursor.getString(2)
            + "\nPriority: " + cursor.getString(3);

        tasks.add(task);
    }

    cursor.close();

    adapter = new ArrayAdapter<>(this,
        android.R.layout.simple_list_item_1, tasks);

    taskList.setAdapter(adapter);
}
}

```

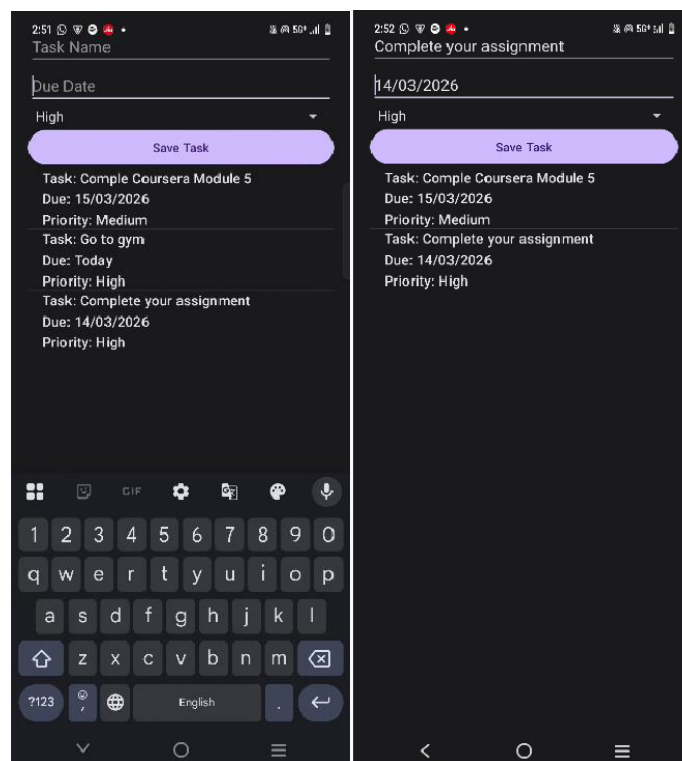


Fig 8.1.2: Application Screenshots

PROBLEM STATEMENT 2

Write a program to add grocery items along with its cost into the database and display the total cost of all the item selected by the user. Items has to be displayed through spinner.

SOLUTION

Activity_main.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Spinner
        android:id="@+id/spinnerItems"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnAdd"
        android:text="Add Item"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/totalCost"
        android:text="Total Cost: 0"
        android:textSize="20sp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

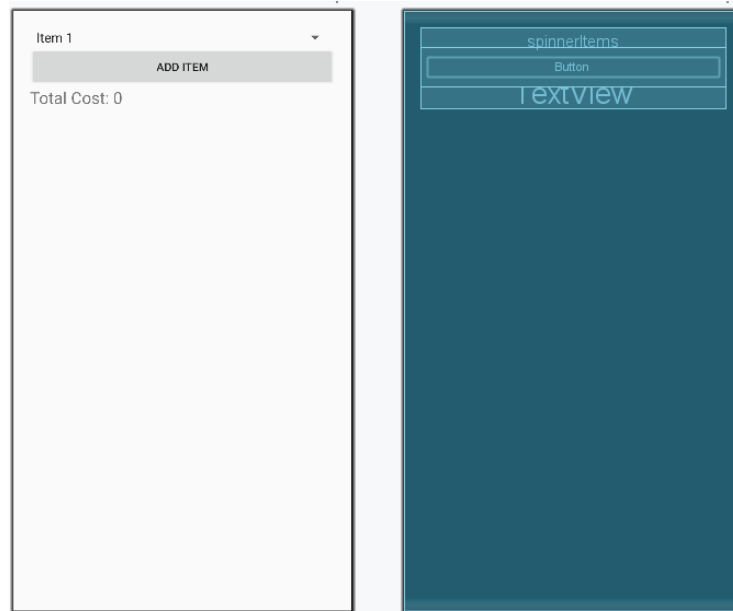


Fig 8.2.1: Main Layout

DBHelper.java

```
package com.example.lab8_q2;

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "GroceryDB";
    public static final int DB_VERSION = 1;

    public DBHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {

        db.execSQL("CREATE TABLE Grocery(id INTEGER PRIMARY KEY
        AUTOINCREMENT, item TEXT, cost INTEGER)");

        db.execSQL("INSERT INTO Grocery(item,cost)
        VALUES('Rice',50)");
        db.execSQL("INSERT INTO Grocery(item,cost)
```

```

VALUES('Milk',30)");
        db.execSQL("INSERT INTO Grocery(item,cost)
VALUES('Bread',25)");
        db.execSQL("INSERT INTO Grocery(item,cost)
VALUES('Eggs',60)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {

        db.execSQL("DROP TABLE IF EXISTS Grocery");
        onCreate(db);
    }
}

```

MainActivity.java

```

package com.example.lab8_q2;

import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    Spinner spinnerItems;
    Button btnAdd;
    TextView totalCost;

    DBHelper dbHelper;
    SQLiteDatabase db;

    ArrayList<String> items = new ArrayList<>();
    ArrayList<Integer> costs = new ArrayList<>();

    int total = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```



```

        spinnerItems = findViewById(R.id.spinnerItems);
        btnAdd = findViewById(R.id.btnAdd);
        totalCost = findViewById(R.id.totalCost);

        dbHelper = new DBHelper(this);
        db = dbHelper.getWritableDatabase();

        loadItems();

        btnAdd.setOnClickListener(v -> {

            int position =
spinnerItems.getSelectedItemPosition();

            int cost = costs.get(position);

            total += cost;

            totalCost.setText("Total Cost: " + total);
        });
    }

    private void loadItems() {

        Cursor cursor = db.rawQuery("SELECT * FROM Grocery",
null);

        items.clear();
        costs.clear();

        while(cursor.moveToNext()) {

            items.add(cursor.getString(1));
            costs.add(cursor.getInt(2));
        }

        cursor.close();

        ArrayAdapter<String> adapter =
            new ArrayAdapter<>(this,
android.R.layout.simple_spinner_dropdown_item,
            items);

        spinnerItems.setAdapter(adapter);
    }
}

```

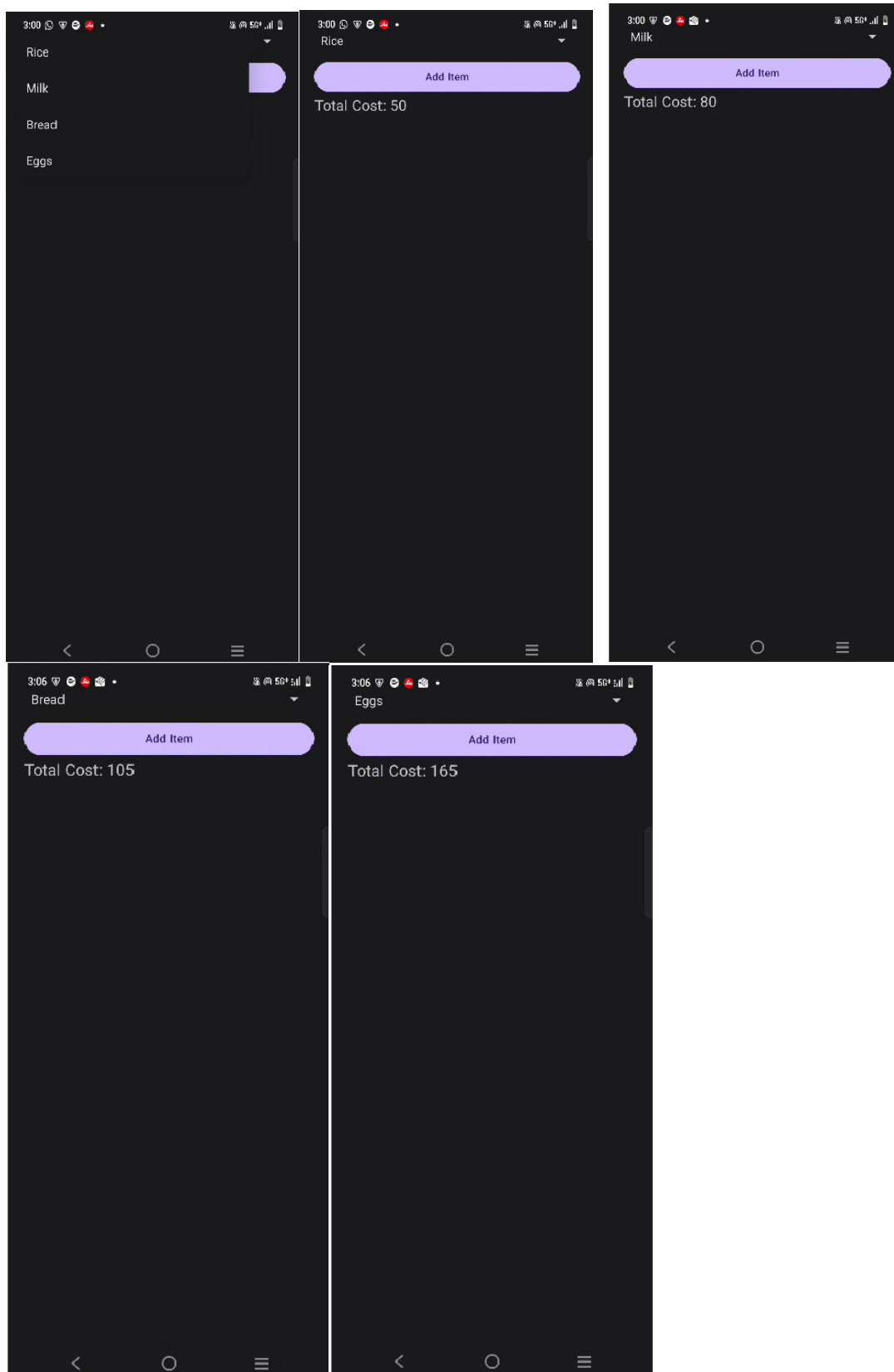


Fig 8.2.2: Application Screenshots

PROBLEM STATEMENT 3

Create an application “Movie Review” to create a movie review to do the following: a) User can write a movie review by stating movie name, year, giving points ranging from 1-5 and save details in a database. b) User can view the movie review for which review are already defined .Movie names has to be displayed using

SOLUTION

Activity_main.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/movieName"
        android:hint="Movie Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/movieYear"
        android:hint="Year"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Spinner
        android:id="@+id/ratingSpinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/saveBtn"
        android:text="Save Review"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Spinner
        android:id="@+id/movieSpinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
```

```

        android:id="@+id/viewBtn"
        android:text="View Review"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

<TextView
    android:id="@+id/result"
    android:textSize="18sp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

</LinearLayout>

```

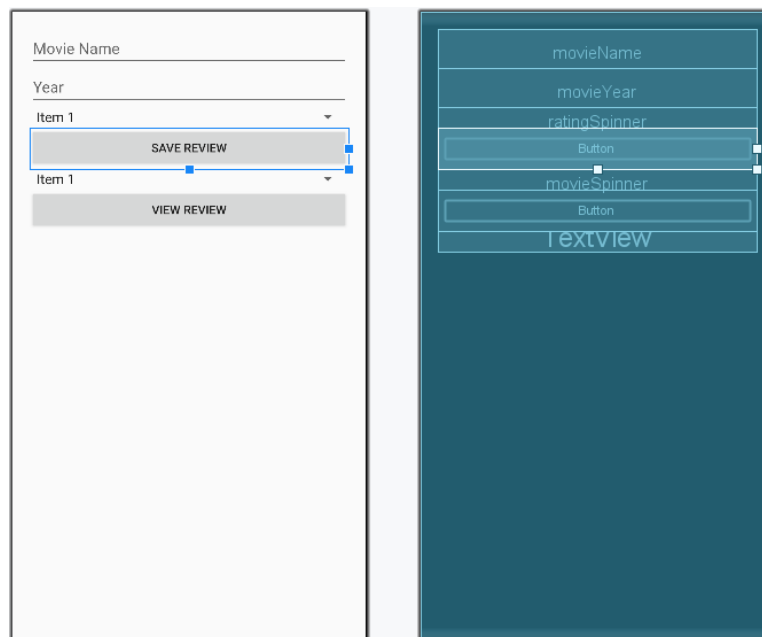


Fig 8.3.1: Main Layout

DBHelper.java

```

package com.example.lab8_q3;

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "MovieDB";
    public static final int DB_VERSION = 1;

    public DBHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }
}

```

```

@Override
public void onCreate(SQLiteDatabase db) {

    db.execSQL("CREATE TABLE Movies(" +
        "id INTEGER PRIMARY KEY AUTOINCREMENT," +
        "name TEXT," +
        "year TEXT," +
        "rating INTEGER)");
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int
newVersion) {

    db.execSQL("DROP TABLE IF EXISTS Movies");
    onCreate(db);
}
}

```

MainActivity.java

```

package      com.example.lab8_q3;

import android.content.ContentValues;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    EditText movieName, movieYear;
    Spinner ratingSpinner, movieSpinner;
    Button saveBtn, viewBtn;
    TextView result;

    DBHelper dbHelper;
    SQLiteDatabase db;

    ArrayList<String> movieList = new ArrayList<>();

    @Override
    protected void onCreate(Bundle savedInstanceState) {

```

```

super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

movieName = findViewById(R.id.movieName);
movieYear = findViewById(R.id.movieYear);
ratingSpinner = findViewById(R.id.ratingSpinner);
movieSpinner = findViewById(R.id.movieSpinner);
saveBtn = findViewById(R.id.saveBtn);
viewBtn = findViewById(R.id.viewBtn);
result = findViewById(R.id.result);

dbHelper = new DBHelper(this);
db = dbHelper.getWritableDatabase();

String[] ratings = {"1", "2", "3", "4", "5"};

ArrayAdapter<String> ratingAdapter =
    new ArrayAdapter<>(this,
android.R.layout.simple_spinner_dropdown_item,
    ratings);

ratingSpinner.setAdapter(ratingAdapter);

loadMovies();

saveBtn.setOnClickListener(v -> {

    String name = movieName.getText().toString();
    String year = movieYear.getText().toString();
    int rating =
Integer.parseInt(ratingSpinner.getSelectedItem().toString());

    ContentValues values = new ContentValues();
    values.put("name", name);
    values.put("year", year);
    values.put("rating", rating);

    db.insert("Movies", null, values);

    Toast.makeText(this, "Review
Saved", Toast.LENGTH_SHORT).show();

    loadMovies();
});

viewBtn.setOnClickListener(v -> {

```

```

        String selectedMovie =
movieSpinner.getSelectedItemAt().toString();

        Cursor cursor = db.rawQuery(
            "SELECT * FROM Movies WHERE name=?",
            new String[]{selectedMovie});

        if(cursor.moveToFirst()){

            String year = cursor.getString(2);
            int rating = cursor.getInt(3);

            result.setText("Movie: "+selectedMovie +
                "\nYear: "+year +
                "\nRating: "+rating+"/5");
        }

        cursor.close();
    });
}

private void loadMovies(){

    movieList.clear();

    Cursor cursor = db.rawQuery("SELECT name FROM
Movies",null);

    while(cursor.moveToNext()){
        movieList.add(cursor.getString(0));
    }

    cursor.close();

    ArrayAdapter<String> adapter =
        new ArrayAdapter<>(this,
android.R.layout.simple_spinner_dropdown_item,
        movieList);

    movieSpinner.setAdapter(adapter);
}
}

```

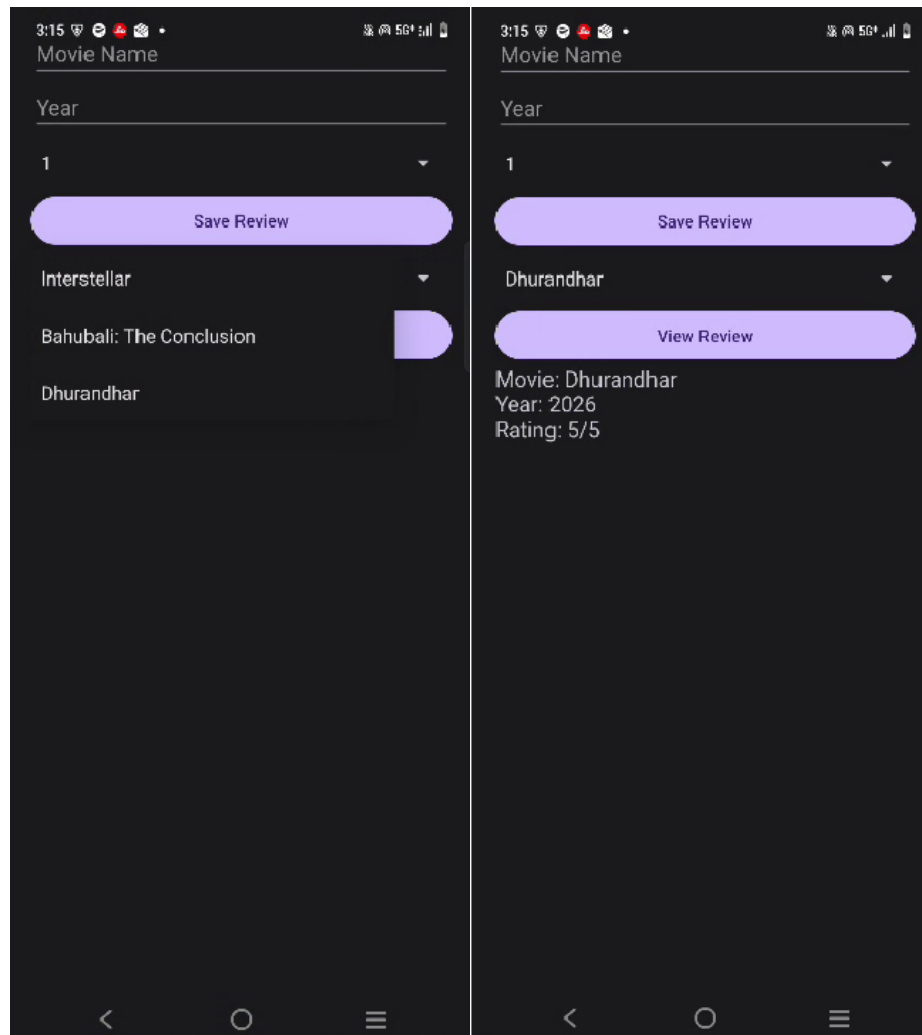


Fig 8.3.2: Application Screenshots

PROBLEM STATEMENT 4

Make use of SQLite browser to view the database ,tables created.Modify the tables and upload the same into your project and view the modified data through your app.

SOLUTION

Opening MovieDB using DB Browser for SQLite

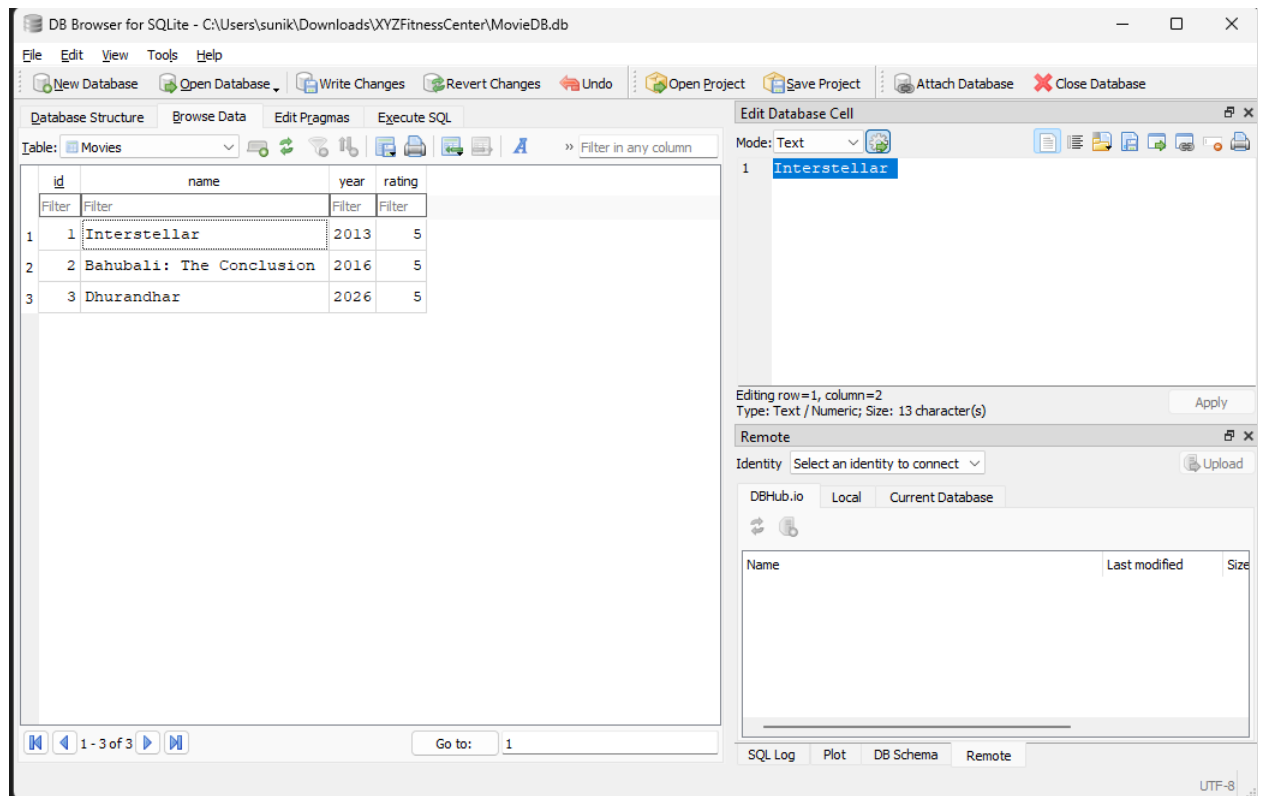


Fig 8.4.1: MovieDB contents displayed in DB Browser for SQLite

Inserting a new record in MovieDB using DB Browser for SQLite

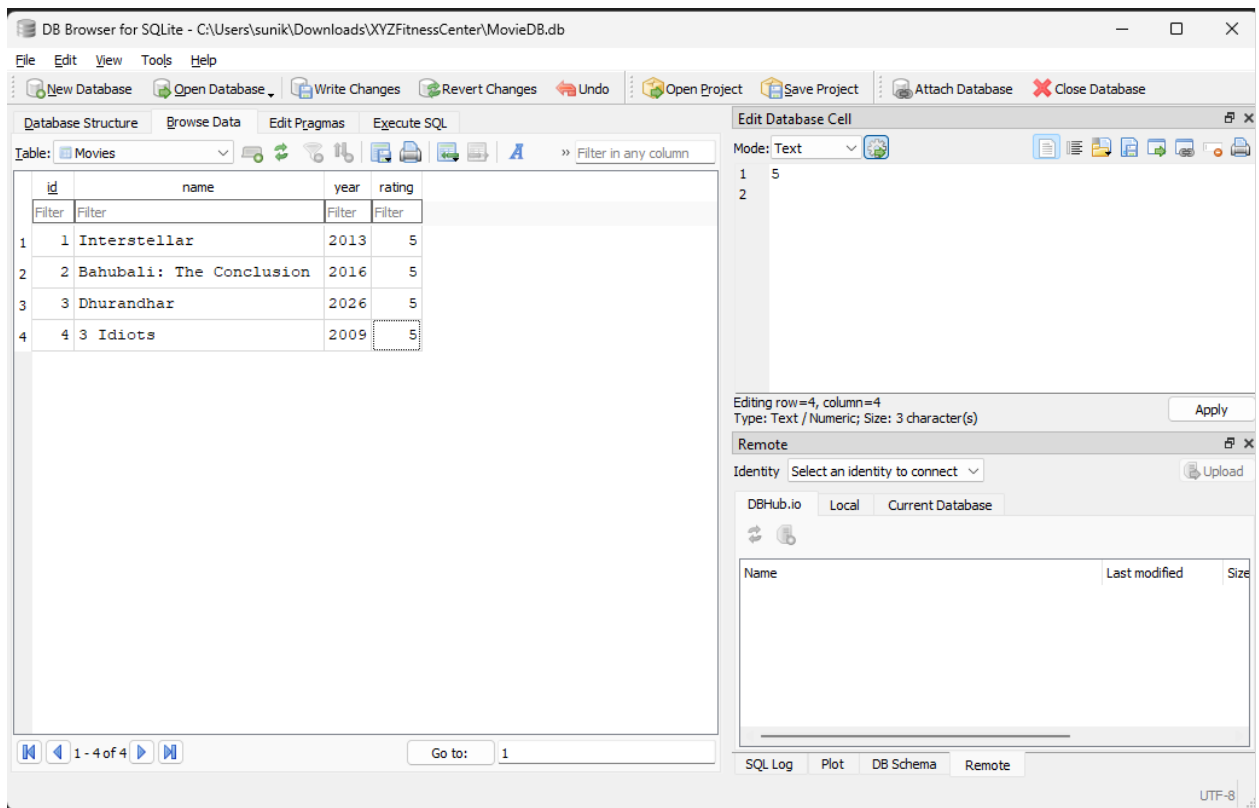


Fig 8.4.2: A new review for 3 Idiots added using DB Browser for SQLite

Viewing the newly added record in the application

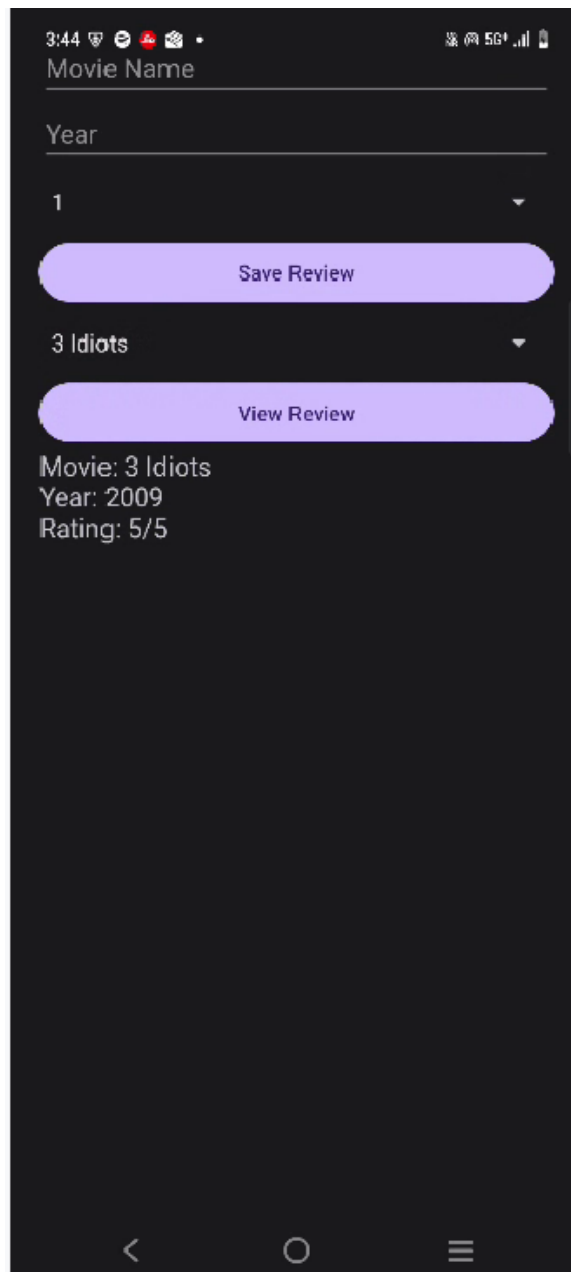


Fig 8.4.3: We can now access the reviews for 3 Idiots through our app

PROBLEM STATEMENT 5

Demonstrates the use of the Shared Preferences through an android application which displays a screen with some text fields. Save the value when the application is closed and brought back when it is opened again.

SOLUTION

Activity_main.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/name"
        android:hint="Enter Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/email"
        android:hint="Enter Email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/age"
        android:hint="Enter Age"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

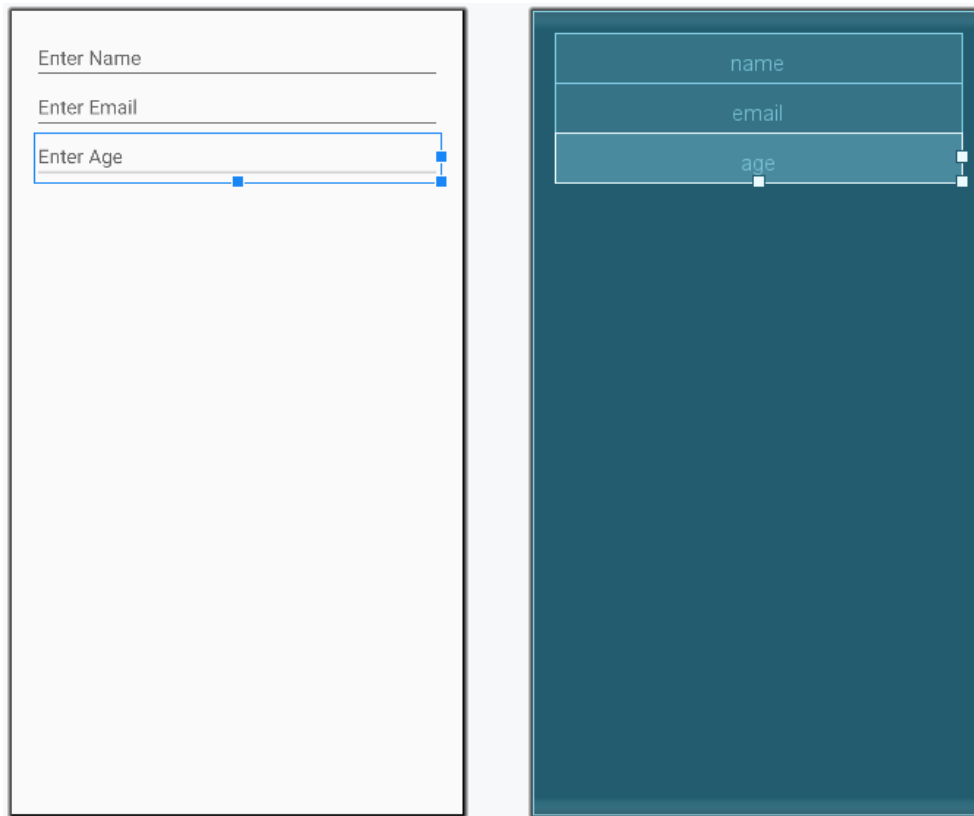


Fig 8.5.1: Main Layout

MainActivity.java

```
package com.example.lab8_q5;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText name, email, age;
    SharedPreferences sharedPreferences;

    @Override
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

        name = findViewById(R.id.name);
        email = findViewById(R.id.email);
        age = findViewById(R.id.age);

        sharedPreferences = getSharedPreferences("MyPrefs",
MODE_PRIVATE);

        // Retrieve saved values
        name.setText(sharedPreferences.getString("name", ""));
        email.setText(sharedPreferences.getString("email", ""));
        age.setText(sharedPreferences.getString("age", ""));
    }

    @Override
    protected void onPause() {

        super.onPause();

        SharedPreferences.Editor editor =
sharedPreferences.edit();

        editor.putString("name", name.getText().toString());
        editor.putString("email", email.getText().toString());
        editor.putString("age", age.getText().toString());

        editor.apply();
    }
}

```

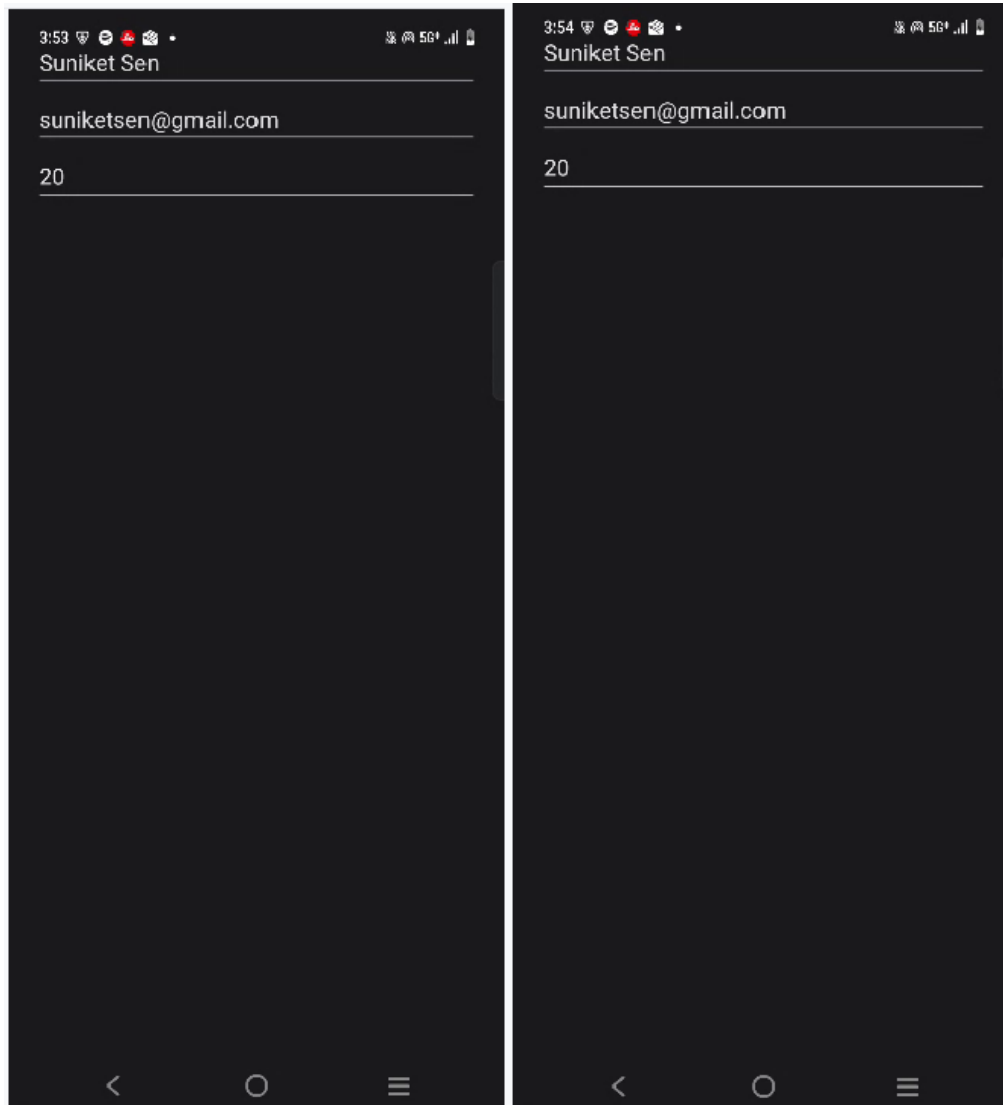


Fig 8.5.2: After the application is closed and reopened, the data is still there as the saved values are retrieved which were stored using SharedPreferences